

Oral Presentations

O1

TUMOR BIOLOGY AND METASTASIS

O1.1 PAR genes (PAR1 & 2): Probes for pathological assessment and targets for therapy

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Background: Taking the issue of breast tumor categorization a step forward and establish molecular imprints to accompany histopathological assessment is an important and timely task. Mammalian protease-activated receptors (PARs) comprise a family of four genes, uniquely activated by proteolytic cleavage. The central role played by PAR1, and PAR2 in tumor biology is currently emerging. We have identified PAR1 and PAR2 C-tails as scaffold sites for the immobilization of a PH-domain signal protein (e.g., Etk/Bmx) and established the minimal binding region within these tails.

Indeed, a mutant of consecutive A residues inserted into the proposed Etk/Bmx binding region of PAR1 C-tail (e.g., hPar1-7A) abolished PAR1 induced tumor growth, *in vivo*. Therefore, these sites provide a good platform to develop therapeutic modalities.

Aims: To elucidate molecular mechanism of PAR1 and PAR2 in breast cancer

Methods: Colony formation, immunoblots, immunofluorescence confocal analyses, xenograft mouse model for mammary gland tumor development.

Results: We have demonstrated that PAR1 and PAR2 act as one functional heterodimeric unit in breast cancer progression. As a consequence knocking-down the expression of PAR2 inhibits the function of PAR1. Silencing hPar2 with

sh-RNA attenuated PAR1 -instigated colony transformation, Matrigel invasion and abrogated the formation of a complex between Etk/Bmx or Akt and the PAR1-C-tail. In-addition when we used a non-functional PAR2 that lacks the entire cytoplasmic C-tail we showed that the functional form of PAR2 is required for PAR1-PAR2 heterodimer formation. In a xenograft mouse model, the non-functional PAR2 inhibited PAR1-PAR2-driven tumor growth, *in vivo* and the interaction of Etk/Bmx with the PAR1-C-tail, *in vitro*. Confocal images demonstrated the co-localization of PAR1 and PAR2 in MDA-231 breast cancer cells and co-immunoprecipitation analyses showed stable PAR1-PAR2 complex formation in HEK293T cells.

Conclusion: These observations confirm that PAR1 and PAR2 act together as a heterooligomer/heterodimer during tumor development. The functional requirement for heterodimerization between these PARs broadens the spectrum of future therapeutic modalities.

O1.2 PARP inhibitors: New agents to treat cancer?

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Background: The PARPs are a family of enzymes, best known for their role in single strand DNA repair. Inhibition of PARPs is currently undergoing intensive investigation for the treatment of several different types of malignancy. To date however, the efficacy of PARP inhibitors as monotherapy, has been mostly confined to patients with BRCA1/2-related breast and ovarian cancer. Because of the multiple biological similarities between BRCA1/2-associated and triple-negative breast cancer (TNBC), we hypothesised that

the latter mentioned subgroup of breast cancer patients may also be sensitive to PARP inhibitors.

Aims: The aim of this investigation was therefore to evaluate the anti-proliferative effects of 2 PARP or PARP-like inhibitors, ie, olaparib and iniparib on a panel of TNBC and non-TNBC breast cancer cell lines.

Methods: The antiproliferative activity of olaparib and iniparib was evaluated using MTT and colony formation assays.

Results: Using monolayer cultures and the MTT assay to assess cytotoxicity, IC50 concentrations for olaparib across the 12 cell lines ranged from 3.7–31 μ M, while IC50 concentrations for iniparib varied from 13–70 μ M. No difference in sensitivity was seen between the TN and non-TN cell lines. These results with olaparib in monolayer cultures were confirmed using clonogenic growth assays. However, with the latter, lower IC50 values were found, for olaparib, ie, ranging from <0.01–2.5 μ M.

Using immunohistochemistry, high nuclear staining intensity for PARP1 was more frequently present in ER-negative than ER-positive tumors, in high grade vis-a-vis low grade tumors and in samples with high Ki-67 values than those with low values. In contrast, nuclear PARP1 staining intensity was unrelated to tumor size, lymph node status, lymphovascular invasion or patient age at diagnosis. Of potential importance, high intensity staining for both nuclear PARP1 was found significantly more frequently in TN than non-TN cancers.

Conclusion: We conclude that: a) olaparib is a more potent inhibitor of in the vitro growth of breast cancer cells, b) that TN and non-TN breast cancer cells exhibit similar sensitivity to olaparib and iniparib and that c) PARP1 is widely expressed in breast cancer.

*Contributed equally to this work.

O1.3 L1 confers metastasis in colon cancer cells by activating NF- κ B signaling

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Background: Aberrant activation of β -catenin-T cell factor (TCF) gene expression is a common event in most human colorectal cancer (CRC) patients. Our studies identified the neuronal cell adhesion receptor L1 as a target gene of β -catenin-TCF signaling in CRC cells and tissue and showed

that L1 over-expressing human CRC cells when injected into the spleen of mice, exhibit liver metastasis.

Aims: To discover the molecular pathways whereby L1 promotes colon cancer metastasis.

Methods: Using human colon cancer cell lines, human colon cancer tissue samples, gene expression microarray analysis and mouse models, putative upstream and targets of L1 signaling were manipulated to identify molecules and signaling pathways essential for L1-mediated metastasis.

Results: We found that L1 is exclusively localized at the invasive front of human CRC tissue together with p65, a member of the NF- κ B pathway, and ezrin, that links cell-surface receptors to the actin cytoskeleton. L1 activates NF- κ B by inducing the phosphorylation and degradation of I κ B (the NF- κ B inhibitor), by an interaction with ezrin. The knockdown of NF- κ B activity and of ezrin levels abrogated L1-mediated metastasis. Global transcriptome analysis of CRC cells expressing L1, ezrin and modulators of NF- κ B signaling, identified up and down-regulated genes. Insulin-like growth factor binding protein 2 (IGFBP-2) was increased by L1 and by ezrin via NF- κ B-mediated transactivation of the IGFBP-2 gene promoter. IGFBP-2 increased cell proliferation, motility, tumorigenesis and induced liver metastasis, similar to L1. Suppression of endogenous IGFBP-2 in L1-transfected cells inhibited these properties. In addition, c-KIT, was downregulated by L1 and p65. Re-introduction of c-KIT abrogated L1-mediated metastasis by changes in E-cadherin expression and cell adhesion.

Conclusion: L1 promotes metastasis in CRC cells by activating the NF- κ B pathway, leading to changes in key cell adhesion and other molecules that confer metastasis in CRC.

O1.4 Splenectomy inhibits tumor growth causing regression and reduced metastases in murine lung cancer models

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Background: Novel treatments for lung cancer are desperately needed. During the last decade, it has been shown that inhibitors of the immune system (e.g. myeloid derived suppressor cells, MDSC) reside in the spleen and inhibit the endogenous anti-tumor effects of the immune system. Although many patients with cancer have relatively large numbers of cytotoxic T lymphocytes (CTL) directed against tumor antigens, they were not able to control the tumor

Aims: We hypothesized that excision of the spleen (splenectomy) can inhibit the growth of relatively big tumors, by modulating the systemic inhibition of the immune system. We aimed to find the effect of splenectomy on tumor growth and metastases, and elucidate the specific mechanisms of this effect. Our long-term goal is to possibly implement the approaches and mechanisms elucidated in these translational studies into clinical trials in patients with thoracic malignancies

Methods: We compared immunological properties of the blood and in tumor micro-environment following splenectomy compared to sham operation in tumor-bearing mice

Results: We found in several models that splenectomy reduces tumor growth, and even induces their regression, and decreases the development of metastases. This effect disappeared in NOD/SCID mice, suggesting that it is dependent on lymphocytes and myeloid cells. Although no significant changes in the different cell types was found in the blood, the percentage of intratumoral neutrophils out of total tumor cells in splenectomized mice was increased as well as the percentage of macrophages which tended to be less immune-inhibitory. We further noted increased activation of CD8⁺ CTL. The milieu of tumor microenvironment was found to be more pro-inflammatory following splenectomy with upregulation of chemokines such as MIG, TNF- α , and IFN- γ . We further performed studies with specific depletion of cells in order to find the specific role of each cell in the effect of splenectomy. In preliminary studies we found that splenectomy augments the effects of immunotherapy.

Conclusion: Splenectomy reduces tumor growth and metastases and can even cause tumor regression in murine models of lung cancer, by changing the amount and characteristics of myeloid cells, and enhancing activation of CD8⁺ T-cells.

exists that such cells are tumoral, and EMT in human cancer remains uncertain.

Aims: To demonstrate the existence of EMT-neoplastic cells in mesenchymal disguise in the stroma of human colorectal cancer (CRC).

Methods: In vitro, gene expression profile analysis of 18 CRC cells by microarray. Transfection/silencing of TWIST1 in CRC cells and invasion and migration assay. Immunohistochemistry for TWIST1 of 11 adenomas and of 201 CRC specimens. Combined TWIST1 immunofluorescence and in situ hybridization (iFISH) for CEP7 and 20 of 20 CRC specimens. In vivo, orthotopic graft of murine CRC cells CT26 in the rectal submucosa of Balb/c mice.

Results: We documented EMT in a human cancer cell-line, CoLo741, with mesenchymal phenotype (TWIST1⁺, CDH2⁺ and Vimentin⁺, CDH1-negative). TWIST1 is crucial in maintaining EMT and in increasing invasiveness of CRC cells. In 201 CRC specimens, TWIST1 expression in the stromal compartment was associated with advanced stage, and worse survival ($p=0.007$). None of 11 adenoma expressed TWIST1 ($p<0.001$ vs CRC). Out of 20 CRC, 17 showed chromosome 7 and 20 trisomic epithelial neoplastic cells, and in all these 17 cases, we traced the same trisomies only in TWIST1⁺ cells in the stroma. In a murine model of rectal cancer, Twist1⁺ CT26 cells switched to fibroblast shape once invading the stroma.

Conclusion: TWIST1 plays a crucial role in maintaining EMT in human CRC. By using TWIST1 as marker of EMT cells, we demonstrated that stromal cells, disguised as fibroblasts, are actually neoplastic. Targeting TWIST1⁺ stromal cells might open new scenarios for interfering with the metastatic cascade of CRC

O1.5 TWIST1 positive cells in the stroma of colorectal cancer: tumor cells in disguise

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Background: Epithelial-to-mesenchymal transition (EMT) increases cancer cells invasiveness in animal models, and expression of EMT markers (TWIST1, SNAI1 and ZEB2) in stromal cells has been associated with worse outcome in human cancers. Yet, no proof

O1.6 Rational design of heparanase inhibitors for cancer therapy

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Background: Heparan sulfate proteoglycans (HSPGs) not only provide a storage depot for heparin-binding molecules in the cell microenvironment, but also regulate their accessibility, function and mode of action. As such, they are intimately involved in modulating cell invasion and signaling loops that are critical for tumor growth. Heparanase, the sole heparan sulfate degrading

endoglycosidase, is causally involved in cancer progression and hence is a valid target for drug development.

Aims: Develop heparanase-inhibiting heparin-based compounds, small molecules and neutralizing monoclonal antibodies (mAb).

Methods: i) chemical modification of low molecular weight heparin aimed at abolishing its anti-coagulant activity and improving its heparanase-inhibiting capacity; ii) structure-based virtual screening of small molecules directed against the heparanase active site; and iii) generation of neutralizing anti-heparanase mAb.

Results: We have generated a novel chemically modified non-anticoagulant heparin (termed SST0001) that potently inhibits heparanase enzymatic activity and tumor growth and metastasis in xenograft cancer models. Compound SST0001 is a property of Sigma-Tau Research Switzerland S.A and is being subjected to phase I/II clinical trial in myeloma patients.

We are developing second generation oligosaccharide-based heparanase-inhibiting compounds that exhibit therapeutic along with pharmacological properties that will further advance translation of this therapy to the clinic. We have generated heparanase neutralizing mAb directed against the substrate binding domain of heparanase. These mAb inhibit heparanase enzymatic activity, cell invasion and experimental metastasis. Newly resolved structural features of the heparanase protein were applied for virtual screening and docking of small molecules directed against the heparanase active site. Several compounds were found to inhibit heparanase enzymatic activity, providing a basis for further development.

Conclusion: Rational design of heparanase inhibiting species of heparin, mAb and small molecules is ongoing. Selected compounds are being tested in animal models of cancer progression, alone and in combination with other anti-cancer drugs.

O2

CANCER STEM CELLS, CIRCULATING TUMOR CELLS

O2.1 Harnessing human es biomarkers for cancer therapeutics

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Background: On the cell surface, more than 85 % of proteins are glycosylated. Recent studies also present glycosphingolipids (GSLs) which are ubiquitous components of cell membranes and, similar to surface glycoproteins, which can act as mediators of cell adhesion and signal transduction.

Aims: A systematic survey of expression profiles of N-linked sialoglycoproteins and GSLs in human embryonic stem cells (hESCs) and various differentiated derivatives was carried out.

Methods: Glycoproteomics and glycan analysis as well as immunofluorescence and flow cytometry, were used to analyze the N-linked sialoglycoprotein/glycan expression patterns for hESCs.

Results: It was shown that seven newly found surface glycoproteins are expressed abundantly in hESCs and their expressions disappeared after hESC differentiation. The expression levels of these glycoproteins found in hESCs are many folds higher in cancer stem cells (CSC) as compared to non-CSC. Among them, ESC02 has been characterized in more details and the results warrant further development of ESC02 as therapeutic targets and detection agents. Based on MALDI-MS and MS/MS analyses, we also found a close association of the expressions of glycosphingolipids in hESCs with differentiation. In addition, Globo H, a known biomarker for cancer, was highly expressed uniquely in hESCs. We also showed that there was a striking switch in the core structures for globo- and lacto-series GSLs during differentiation of hESCs into embryoid body and three different germ layer cells (i.e. neural progenitors, endodermic, or mesoderm cells), suggesting a close association of GSLs in hESCs with lineage-specific differentiation.

Conclusion: Our studies suggest a close association of glycoproteins and GSLs with lineage specificity and differentiation of hESCs. Thus these results provide new insights into the unique stage-specific transition and mechanism for alterations of glycoproteins and GSL core structures during hESC differentiation. We believe that these newly found glycans present in hESCs could be candidates for cancer detection and glycan-targeted therapy of human tumors.

O2.2 Circulating tumor cell (CTC) counts and CTC telomerase activity (TA) are prognostic of overall survival (OS) in SWOG 0421: a Ph 3 trial in metastatic castration resistant prostate cancer (mCRPC)

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Background: CTCs are promising biomarkers in mCRPC, and telomerase activity (TA) is a recognized cancer marker. We explored their prognostic utility in S0421, a trial of docetaxel +/-atrasentan for men with mCRPC.

Aims: We investigated whether CTCs were prognostic of OS using 2 strategies: CellSearch for fixed cell enumeration, and a novel Parylene-C slot microfilter for live CTC capture and TA measurement.

Methods: Blood samples (7.5 ml) were drawn at baseline (d1) & pre-cycle 2 (d21). For CellSearch enumeration, Cox regression was used to evaluate the association between OS, d1 CTC counts, & CTC dynamics (d1 to d21). For TA, microfilter-trapped cells were lysed and assayed using qPCR-based telomeric repeat amplification. Cox regression was used to evaluate the association between OS and TA overall and in subgroups with good vs. poor prognosis (<5 vs. ≥5) d1 CTC counts. For all data, receiver operator characteristic (ROC) analysis and characteristics & regression trees (CART) were used to explore further prognostic cutpoints.

Results: Samples were obtained from 263 men. Median d1 CTC count was 5, and there was a significant difference in OS for d1 CTCs < vs. ≥5 with hazard ratio (HR) 2.92 (p<0.001) adjusting for other factors. In men with low d1 CTCs (< 5), an increase in CTCs was associated with shorter OS, HR 4.04 (p=0.004), whereas in men with high d1 CTCs (≥5), a ≥2-fold decrease in CTCs was associated with longer OS, HR 0.45 (p=0.012) adjusting for other factors. For TA, men with baseline CTCs ≥5 (41 % of cohort) who had high CTC TA had HR 1.14 (p<0.005) for OS adjusting for other factors. CART identified additional risk subgroups.

Conclusion: D1 CTC counts and d1 to d21 CTC dynamics were prognostic of OS, comprising the largest docetaxel-based prospective cohort to date validating this approach. In men with d1 CTCs ≥5, TA from cells live-captured on a new slot microfilter constitute the first CTC-derived biomarker prognostic of OS in a large Ph 3 clinical trial.

O2.3 Circulating tumor cells (CTCs) in breast cancer post-operative follow-up: a preliminary comparison with serum CEA-TPA-CA15.3 tumor marker (TM) panel based monitoring

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Background: In our Center, for many many years we have post-operatively monitored breast cancer patients with regular serum CEA-TPA-CA15.3 tumor marker (TM) panel in addition to clinical and instrumental exam. Recently, circulating tumor cells (CTCs) have been proposed as a potential tool in breast cancer monitoring.

Aims: We compared serum CEA-TPA-CA15.3 TM panel and CTCs in a cohort of breast cancer patients submitted to intensive follow-up.

Methods: Between November 2011 and June 2012, 18 breast cancer patients, aged 45 to 72 (56+11) years were enrolled. The stage was I in 2, II in 5 and III in 6 patients respectively. Five patients had diffuse metastatic disease at the enrolment and were treated with systemic therapy. The mean follow-up was 8.2+2.5 months (range 6-12). CTCs determination was performed every 3 months by The Cell-Search System. Cut-off value was 5 cells/7.5 mL.

Results: So far, 3 patients relapsed during the follow-up. In 1 (33 %) of them at the time of the instrumental diagnosis, CTCs value was between 1 and 4 cells/7.5 mL. In the 2 remaining, CTCs were negative. In these 3 patients, an increase in TM panel preceded the diagnosis of distant metastases. Three of the 10 other ones who remained disease-free had negative CTC values and 7 had CTC between 1 and 4 cells/7.5 mL in at least one sample. In these patients, TMs were constantly in the normal range. As to the 5 patients with metastases at the beginning of follow-up, in 2 (40 %) CTCs were negative or between 1 and 4 cells/7.5 mL and serum TMs reflected a stable disease. In the 3 (60 %) remaining CTCs were >4 cells/7.5 mL. These 3 patients had diffuse metastatic disease. In 2 of them CTCs and TMs increased during the follow-up and instrumental exams documented a progressive disease (PD). In another one with partial response (PR) CTCs and TMs progressively decreased. In patients with PR and PD, the TM variation preceded 1 to 3 months the CTC variation.

Conclusion: CTC determination had lower sensitivity than CEA-TPA-CA15.3 TM panel for early detection of relapse. As to metastatic patients, our preliminary data suggest it can be an helpful tool in addition to the serum TM panel.

O2.4 Expression of CD133 and nestin in sarcomas.

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Background: CD133 (prominin-1), a plasma membrane glycoprotein, usually in combination with other specific markers, is widely used to identify stem cells in various human tissues including neoplasias. Similarly, nestin, a class VI intermediate filament protein was previously described to be marker of neural stem cells and is expressed in many types of human tumors.

Aims: Our study was aimed at a detailed study of expression CD133 and nestin in osteosarcomas and rhabdomyosarcomas, as well as in cell lines derived from these tumors.

Methods: 45 samples of osteosarcoma tissues and 10 samples of rhabdomyosarcoma tissues were included in this study. Furthermore, we analyzed 5 osteosarcoma and 5 rhabdomyosarcoma cell lines. Immunohistochemistry, immunofluorescence, flow cytometry and immunoblotting were employed for detection of CD133 and nestin. To confirm a presence of tumor-initiating cells in cell lines, clonogenic assay and test of tumorigenicity in vivo were performed.

Results: In all osteosarcoma and rhabdomyosarcoma samples both CD133 and nestin were detected, although the frequency and intensity of staining varied among samples. The presence of CD133 and/or nestin positive cell was also clearly confirmed in all sarcoma cell lines included in this study. Moreover, functional assays showed a presence of tumor-initiating cells in sarcoma cell lines.

Conclusion: Although a co-expression of these two markers is recognized as a phenotype of cancer stem cells particularly in neurogenic tumors, we identified the expression of CD 133 and nestin in all sarcoma samples and in all cell lines derived from them. The low incidence of CD133-positive cells in tumor tissues and of CD133/nestin-positive cells in cell cultures, as well as results of preliminary functional assays suggest a possible stem-like phenotype of cells showing co-expression of these markers.

Supported by grants IGA MZCR NT13443-4, VZ MSM 0021622430 and CZ.1.07/2.3.00/20.0183

O2.5 Tumor stem cells: Do they really exist?

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Background: There is considerable debate in the cancer literature as to whether or not tumor stem cells exist. This in part is due to a lack of an accepted definition for a tumor stem cell, what molecular markers they express, and what cellular characteristics they express.

Aims: This study is directed at assessing the available data that supports the contentious idea that tumors are made of both tumor cells and tumor stem cells.

Methods: A review of the literature was conducted. The High Nitric Oxide (HNO) adapted cell line model (A549 and A549-HNO) was used to assess tumor stem cell markers like CD antigen expression using immunostaining. Drug library screening (via MTT assay) was used to determine if there were molecules selective for either tumor cells and/or tumor stem cells. Scratch and adhesion tests were conducted to assess the metastatic potential of these cells. ALDH levels of expression (via gene chip analysis and immunostaining) were also studied. ALDH activity was also assessed (via an enzymatic assay).

Results: Emerging data to support at least part of the tumor stem cell theory was found in the literature. The immunohistochemical analysis indicated cells were positive for CD antigen expression. As reported in the literature, other researchers have used drug libraries to screen for compounds that selectively kill tumor stem cells, as did our drug library screen. Scratch test showed that HNO cells migrate faster than the parent cells, and adhesion cell assays showed that the HNO cells were less adherent to each other when compared to the parent cells. ALDH expression was up regulated in gene chip experiments, and up-regulated in HNO cells when compared to the parent cell line, using immunohistochemical staining or by enzymatic activity.

Conclusion: There is a growing body of data to support the idea that there are tumor stem cells. The question still remains, what are the final tests and markers that will be used to define these cells.

O2.6 Mouse models for evaluation of the risk factors for human hepatocellular carcinoma

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Background: The ability to knock out or knock in genes in mice provides many models for study of human diseases. Such mouse models of cancer are essential for understanding the origin, progression, treatment and prevention of cancer. Epidemiologic evidence indicates that the major risk factors for human HCC are male gender, age, hepatitis injury, aflatoxin (AFB1) exposure and alcohol consumption. In contrast to humans, rats, ducks and other AFB1 sensitive species, adult mice are highly resistant to AFB1. However, up to about 3 weeks of age mice are very sensitive to AFB1, but older wild type mice are very resistant.

Aims: To use transgenic and knockout mice to develop models for analyzing the risk factors for human liver cancer.

Methods: The effect of aflatoxin exposure is tested on transgenic models of hepatitis, as well as mice lacking p53 (knockout) and carrying the p53 mutation (p53ser246) analogous to the p53 mutation associated with human HCC (p53ser249). We now report results using a new mouse lacking the critical enzyme for detoxifying aflatoxin (mouse glutathione-S-transferase A3; mGSTA3).

Results: : Using a large cohort of mice bred to carry various risk factors we found that male gender, age, p53 loss or mutation and hepatitis injury each increased AFB1 hepatocarcinogenicity. On careful analysis of the states of the increased sensitivity a common factor found was increased proliferation of hepatocytes; especially in the sensitivity of newborn mice. A major difference between newborn and adult mice is the low level of glutathione-S-transferase, namely mGSTA3 in newborn, as compared to adult mice. To test this further we produced an mGSTA3 knockout (KO) mouse line. This is essentially a “humanized” mouse in respect to AFB1 metabolism as humans do not have the equivalent of mGSTA3. Our preliminary studies show that the KO mice are highly sensitive to AFB1 by DNA-AFBO adduct formation, oval cell generation and production of HCCs.

Conclusion: The mGSTA3 KO mouse provides a model to test the hypothesis that low levels of AFB1 exposure combined with alcohol exposure may act synergistically to cause HCC. In addition, the finding of large numbers of oval cells after AFB1 exposure in the KO mouse will allow us to determine the role of oval cells in development of HCC.

O3

CANCER IMMUNOLOGY

O3.1 Development of a signature to predict response to cancer immunotherapy

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Background: The unique feature of cancer immunotherapy is that the clinical outcome integrates biological traits of the tumor and immune competence of the host. Although the literature contains a large volume of data on tumor biomarkers, only few have been validated as predictors of outcome in melanoma patients undergoing immunotherapy. This issue has become critical with the US FDA approval of ipilimumab (Yervoy), a monoclonal antibody blocking CTLA-4 receptor on lymphocytes. The issue of patient selection for ipilimumab is critical because of its expense, toxicity and the relatively limited proportion of patients deriving clinical benefit.

Aims: In this study we aim at identifying a selected group of biomarkers derived from tumor and peripheral blood, which taken together can predict the chances of deriving benefit from immunotherapy.

Methods: We based the initial signature on a homogeneous group of patients receiving autologous melanoma vaccine following tumor resection (adjuvant cohort). Melanoma lines from 35 immunized patients, consisting of good- and poor responders, were used to generate a gene expression array. Gene expression profiling was performed using a Luminex-based assay that was developed at the Broad institute. The genes were tested separately employing a bioconductor (R) based test (SSAT) followed by a second test based on rank statistics that determines the best cut-off value that separates expression values into those associated with good versus bad survival.

Results: Based on the gene expression data from good versus poor responders, an initial predictive signature has been established. This signature includes 16 genes related to cancer-testis antigens, NK cell receptors/ligands and transcription factors. The expression profile has been validated by qPCR and, at least for some of them, on a protein level.

Conclusion: The initial predictive signature will be validated and enlarged by a retrospective cohort of ipilimumab

treated patients, based on analyses of tumor and peripheral blood.

O3.2 Clinical utility of circulating immunogenic cell death biomarkers in cancer patients

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Background: Cellular products that are released during apoptosis or necrosis as a consequence of disease or cytotoxic therapy may lead to a sustained stimulation of immune cells. Modulation of this immunogenic effect in the tumor microenvironment plays a major role in cancer development and in the persistent response to cytotoxic therapies in cancer.

Aims: In diverse prospective trials on 49 patients with colorectal cancer with liver metastases undergoing selective internal radiation therapy (SIRT), on 50 patients with primary liver cancer undergoing transarterial chemoembolization (TACE), and on 44 patients with localized breast cancer receiving neoadjuvant systemic chemotherapy, blood samples were taken before as well as during the course of the treatments. In local SIRT and TACE trials, blood was taken daily after application to see the immediate effect of therapy on marker levels.

Methods: High mobility group box 1(HMGB1) and receptor of advanced glycation end products (RAGE) were quantified by ELISA. Therapy response was quantified by radiology, outcome was objectified as progression-free and overall survival.

Results: In cancer patients, levels of HMGB1 were generally higher than in control groups while RAGE levels were decrease. Further, HMGB1 levels rapidly increased already 24 hours after application of local SIRT or TACE therapies whereas RAGE levels were further decreased. In CRC patients undergoing SIRT, elevated HMGB1 concentrations were associated with poor therapy response (24 h value) and poor overall survival (0 h and 24 h values). In liver cancer patients undergoing TACE, low 0 h and 24 h RAGE levels were associated with insufficient therapy response and unfavourable overall survival. In breast cancer patients, pretherapeutic high HMGB1 and low RAGE concentrations were associated with poor response to neoadjuvant chemotherapy as well, while CEA and CA 15-3 were not.

Conclusion: Proimmunogenic cell death serum marker HMGB1 and antiimmunogenic soluble RAGE are new and valuable biomarkers for the prediction of therapy response and prognosis in cancer patients.

O3.3 Tumor-associated neutrophils – Friend or Foe?

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Background: We identified two phenotypes of tumor-associated neutrophils (TAN) in murine models dependent on TGF-beta. TGF-beta blockade increased neutrophil-attracting chemokines resulting in an influx of CD11b+/Ly6G + TAN that were hypersegmented, more cytotoxic to tumor cells, and expressed higher levels of pro-inflammatory cytokines. Depletion of these TAN blunted anti-tumor effects of anti-TGF-beta. In control tumors, neutrophil depletion decreased tumor growth and activated CD8+ T-cells. We suggested different populations of TAN – pro-tumor (N2) and anti-tumor (N1) phenotype.

Aims: We aimed to further characterize differences between populations of neutrophils related to cancer, i.e Myeloid derived suppressor cells (MDSC), and N1/N2-TAN. We further hypothesized that TAN phenotype is dependent on the stage of tumor development. Lastly we preliminarily evaluated the presence of TAN in human cancer.

Methods: Using a transcriptomics approach, we compared N1/N2 TAN to naïve neutrophils (NN) and to the granulocytic MDSC. We compared TAN found in tumors at early and late stages of tumor development using FACS, cytotoxicity assays, ELISA and RT-PCR.

Results: NN, TAN and MDSC are significantly different in their mRNA profiles. Structural genes and genes related to cell-cytotoxicity were significantly down-regulated in TAN. Many immune-related genes, including cytokines, were up-regulated in G-MDSC, and further-up in TAN. Of 25 chemokines, 13 were markedly up-regulated in TAN compared to NN, including chemoattractants for T/B-cells, and macs. N1 TAN appear to make more macrophage chemokines (e.g. CCL2) and antigen processing/presentation genes. Only 1 major chemokine, the T-regs chemoattractant CCL17, was up-regulated and functionally active in N2 compared to N1 TAN. TAN at early stages of tumor growth were more cytotoxic to tumor cells directly and indirectly. Systemic neutrophil depletion reduced tumor growth only after tumor establishment. Preliminary data in human lung cancers indicates that about 5-7 % of the isolated CD45+ cells have neutrophil markers. Their phenotype remains under study.

Conclusion: We added here another layer on the data on TAN. Understanding the effect of tumor on neutrophils, and

the way these cells support or fight cancer, will help developing strategies to direct the immune system against the tumor.

O3.4 The interrelation between cancer, chronic inflammation and immunosuppression: biomarkers and clinical implications

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Background: The chronic inflammation-cancer linkage is bidirectional as chronic inflammation can predispose an individual to cancer, and developing tumors in turn, can induce chronic inflammatory environments. Under both settings, immunosuppression ensues and limits success of immune-based and conventional therapies, skewing the balance in favor of tumor escape.

Aims: We aim at delineating the factors within the chronic inflammatory milieu responsible for the induced immunosuppression and define their impact on the individuals' immune function and tumor development. We also seek at discovering drugs that could combat the immunosuppressive environment and increase immune system potency. Along side, unique biomarkers that sense the chronic inflammatory environment and associated hosts' immune status are tested as predictors of therapy success, evaluators of therapeutic strategies and disease regression.

Methods: Mouse models for spontaneously occurring inflammation-based colon cancer, and chronic inflammation are used. Blood samples from cancer patients are analyzed for the relationship between the patients' immune status and success rates of applied chemo- and immune-based therapies, based on novel biomarkers.

Results: We show that myeloid derived suppressor cells mediate chronic inflammation-induced immunosuppression, leading to enhanced tumor growth and affect chemo- and immune-based therapies. Key environmental factors were identified as supporting the immunosuppressive cells. The novel biomarkers used herein were shown as efficient in monitoring the immune status of tumor-bearing hosts, the success of applied therapies and disease progression.

Conclusion: Delineating the molecular basis for tumor-associated immunosuppression along side with monitoring the patient's immune status, are expected to lead to a well-

designed and fine-tuned immune intervention that could resolve inflammatory responses and associated immunosuppression, leading to enhanced disease regression and reinforced successful responses to given therapies. These could facilitate designing of innovative personalized therapeutic strategies for cancer patients.

O3.5 Harnessing the innate immunity (ivig) to treat autoimmunity and malignancy – novel mechanisms

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Background: IVIG is a human serum Ig fraction that is mainly composed of IgG prepared from plasma pools of over 15,000 healthy blood donors and is suitable for I.V. use. IVIG is employed to treat several autoimmune conditions. The effect of IVIG in autoimmunity is multi-mechanistic and entails anti-idiotypes, cytokine effect, anti-complement and others.

Aims: Due to the similarity and overlap between autoimmunity and cancer, we had evaluated the efficacy of IVIG in prevention of metastatic spread.

Methods: The impact of IVIg on the invasive and metastatic capacity of murine cell carcinoma using in vivo and in vitro model systems, was delineated;

Results: Systemic treatment with IVIg showed the inhibition of tumor lung metastatic potential of CT26 cells. Reduction in size of tumor nodules and lung weight (38.9 %, 56.4 %, and 51.5 % in mice treated with 2, 0.2 and 0.04 mg/mouse, respectively) was observed. Incubation of tumor cells with 30–100 mg/well of IVIg for 72 h resulted in 40–60 % inhibition of cell invasion across a layer of extracellular matrix. Furthermore, treatment with IVIg attenuated the proliferation and/or growth potential of tumor cells in a dose-dependent manner.

Conclusion: IVIG was effective in reducing significantly metastases of carcinoma, melanoma and sarcoma. The mechanisms involved in these effects include:

- 1) Neutralization of pathogenic autoantibodies via idiotypic and anti-idiotypic antibodies.
- 2) Modulation of cytokines secretion.
- 3) Effects on the Fas apoptotic pathway via agonistic and antagonistic anti-Fas autoantibodies.
- 4) Regulation of complement proteins.
- 5) Fc-receptor blockade.
- 6) Inhibition of NF- κ B activation.
- 7) Cell-cycle alteration

- 8) Effect on Treg cell.
- 9) Inhibition of matrix metallo proteinase 9.
- 10) Modulation of cell-proliferation via cytokines.

Many of the mechanisms may explain the dual beneficial effects of IVIG in autoimmunity and cancer. IVIG serves as an example of harnessing the innate immunity for treating incapacitating medical conditions.

O3.6 Immunotherapy of cancer by IL-2 activated intentionally mismatched T & NK cells targeted by monoclonal or bispecific antibodies against tumor markers

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Background: We tested the hypothesis that intentionally mismatched alloreactive T & NK cells activated with IL-2 can eliminate malignant cells resistant to anti-cancer modalities. Graft-vs-tumor effects following allogeneic stem cell transplantation (SCT) are well established, however, SCT is associated with unavoidable hazardous graft-vs-host disease (GVHD).

Aims: Investigate the potential therapeutic effects of short-term immunotherapy induced by intentionally mismatched alloreactive killer lymphocytes (IMAK).

Methods: Our experimental model was based on the use of trifunctional bispecific antibody (BSA) directed against murine CD3 and human epithelial-cell adhesion molecule (EpCAM) testing the feasibility of cell-mediated adoptive immunotherapy in a murine model of B16 melanoma cells transfected with human EpCAM. We confirmed that IMAK guided by BSA can eliminate residual cancer in vivo while avoiding GVHD. Furthermore, successfully treated mice could resist a fresh tumor challenge, suggesting development of anti-cancer immunity probably by binding of Fc of BSA to host dendritic cells. Assuming that the efficacy and selectivity of anti-cancer immunotherapy mediated by IMAK can be improved by targeting such effector cells with monoclonal or bispecific antibodies against antigens over-expressed on the surface of the malignant cells, we pioneered the use of IMAK guided by monoclonal antibodies against EGFR, VEGF, Her-2/neu, CD20 or BSA against EpCAM using commercially available Erbitux, Avastin, Herceptin, MabThera or bispecific anti-EpCAM.

Results: We have confirmed that treatment with IMAK with or without (n=40) targeting antibody is reasonably safe and can result in elimination of minimal residual disease with no clinically significant GVHD in patients with hematological malignancies, while resulting in progression-free survival in a proportion of patients with

metastatic solid tumors fully resistant to conventional anti-cancer modalities.

Conclusion: Based on successful experiments in mice and pilot clinical experience we conclude that treatment with short-lived IMAK $\pm$ targeting antibodies against malignant cells may represent an effective future approach for cancer immunotherapy

O3.7 IL-1 is a major molecule that controls the balance between inflammation and immunity at tumor sites

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Background: Both IL 1 β and IL 1 α are produced by normal as well as malignant cells in the microenvironment of tumors. However, IL 1 β and IL 1 α differ in their effect on the malignant process. IL 1 α is expressed as a transmembrane cytokine and activates adjacent cells via a juxtacrine mechanism. In contrast, IL 1 β is active only in its mature secreted form following cleavage by caspase 1.

Aims: The mechanisms by which microenvironment-derived IL-1 perpetuates the balance between invasiveness and immunity have been assessed.

Methods: By using IL-1 knockout (KO) mice, we have studied the role of IL 1 during the process of chemical carcinogenesis induced by 3-methylcholanthrene (3-MCA).

Results: Mice deficient in the naturally occurring IL 1Ra develop more tumors, multiple metastases and prominent angiogenesis, while, mice deficient in IL 1 β exhibited markedly reduced tumor development and invasiveness. The patterns of tumor growth were similar in IL-1 α and control mice. Thus, a role for host-derived IL 1 β in promoting the malignant process has been established, where, host-derived IL-1 α was essential for the process of immunomodulating the repertoire of the newly formed neoplastic cells. In IL-1 α KO mice, 3-MCA-induced tumors were highly immunogenic compared to WT mice. IL-1 β of host origin was shown to mediate infiltration of myeloid cells of bone marrow origin into tumors, recruit endothelial precursors and activate local endothelial cells for VEGF production. IL-1 β of host myeloid origin is pro-inflammatory and increases cell adhesion, tumor invasiveness and production of a local network of cytokines/chemokines. Secreted IL 1 β also induces immune anergy, mediated by myeloid-derived suppressor cells recruited from the bone marrow.

Conclusion: Thus, IL-1 serves as a major molecule that tilts the balance between destructive inflammation and activation of anti-tumor immunity. Treatment of tumor-bearing mice with the IL-1Ra reduces tumor angiogenesis and invasiveness, supporting the possible therapeutic use of IL 1 β in tumor therapy.

O4 EPIGENOMICS / PROTEOMICS

O4.1 Exploring proteomics in bladder cancer: functional and biomarker studies

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Background: Proteomic approaches are accelerating the discovery of the molecular alterations involved in tumorigenesis and bladder cancer progression by the combination of analyses characterizing cultured cancer cells in vitro and also human clinical material both using tissue and body fluids. Such alterations may represent critical molecular events that could be developed into potential targets for therapeutic interventions and also be translated into cancer biomarkers.

Aims: To describe proteomic techniques through applications for functional and biomarker identification analyses in bladder cancer.

Methods: Several proteomic strategies to identify critical pathways and candidates involved in bladder cancer initiation, progression and metastasis will be discussed such as iTRAQ and SILAC approaches. With the aim of diagnosing and predicting clinical behavior for patients with bladder cancer, proteomic technologies will be described using clinical material. Discussed tissue strategies will include tissue and reverse phase arrays, serum approaches will refer to antibody and protein arrays, while DIGE and MS technologies applications will be described for urine samples.

Results: Quantitative proteomics is strongly contributing to the discovery of candidate disease-specific target and biomarkers. While protein and antibody arrays permit differential quantification of known proteins, mass spectrometry techniques are leading for protein identification. The potential for expanding our understanding of bladder carcinogenesis can be explored in cancer cells in vitro and compared to the molecular alterations found in tumor specimens. Furthermore, the application and optimization of high-throughput proteomic technologies in non-invasive specimens such as blood and urine samples would contribute to translate proteomic alterations characteristic of bladder tumors into potential biomarkers that could be applied to the clinics.

Conclusion: Application of current proteomic technologies to in vitro models, and human tissue and body fluids samples shows promise as strategies enabling pathway characterization for functional analyses and biomarker identification

towards improving the management of bladder cancer patients.

O4.2 Histone modifications on circulating nucleosomes as a new class of cancer biomarkers

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Background: Post-translational modifications and alternative variants of histones play an essential role in diverse cellular processes like gene expression, DNA replication and repair, chromosomal condensation, segregation and apoptosis. Thereby, the concerted interaction with other epigenetic changes like DNA methylation and histone-binding proteins is highly important. A dysregulation of the histone code is involved in the development and progression of tumor diseases

Aims: Modified histones can be detected in tumor tissue and on circulating nucleosomes in the blood suggesting altered histones in the serum as new biomarker class for cancer detection

Methods: Typical changes found in cancer tissue are a global loss of the acetylation of lysine 16 on histone 4 (H4K16ac) and of trimethylation of lysine 20 on H4 (H4K20me3) leading to an increased DNA instability. Loss of the activating H3K4 methylation and H3K9 acetylation as well as increase of suppressive H3K9 and H3K27 methylation result in a reduced transcriptional activity that is particularly relevant in tumor-repressive gene regions. However, findings showed discrepancies in diverse tumor entities. Concerning histone variants, H2A.1 was found to be overexpressed in cancer while H2A.2 as well as MacroH2A1.1 and MacroH2A2 were underexpressed and correlated with poor prognosis.

Results: In blood plasma, levels of H3K9me3 and H3K20me3, measured by chromatin immunoprecipitation and subsequent qPCR, were found to be elevated in breast cancer and myeloma but decreased in colorectal cancer pointing on diverse pathogenetic mechanisms in these cancers. Recently, immunological assays for histone modifications and variants have shown good analytical performance and good discriminative potential for cancer detection.

Conclusion: Current studies that compare the new histone modification markers with established oncological biomarkers in diverse cancers will be presented.

O4.3 proteomics analysis of human colorectal carcinoma and polyps

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Background: Early diagnosis of colon cancer, which enables efficient full dissection, provides the greatest improvement in survival rates. The availability of tumor markers on the tissue and in the peripheral blood is major goal of modern biomedical research.

Aims: In order to identify colorectal carcinoma (CRC) proteins that can serve as biomarkers or as targets for therapeutics, a large scale analysis was performed while looking for proteins whose levels of expression are elevated in the tumor tissues relative to the tumor margins.

Methods: A large scale proteomics analysis was performed while looking for proteins whose levels of expression are elevated in the tumor tissues relative to the tumor margins. The comparative proteomics analysis included CRC tumors and polyps collected from specimens of 52 patients undergoing surgery. The assays were based on both stable isotope labeling by reductive dimethylation and on label-free analyses, using two-dimensional capillary chromatography and tandem mass spectrometry.

Results: A few thousands proteins were identified, quantified and sorted according to their expression levels in the polyps or the tumors, relative to the ‘healthy’ margins. Above fifty proteins were detected with at least three fold higher levels in more than half the tumors and similar numbers were detected at significantly lower levels. Clustering of the data, according to the protein repertoires, points to the existence of subtypes of CRC and to strong similarities in the proteomes of the different polyps of the same patients. Some of the proteins of interest were further validated by specific antibodies.

Conclusion: This identification of a number of differentially expressed proteins in the tumors and polyps relative to the margins, include some potentially useful tumor biomarkers and candidates for targeted therapy. Furthermore, the protein patterns segregate into clusters that suggest the potential association with subtypes of the disease.

O4.4 Tissue tropism of oncolytic viruses and potential application in head and neck cancer

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Background: Oncolytic viruses preferentially infect and kill cancer cells while sparing normal ones. Oncolytic viruses can also be used as vectors for gene expression of anticancer proteins to be delivered to the tumor site. Advanced Head and Neck cancer is currently treated by a combination of surgery and chemo radiotherapy. Despite this aggressive treatment the quality of life of patients and survival are still very low.

Aims: To develop ex vivo organ culture model for human head and neck cancer. To apply the organ culture technique for the purpose of assessing the tissue tropism of oncolytic viruses using human head and neck tumor tissue. To apply the ex vivo model for the prediction of individual patient treatment response to current treatments and to oncolytic virus therapy.

Methods: After receiving informed consent before surgery, fresh human tumor tissues are grown as organ culture by a method developed in our lab. Tissue viability is tested by histology analysis and glucose uptake. Tissues are infected ex vivo by oncolytic Newcastle disease virus (NDV-HUJ) and herpes simplex-1 virus (HSV1). Viral replication and oncolytic activity are assessed by molecular technologies, histopathology and immunohistochemistry.

Results: Human head and neck tumor tissues were maintained viable for up to four days after surgery in organ culture, as indicated by glucose uptake and preservation of histological appearance. NDV-HUJ infection of the corresponding tissues showed a differential infection pattern between normal and cancer tissues with high preference to the cancer tissue. The oncolytic activity corresponded with a high rate of apoptosis and a low rate of proliferation, as shown by immunohistochemistry.

Conclusion: Human Head and Neck tumor tissue can be maintained viable ex vivo and can be used for the evaluation of new treatment modalities. We developed a new model for predicting personal treatment response that can be applied for selecting specific treatments as well as viral strains for adjuvant anticancer therapy.

O4.5 Epigenome deregulation in hepatocellular carcinoma: elucidating molecular mechanisms and discovering novel biomarkers

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Background: Liver cancer is one of the most frequent human cancer and a major cause of cancer-related death worldwide, owing to late diagnosis and resistance to therapies. Molecular mechanism by which specific risk factors promote the development and progression of liver cancer is still a mystery. Curiously, there are no consistent genetic changes associated with hepatocarcinogenesis, and only a handful of genes are known to be frequently mutated in liver cancer. This has led to research into alternative mechanisms by which viral infection and excessive alcohol intake make promote hepatocarcinogenesis.

Aims: We aim to identify epigenetic changes (both DNA methylation and non-coding RNAs) in hepatocellular carcinoma (HCC, the most common liver cancer) associated with viral HBV infection.

Methods: We combined the advantages of genome-wide and quantitative measurement of DNA methylation levels and non-coding RNAs in series of HCC tumours associated with primary risk factors and collected from different geographical areas.

Results: Our analyses revealed a high frequency of aberrant hypermethylation of specific genes and pathways as well as non-coding RNAs in liver tumours as compared to control cirrhotic or normal liver tissues, consistent with the notion that HBV infection and/or integration of HBV genome may trigger aberrant epigenetic events during cancer development and progression. Importantly, we also observed that promoter hypermethylation occurred concurrently in specific set of genes in liver tumours and cirrhotic tissues, indicating that concurrent hypermethylation of certain genes in liver cancer may be specific to HBV infection. We have further carried out detailed functional studies using primary hepatocytes and putative liver progenitor/stem cells to define the mechanism by which HBV-dependent epigenetic disruption in liver cells may promote the development of liver cancer and the results of these analyses will be reported.

Conclusion: Together, our study identifies aberrant epigenetic changes in specific genes and pathways in liver cancer induced by HBV infection, providing information

that could be exploited in development of novel strategies for early detection, targeted therapies, and prevention of liver cancer.

O4.6 A novel panel of tumor suppressor genes for non-invasive diagnosis and prognosis of bladder cancer: a two center prospective study

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Background: Changes in DNA methylation of tumor suppressor genes (TSGs) occurring early in carcinogenesis, represent potential indicators for cancer detection and disease evolution.

Aims: This study examined the role of the methylation of a panel of TSGs measured using a multiplexed approach as tumor stratification, diagnostic and prognostic markers for bladder cancer.

Methods: The methylation of 18 TSGs was evaluated using a methylation-specific multiplex ligation-dependent probe amplification assay (MS-MLPA) in bladder cancer cell lines (n=14), fresh-frozen primary bladder tumor specimens (n=61), and two prospective independent sets of urinary specimens (training, n=120; and validation, n=128).

Results: Nine genes were discovered methylated in bladder cancer. In tumors, RUNX3 (p=0.026), TWIST1 (p=0.009), SFRP4 (p=0.002) and CCND2 (p=0.027) methylation correlated to stage. Univariate analyses indicated prognostic correlations for recurrence, disease-specific, and overall survival, remaining independent prognosticators in multivariate analyses in tumors. CCND2, SCGB3A1 and RUNX3 were the most frequently methylated TSGs in both urinary sets. ROC analyses revealed significant diagnostic accuracies for RUNX3 in both series. Using univariate and multivariate analyses, methylation correlated to recurrence in the training set, while methylation was associated with recurrence and disease-specific survival in the validation set.

Conclusion: In tumors, TSGs methylation allowed histopathologic and outcome stratification. Urinary methylation offered a non-invasive approach for patient clinical management, not only for diagnosis but also for prognosis.

O5

PATHOLOGY & IMAGING

O5.1 Radiological evaluation of patients with hcc: mri and MSCT

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Background: Worldwide, hepatocellular carcinoma (HCC) is one of the most common causes of cancer-related death with over 500,000 new cases occurring per year. Risk factors associated with HCC include hepatitis B and C, cirrhosis, alcoholic liver disease, and aflatoxin exposure.

Multislice-computed tomography (MSCT) and magnetic resonance imaging (MRI) are commonly used in the diagnosis, staging, and surveillance of these patients

Aims: The purpose of this educational presentation is to demonstrate a stepwise approach to the radiological evaluation of patients suspected of HCC.

Methods: This educational presentation will review the radiological literature of HCC. Utilizing examples from our institutions the role of MRI and MSCT images on the management of HCC will be presented.

Results: The presentation will discuss the importance of the optimum imaging protocol and the common and uncommon imaging patterns of HCC. The imaging features that may distinguish HCC from other primary liver masses will be presented. There are various staging criteria for HCC and the role of imaging in these various criteria will be presented.

Relevant functional and anatomical information will be demonstrated. Among these, examples of the variant anatomy, cirrhosis, fatty infiltration, and portal hypertension will be shown.

Following the treatment of HCC, the detection of recurrent disease also depends on optimum imaging findings. The imaging patterns that distinguish post-treatment changes versus recurrent disease will be presented.

The limitations and potential pitfalls MSCT and MRI in the evaluation of HCC will also be discussed.

Conclusion: The goals of this presentation are for the participants to (i) become familiar with the information that should be expected from the radiological report of patients with HCC (ii) to learn the importance of optimum imaging protocols, (iii) to become familiar with the imaging features of HCC, (iv) to recognize the imaging patterns of recurrence following treatment, and (v) finally to learn the limitations and potential pitfalls of these imaging techniques.

O5.2 Primary salivary gland–type lung cancer: imaging and clinical predictors of outcome

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Background: The staging of primary salivary gland-type tumors of the lung (PSGTLs), adenoid cystic carcinomas, mucoepidermoid carcinomas, is usually most affected by the primary local tumor invasion as they usually present without metastatic disease. Failure to identify the PSGTLs growth along the airways and/or submucosally, leads to incorrect tumor delineation and staging, which impacts treatment planning: surgical and radiation planning, and affects patients negatively because local recurrence is likely if complete surgical resection is not achieved.

Aims: To assess whether CT and FDG uptake of primary salivary gland–type tumors of the lung are associated with tumor type, disease stage, or survival.

Methods: CT and PET data of 30 consecutive patients with primary salivary gland–type tumors of the lung were evaluated for: tumor size, location, homogeneity, lymphadenopathy, pleural effusions, and metastases. Maximum and volumetric FDG uptake of the tumors of 15 patients was recorded. Wilcoxon rank-sum, Fisher exact tests, and univariate Cox regression were used for statistical calculations.

Results: Compared with mucoepidermoid carcinomas, adenoid cystic carcinomas (57 %) were larger (3.5 cm vs 2.2 cm, $p=0.03$), more frequently involved in central airways (94 % vs 63 %, $p=0.002$), and had a higher median FDG uptake ($p=0.0264$). Higher FDG uptake of the primary tumor was associated with nodal tumor involvement ($p=0.05$). The median overall survival times for patients with adenoid cystic carcinoma and mucoepidermoid carcinoma were 7.7 and 4.0 years, respectively. Imaging features that significantly affected overall survival included presence of mediastinal or hilar lymphadenopathy (hazard ratio [HR]: 4.33; 95 % confidence interval [CI]: 1.15–16.26; $p=0.03$), suspected metastatic disease (HR: 5.10; 95 % CI: 1.27–20.47; $p=0.02$), and primary tumor heterogeneity (HR: 3.46; 95 % CI: 1.04–11.55; $p=0.04$).

Conclusion: Higher FDG uptake is associated with nodal disease in patients with primary salivary gland–type tumors of the lung but is not predictive of their survival, whereas CT features suggestive of advanced disease correlate with worse outcome.

O5.3 Folate Receptor Alpha (FRA) expression in Triple Negative Breast Cancer (TNBC)

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Background: TNBC represents ~20 % of breast cancers. It is an aggressive phenotype characterized by lack of expression of estrogen and progesterone receptors as well as an absence of human epidermal growth factor receptor-2 (Her2) overexpression. Compared to other subtypes of breast cancer, TNBC has a poorer overall survival. Treatment options are limited since the hormonal receptors and HER2 antagonists are ineffective. Traditional cytotoxic chemotherapy is the mainstay of treatment. Research aimed at identifying the unique genetic and molecular characteristics of TNBC may allow for the development of targeted treatment options. Since folate is a necessary cofactor for the synthesis of purines and pyrimidines, anti-folates are commonly used as anticancer agents. A better understanding of folate cellular accumulation has also led to the development of agents that target folate receptors as potential cancer treatments. FRA is a GPI anchored protein that binds and transports folate, however its expression is not mandatory for folate entry into a cell as there are multiple portals of entry including the reduced folate carrier. Thus blocking FRA will not necessarily deprive the cell of needed folate. However, FRA has been reported to impart a growth advantage to cells in a low folate state, one that is hypothesized to be present in the extracellular milieu of a rapidly growing tumor such as TNBC. FRA is expressed on a number of epithelial malignancies compared to normal tissue and provides a rational for use as a potential target. FRA is regulated by steroid hormones. Estrogen down regulates FRA expression by direct action of the estrogen receptor on the P4 promoter of FRA suggesting a negative correlation between the expression of ER and FRA.

Aims: An immunohistochemical (IHC) evaluation of FRA expression in breast cancer samples.

Methods: IHC performed on breast cancer TMA.

Results: Approximately 30 % of all invasive samples were found to express FRA and seemed to associate with TNBC tumors. In separate cohort of metastatic Her2- breast cancer, FRA was expressed in 86 % of TNBC patients.

Conclusion: FRA expressing breast cancer may represent an important and clinically significant subset of TNBC where FRA targeted therapies may merit investigation. Such treatments, alone or in combination with chemotherapy, may represent a targeted approach that TNBC has needed.

O5.4 [18F]Fluorocholine PET (FCH-PET) during treatment may serve as a predictive biomarker of early response to radiation (RT)/ chemoradiation (CRT): results of a pilot study

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Background: [18F] (FCH-PET) employs choline as a proliferative marker. [18](FDG-PET/CT), commonly used to assess response in patients treated with RT/CRT, is obtained 8-12 weeks after RT/CRT so as to limit false-positive and false-negative results.

Aims: The intent of this pilot study is to explore whether FCH-PET can serve as a predictive biomarker for early detection of response to RT/CRT

Methods: Patients with tumors of HN, GYN, GI, CNS, GU, Breast and Lung with lesions (of base of tongue (3), tonsil (1), neck nodes (1), hypopharynx (1), maxilla (1), palate (1), lung (4), pancreas (1), brain (1), uterus (1), breast (1) and rectum (1) were accrued. All patients underwent a pre-RT FCH-PET/CT and second one performed 2-4 weeks into RT/CRT. There were 16 lesions in 14 patients that were considered target lesions and were followed. Median RT dose was 66 Gy (range, 36-70 Gy). Clinical outcome was determined by clinical exam and/or imaging as well as pathological assessment of resected specimens when possible. Response was defined as complete response (CR), less than complete response-partial response (PR), or no change/progressive disease - no response (NR).

Results: The median FCH-PET SUVmax for the 1st and 2nd scans was 6.15 and 4.65, respectively. 7/16 (44 %) of the lesions had CR, 7/16 (44 %) had PR, and 2/16 (12 %) had NR. 14/16 (87.5 %) lesions showed a reduction in SUVmax all of which were determined to have CR or PR (Positive Predictive Value of 100 %) and 2 lesions showed an increase in SUVmax both of which were determined to be NR. The overall median percentage Δ SUVmax between the 1st and 2nd scan was -19.5 %. Wilcoxon rank sum test comparing percentage Δ SUVmax between NR vs CR and NR vs PR each approached statistical significance ($p=0.055$) thereby suggesting a potential association between Δ SUVmax in FCH-PET and clinical outcome.

Conclusion: FCH-PET scan performed to evaluate early prediction of RT response demonstrated a high positive predictive value in assessing clinical outcome. This novel tracer therefore holds potential for future use in early treatment evaluation of radiation response.

O5.5 Pre-analytic conditions in tissue related markers. How well do we know our sources?

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Background: For more than 100 years morphology of cells and tissue has been the key feature by which pathologists described characteristics of normal and diseased elements of the human corps. Even after death the basic shapes and protein distributions remain sufficiently intact to allow the recognition of tumors, inflammation, ischaemia or trauma. Histochemistry, enzyme histochemistry and later immunohistochemistry became more dependant on the quality of tissue preservation. Nevertheless histopathology was not yet a profession of exact measurements but more a field of knowledgeable interpretation of visible changes.

With the introduction of more refined methods for the identification of genetic disorders, tissues have now become part of the analytes that are involved in the reactions and results. Registration of the steps followed in tissue preservation does not necessarily reflect the effects of it.

Aims: Systematic analysis of tissue quality on individual paraffin blocks before molecular tests are run is of crucial importance, but expensive in terms of reagents and labour and therefore hard to realise.

Methods: A combination of whole slide imaging (WSI) together with tissue micro array (TMA) robotics has opened new possibilities. Annotation in digital microscopy images allows identification of specific areas that can be located in the paraffin donor blocks and punched by a TMA robot.

Results: These samples, collected in a TMA recipient block, can be tested in a very economical way for the preserved quality of proteins and nucleic acid components.

Conclusion: In a situation where optimal preservation was attempted but not achieved, the degree of tissue degradation can be documented and appropriate caution can be introduced when interpreting the results of highly sensitive and specific tests that play a crucial role in the choice of further diagnostics, treatment or follow-up.

O5.6 Implementation of TMA and digitalization in routine diagnostics of breast pathology

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Background: To ensure optimal treatment of breast cancer patients, breast tumours are classified based on clinicopathological features. As part of this process, routine diagnostics of breast tumours includes histological typing and grading, as well as profiling by use of an immunohistochemistry (IHC) panel of antibodies, probes and fluorescence in situ hybridization (FISH). This will, as a minimum, include assessment of oestrogen receptor (OR) and human epidermal growth factor receptor 2 (HER2). The individual preparation and staining of many breast tumours in a large laboratory with this standard panel is thus time consuming and costly.

Aims: To show that in breast cancer routine diagnostics the use of the tissue microarray (TMA) technique in combination with digitalization of the stained multi-slides is not only economical, with a considerable cost reduction, but it also enhances standardization of tumour profiling, by elimination of day to day variation.

Methods: Ninety-three consecutively collected invasive breast tumour samples are included, routinely formalin fixed and paraffin embedded. TMA technique is used, designed to hold 40 tissue cores of 2 mm in diameter each. TMA sections are used for staining/hybridizing with the biomarkers in our panel consisting of HER2, OR and Topoisomerase IIa. Slides are digitalized, and scored by three independent pathologists.

Results: By use of the TMA technique in combination with digitalization we demonstrate that 2 mm breast tumour cores correlate with the corresponding tumour on whole mount slides, regarding staining/hybridizing results. Furthermore, we show that simultaneous staining/hybridizing of multiple breast tumour specimens reduces variation of staining/hybridizing quality, hereby increasing reliability of interpretation. By scanning and digitalization of the stained and hybridized multi-slides, we could optimize documentation and filing of the results.

Conclusion: Our work is an example of translational research by implementing a tool in routine diagnostics, originally developed for high throughput analyses in the search for prognostic and predictive markers in targeted medicine.

O5.7 Epithelial Columnar breast lesions: Histopathology and molecular markers

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Background: Columnar cell lesions of the breast (CCLs) are characterized by the presence of tightly packed columnar

cells lining distended TDLUs, apical snouts and intraluminal microcalcifications.

Aims: There have been several reports describing the association of CCLs with low grade ductal carcinoma in situ or lobular carcinoma in situ and invasive tubular carcinoma (this coexistent triad has been proposed as “Rosen triad”). Despite these associations, the risk of developing subsequent carcinoma after the diagnosis of CCLs with atypia is not clear yet.

Methods: In the initial classification CCLs were divided into two categories: columnar cell change (CCC) and columnar cell hyperplasia (CCH). CCC is characterized by one to two layers of cells; when cellular stratification with more than two layers, with crowding, overlapping, tufts, fronds, short micropapillae or rigid bridges is present, the term CCH is used. Accordingly to the degree of architectural and cytological changes, CCLs were subclassified into further categories. In the current WHO classification, CCLs with cytological atypia are referred as “flat epithelial atypia

(FEA)”. In the latest revision of DIN (ductal intraepithelial neoplasia) system, FEA is designated as DIN1a.

Results: CCLs show immunopositivity for ER, PR, Bcl2, luminal cytokeratin 19 and cyclin D1. The lesions are negative for p53, HER2/neu and basal cytokeratins 5/6 and 14. Ki67 positivity is low. Genetic/molecular profile of CCLs, particularly these with atypia, resembles that seen in atypical ductal hyperplasia and low grade ductal carcinoma in situ or even in some low grade invasive carcinomas: loss on 9q, 10q, 16q and 17p, gain on 15q, 16p and 19 as well as LOH at 11q, 16q and 3p.

Conclusion: CCLs seems to be biologically significant lesions, since the co-existence with more advanced entities may suggest that CCLs probably represent a very early form of malignant changes. Whether further tissue excision should be recommended for CCLs detected in core biopsies remains controversial. However, there are limited outcome data which indicate that subsequent excision shows a more advanced lesion in 20–30 % of cases, when CCLs with atypia are identified in core biopsy.

O6 BREAST CANCER

O6.1 Tumor STAT3 tyrosine phosphorylation status, as a predictor of benefit from adjuvant chemotherapy for breast cancer

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Background: Signal transducer and activator of transcription 3 (STAT3) is a point of convergence for numerous oncogenic signaling pathways. In breast cancer cell lines and xenograft models activated STAT3 participates in breast tumorigenesis, while studies in humans have demonstrated that phosphorylated (tyrosine705)-STAT3 is a marker of good prognosis in breast cancer.

Aims: The goal of this study was to determine the usefulness of phospho-STAT3 status as a predictor of benefit from adjuvant chemotherapy in breast cancer patients.

Methods: Immunohistochemical analysis of Phospho-STAT3 was performed on a tissue microarray of breast cancer specimens. The expression pattern of Phospho-STAT3 was retrospectively correlated with pathological parameters and overall survival in patients who were or were not treated with adjuvant chemotherapy.

Results: Of 375 tissue specimens interpretable for P-STAT3, 134 (36 % percent) exhibited positive Phospho-STAT3 nuclear expression. Among 234 patients who received adjuvant therapy, those with tumors displaying positive Phospho-STAT3 nuclear expression had a better ten-year rate of overall survival than patients with tumors displaying negative phospho-STAT3 nuclear expression ($P=0.001$). Among patients who did not received adjuvant chemotherapy, positive phospho-STAT3 nuclear status was not correlated with increased overall survival ($P=0.54$). Positive phospho-STAT3 was correlated with improved overall survival only among patients who received adjuvant chemotherapy in a multivariate analysis adjusted for stage, grade, hormonal status, Her2 status, and age, irrespective of the chemotherapy regimen received (hazard ratio for death, 0.35 [95 percent confidence interval 0.188- to 0.667]; $P=0.001$).

Conclusion: These findings support the role of Phospho-STAT3 as a marker of favorable outcome in breast cancer patients treated with adjuvant chemotherapy. Whether

Phospho-STAT3 has a predictive role of benefit from adjuvant chemotherapy has to be validated on prospective, randomized, controlled studies.

O6.2 CEACAM-1 negatively regulates G-CSF production by breast tumor associated macrophages that mediate tumor angiogenesis

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Background: CEACAM1, a cell adhesion molecule expressed on epithelial cells and activated immune cells, is down-regulated in many cancers and plays a role in inhibition of inflammation in part by inhibition of G-CSF production by myeloid cells. Since macrophages are associated with a poor prognosis in breast cancer, but play important roles in normal breast, we hypothesized that CEACAM1 down-regulation would lead to tumor promotion under inflammatory conditions.

Aims: Determine in vitro cytokine profiles and in vivo tumor growth and angiogenesis for breast cancer cell and macrophage co-cultures in the presence and absence of CEACAM1 expression in breast cancer cells.

Methods: Monocytes were differentiated into pro-inflammatory (M1) or anti-inflammatory (M2) macrophages and co-cultured with MCF7 cells expressing or not CEACAM1. Cytokines were measured by Luminex assays and blood vessel density by CD31 staining.

Results: Co-cultures of pro-inflammatory M1 macrophages with CEACAM1 negative MCF7 breast cells produced high levels of G-CSF (10 ng/mL) compared to CEACAM1 transfected MCF7/4S cells (1 ng/mL) or anti-inflammatory M2 macrophage co-cultures (0.5 or 0.1 ng/mL, MCF7 or MCF7/4S, respectively). The expression of CEACAM1 on M1s was much greater than for M2s and was only observed in co-cultures with either MCF7 or MCF7/4S cells. When M1 macrophages were mixed with MCF7 cells and implanted in murine mammary fat pads of NOD/SCID mice, tumor size and blood vessel density were significantly greater than MCF7 or MCF7/4S only tumors which were hardly detected after 8 weeks of growth. In contrast, M1 cells had a much reduced effect on MCF7/4S tumor growth and blood vessel density, indicating that the tumor inhibitory effect of CEACAM1 is most likely related to its anti-inflammatory action on inflammatory macrophages.

Conclusion: These results support our previous finding that CEACAM1 inhibits both G-CSF production by myeloid cells and G-CSF stimulated tumor angiogenesis.

O6.3 Standardized use of serum tumor markers (TMs) for post-operative breast cancer follow-up: an impossible task?

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Background: So far, any post-operative breast cancer follow-up is investigational, due to the assumption of an unfavorable cost-effectiveness ratio. Serum TMs are likely suitable tools to carry out a post-operative follow-up to definitely validate this assumption. However, no prospective multi center randomized trial has been carried out based on serum TMs, due to the lack of standardized criteria for their use.

Aims: In this study, we propose an individual reference limit (IRL) and a related critical change (CC) of a selected serum CEA-TPA-CA15.3 TM panel to standardize the use of serum TMs in breast cancer follow-up.

Methods: Our previous as well as most different criteria applied in other Centers do not allow to standardize the use of serum TMs. In this study, in 129 disease-free breast cancer patients an IRL was calculated with 5 consecutive serum values regularly obtained during 6 (mean; 4-9 range) months at the beginning of the follow-up. IRL was calculated considering mean value of the 5 consecutive determinations plus 2SD for CEA and CA15.3, and 3SD for TPA. Two consecutive values higher than IRL were considered the CC to suspect a relapse. In this instance, immediately a complete instrumental study was carried out to find out and define the site of recurrences.

Results: Using the commercial kit cut-off value, we have reported 31 % CE and/or PI for CEA-TPA-CA15.3 TM panel. In this study, total determinations were 1213, during a total follow-up of 3107 months. Two consecutive values higher than IRL occurred 39 times (6.4 %) in 31 (26 %) of the 122 non relapsed patients. The mean percentage per year of falsely suspected patients was 12 % with IRL specificity of 88 %. In 6 of the 7 patients who relapsed during the follow-up, IRL increase was the first suspicion of recurrence, with 4.5 (m + sd; 1.5 to 17 range) months lead time. In the last patient, IRL was falsely negative with IRL sensitivity of 86 %.

Conclusion: In this study, diagnostic accuracy per year of serum CEA-TPA-CA15.3 TM panel with the proposed IRL and the related CC was 88 %. These preliminary data point out that a selected TM panel, using the proposed IRL and CC permits to standardize the use of serum TMs in breast cancer follow-up and likely to carry out an intensive monitoring of these patients with a favorable cost-effectiveness ratio.

O6.4 Pro-oncogenic factors miR-23b- and miR-27b are regulated by Her2/Neu, EGF, and TNF α in breast cancer

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Background: MicroRNAs (miRs) are a new class of small (21-25 nucleotides) non-coding endogenous RNAs implicated in gene expression regulation. We identified miR-23b and miR-27b as miRNAs that are highly upregulated in human breast cancer.

Aims: Our aims are to understand the important role of miR-23b and miR-27b in breast cancer progression.

Methods: We used several in vivo mouse xenograft assays to address our aims.

Results: We found that engineered knockdown of miR-23b and miR-27b substantially repressed breast cancer growth. Nischarin expression was augmented by knockdown of miR-23b as well as miR-27b. Notably, these miRNAs and Nischarin were inversely expressed in human breast cancers, underscoring their biologic relevance. We comprehensively validated the clinical relevance of the expression of these miRNAs and showed that high expression of miR-23b and miR-27b correlates with poor outcome in breast cancer. Moreover, intraperitoneally delivered anti-miR-27b restored Nischarin expression and decreased tumor burden in a mouse xenograft model of human mammary tumor. Also we report for the first time that HER2/neu, EGF, and TNF α promote miR-23b/27b expression through the AKT-NF- κ B signaling cascade. Furthermore, Nischarin regulates miR-27b/23b expression through a feedback loop mechanism by suppressing NF- κ B phosphorylation.

Conclusion: Since anti-miR-27b compounds that suppress miR-27b inhibit tumor growth, the anti-miR-27b appears to be a good candidate for the development of new anti-tumor therapies.

O6.5 Prognostic significance of cytokeratin markers in breast cancer – a meta analysis

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Background: Cytokeratins, have been reported in various studies to be sensitive markers for Breast Cancer and detecting earlier metastasis formation.

Aims: A meta-analysis evaluating the potential of Cytokeratin Markers (TPS and TPA) as compared to the established TumorMarkersCA15-3, CEA, CA125, for evaluating the clinical status of advanced Breast Cancer patients (pts), their response to therapy and Lead time to metastatic disease.

This Meta-analysis should improve statistical power by combining information from many small studies during the last 25 years, to estimate an overall ‘effect’ by summarizing data via a single quantity, determine ability to draw overall conclusions, random vs. systemic variations.

Methods: A Meta-analysis of 49 studies, dating from 1985 to now, on 3298 advanced Breast Cancer pts. It included descriptive/graphical methods

(L’Abbe plot, Forest plot, Funnel plot), estimation of a common effect and between -studies heterogeneity, mainly using a random effect model.

Results: The parameters Sensitivity (%) of markers from all studies was recalculated (95 %CI) and found: CA15-3- 69.9 % (65.8 / 71.9), CEA -56.1.0 % (52.8 / 57.1), TPS- 74.1 % (71.0/74.2), TPA- 72.8 % (70.3/ 72.9) and the combinations CA15-3+ TPS- 82.6 % (81.5/82.6) or CA15-3 + CEA + TPA -89.0 % (88.9/89.5).

Clinical Correlations to marker levels, overall estimates (95 % CI) were ;

TPS- 0.798(0.594, 0.903), TPA- 0.672 (0.047, 0.898), CA15-3- 0.584 (0.340, 0.713)

CEA - 0.634 (0.405, 0.761). Lead time for local recurrences or Metastasis formation (months, 95 % CI) were; TPS - 4.97 (3.11, 6.82), TPA- 3.8 (1.76, 3.13),

CA15-3- 4.7 (2.93, 5.93), or the panel CEA + TPA + CA15-3 - 11.9 (3.53, 20.89).

Conclusion: We conclude that the Cytokeratins markers TPS/ TPA demonstrated the best response to therapy and the most important information for prognosis in advanced Breast Cancer pts. Using a panel of markers CEA, CA15-3 and TPA revealed the highest sensitivity, followed by TPS and CA15-3 which we recommend to use.

O6.6 Determination of molecular markers for BRCA1 and BRCA2 heterozygosity using gene expression profiling

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Background: Women with a germline mutation in BRCA1 or BRCA2 have markedly increased risk of early-onset breast and ovarian cancer. Identification of the BRCA1 and BRCA2

mutation carriers offers an opportunity to early identify or prevent the development of malignancy; therefore the ability to determine which patients are more likely to carry BRCA1 or BRCA2 mutations is of great importance.

Aims: To develop more indirect methods for BRCA screening that can improve the selection of patients for sequence-based BRCA testing.

Methods: Fresh blood samples were obtained from proven BRCA1 and BRCA2 mutation carriers and mutation-negative women.

And a molecular assay based on the highest ranked genes was constructed. Using expression microarrays we compared gene expression of irradiated lymphocytes from BRCA1 and BRCA2 mutation carriers versus control non-carriers. Real-time PCR was performed on the genes which were most prominently differentially expressed in the microarray assay, and a molecular assay based on the highest ranked genes was constructed.

Results: We identified 137 probe sets in BRCA1 carriers and 1345 in BRCA2 carriers with differential gene expression. Gene Ontology analysis revealed that most of these genes relate to regulation pathways of DNA repair processes, cell cycle regulation and apoptosis. Real-time PCR was performed on the 36 genes which were most prominently differentially expressed in the microarray assay; 21 genes were shown to be significantly differentially expressed in BRCA1 or BRCA2 mutation carriers as compared to controls ($p < 0.05$). Based on a validation study with 40 mutation carriers and 20 non-carriers, a multiplex model that included six or more coincidental genes of 18 selected genes was constructed in order to predict the risk of carrying a mutation. The results using this model showed sensitivity 95 % and specificity 88 %.

Conclusion: Our study provides insight into the biological effect of heterozygous mutations in BRCA1 and BRCA2 genes in response to ionizing irradiation induced DNA damage. We also suggest a set of 18 genes that can be used as a prediction and screening tool for BRCA1 or BRCA2 mutational carriers by using easily obtained lymphocytes.

O6.7 HER 2 positive in early breast cancer

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Background: About 15-20 % of patients with early stage breast cancer present with tumors that have overexpression

or amplification of the human epidermal growth factor receptor 2 (HER2) gene.

Aims: Before 2005, these individuals had an increased risk of recurrence and death, but since then their outcomes have substantially improved with the adoption in most countries of adjuvant trastuzumab as a standard component of therapy for HER2-positive early-stage breast cancer.

Methods: There is continuing interest in new anti-HER2 therapy.

Results: Randomized adjuvant trials continue to show significant reductions in distant recurrence and death for early-stage women treated with adjuvant trastuzumab.

Neoadjuvant treatment with a sequential anthracycline-taxane-based chemotherapy in combination with trastuzumab is currently a preferred therapy for patients with HER2-positive breast cancer.

Conclusion: The need for additional or alternate treatment options is great in patients who do not achieve a pCR. Addition of lapatinib (Tykerb) or pertuzumab to trastuzumab is a therapeutic option.

O6.8 Therapeutic options in HER2 positive metastatic breast cancer (MBC): An embarrassment of riches

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Background: HER2 is a member of the HER 1-4 family of tyrosine kinase cell surface receptors.

Dimerization of the receptors is needed for activation of the intracellular tyrosine kinase and the ensuing signal transduction pathways.

Overexpression or overamplification of HER2 occurs in approximately 20 % of human breast cancers. HER2-positive tumors are more aggressive and, without specific therapy, are associated with worse outcomes.

Blocking the HER2 signal is an established part of treating HER2 positive MBC.

The extracellular Receptor is targeted by monoclonal antibodies (trastuzumab, pertuzumab – newly FDA approved HER dimerization inhibitor).

The intracellular tyrosine kinase (TK) is targeted by TK inhibitors (lapatinib, neratinib, afatinib).

Using these agents has transformed HER2 positive MBC from a disease with a dismal prognosis and few therapeutic options to a disease with a potential for long-term control and multiple therapeutic options.

Initial clinical studies focused on using a single anti HER2 agent combined with chemotherapy. Trastuzumab combined with various chemotherapeutic agents proved to significantly prolong DFS and OS when compared to chemotherapy alone. Lapatinib combined with capecitabine improved DFS in trastuzumab refractory patients when compared to capecitabine alone.

Unfortunately most patients will develop resistance to anti HER2 therapies, thus novel approaches were needed. One of the most promising approaches is double blockade of the HER2 signal using trastuzumab and pertuzumab or trastuzumab and lapatinib.

The combination of trastuzumab, pertuzumab and docetaxel was shown to be superior in terms of DFS and OS when compared to trastuzumab and docetaxel in women with HER2 positive MBC.

Similarly the combination of trastuzumab and lapatinib was superior to lapatinib alone in women with trastuzumab refractory MBC.

Recently the drug-antibody conjugate trastuzumab emtansine (TDM1), was shown to be superior to lapatinib/capecitabine in terms of DFS and OS with a better tolerance in women with trastuzumab refractory MBC.

Studies are ongoing testing the combination of TDM1 and pertuzumab leading to a possible highly effective chemotherapy free regimen.

The development of anti HER2 agents is the most significant development in therapeutic options for women with MBC.

O7 OVARIAN CANCER

O7.1 MicroRNA-mediated regulation of ovarian carcinoma: Role of exosomes

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Background: Exosomes are naturally occurring biological nanovesicles (~30–100 nm) secreted by various cell types in vitro and into body fluids including malignant and pleural effusions, in vivo. Exosomes contain lipids, proteins, enzymes and short nucleic acid sequences including miRNAs. It has been implicated that an “exosomal messenger system” exhibits paracrine bioactivities that facilitates tumor communication within the local tumor microenvironment.

Aims: Ovarian cancer is the leading cause of death from gynecological cancers in western countries. Primary solid tumors, solid metastases, and effusions (ascites) to the peritoneal and pleural cavities characterize the tumor as it progresses. Recently, we reported on differences in the expression of cellular miRNAs from ovarian carcinoma (OC) patients along tumor progression

Methods: We compared the expression of miRNAs present in effusion fluid-derived exosomes from 50 pre-chemo and 50 post-chemo patients of which 37 were chemo-resistant and 31 chemo-sensitive, respectively. The patient cohort was in addition divided into 28 long and 72 short survivors, respectively.

Results: We identified differentially expressed miRNAs in OC effusion-derived exosomes compared to exosomal miRNAs from sera, malignant mesothelioma effusions and reactive effusions. Fifty-seven miRNAs were differentially expressed in specimens from patients with long vs. short survival and 62 differentially expressed miRNAs in specimens obtained before vs. after chemotherapy treatment.

Conclusion: OC effusion derived exosomes might have autocrine and/or paracrine effects on cells, thus enhancing the oncogenic potential of cancer cells, or sensitizing the body to malignant dissemination. This is the first study to evaluate the differential expression of miRNAs in exosomes derived from OC ascites fluid. We believe that the differential expression and the distinct signatures of miRNAs possess diagnostic, prognostic and therapeutic potential

O7.2 Role of dihydrotestosterone (DHT) on the TGF- β 1 receptors expression in epithelial ovarian cancer

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Background: Epithelial ovarian cancer (EOC) accounts for 90 % of ovarian malignancies. This condition presents a silent course, late detection, with pronounced angiogenesis, poor response to therapy, and low survival. There are several hypotheses about the aetiology of EOC, one of them postulates that androgens stimulate proliferation of ovarian epithelial cells within the inclusion cyst and also of the transformed epithelium. Besides, androgens could alter the TGF- β signalling pathway involved in tissue homeostasis of EOC.

Aims: To evaluate the participation of DHT in the expression of TGF- β 1 receptors in a cell line of epithelial ovarian cancer.

Methods: An “in vitro” model was used; a normal human OSE (HOSE cell line) and EOC (A2780 cell line) to determine the presence of androgen receptors (AR), TGF- β 1 and its receptors by RT-PCR, immunocytochemistry and western-blot. A2780 cells were stimulated with 0, 10 and 100 nM DHT for 72 h to evaluate mRNA and protein levels of TGFBR1 and TGFBR2 by RT-PCR and western blot.

Results: A positive immunodetection was found for AR in HOSE and A2780 cell lines; the AR mRNA levels were higher in A2780 cells ($p < 0.05$) than in HOSE cells. For TGF- β 1 and its receptors we also observed positive immunodetection in both cell lines without differences between them. When A2780 cells were stimulated with DHT, the mRNA levels of TGF- β 1 did not change. Nevertheless, the protein levels of the TGF- β 1R1 were reduced 20 % by DHT effect, while levels of TGF- β 1R2 were reduced 30 % by the hormone DHT.

Conclusion: The high detection of AR in A2780 cells compared with HOSE cells with a decrease of TGF- β 1 receptors expression by DHT stimulation could indicate that androgens would alter the TGF- β 1 signalling with the concomitant increase of A2780 cell proliferation.

This work was funded by grant FODECYT 1110372

O7.3 Global microRNA profile of primary ovarian cancer

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Background: microRNAs (miRs) have been recently associated with TP53-dependent pathways. Changes in these pathways are critically important in the pathogenesis of ovarian cancer (OvCa).

Aims: We aimed to study the global miR expression in the clinical samples of primary serous OvCa of different TP53 status and tumour grade, to identify miRs linked to the pathogenesis of OvCa and of biomarker potential.

Methods: Four series of OvCa samples containing $\geq 75\%$ of cancer cells were analysed: 1) grade 3+4 and the wild type (wt)TP53; 2) grade 1+2 and the wt TP53, 3) grade 3+4 and missense TP53 mutations; 4) grade 3+4 and TP53 null mutations. Nearly 1300 miRs were assessed with Exiqon miRCURY LNATM microRNA Array, 6th gen-hsa. Significance was set at adj. $p < 0.05$, $FC > 2$. Normal human ovarian surface epithelium served as a calibrator. Selected results were verified by RT-qPCR.

Results: High grade OvCa (G-3+G-4) and OvCa with TP53 mutations expressed increased levels of miR-7, miR-103a, miR-107 and miR-4289 as compared to low grade (G-1+G-2) and wtTP53, respectively. High vs. low grade tumours, showed decreased miR-10a and miR-34a.

Within the high grade tumour samples, those with TP53 null mutations vs. wtTP53 overexpressed miR-517c.

Within the wtTP53 tumour series, there was a decreased expression (11–1500FC) of 19 microRNAs in high grade vs. low grade samples.

Conclusion:

1. 25 miRs were identified as potentially associated with TP53 status and tumour grade.
2. miR-7 and miR-4289 were for the first time found to relate to the TP53 pathway.
3. miR-7, miR-103a, miR-107 and miR-4289 expression correlates with tumour aggressiveness, assessed both by the histopathological grade and by TP53 mutations. This also points to the relationship between tumour grade and the TP53 status.
4. Depending on the TP53 status, different miRs are linked to OvCa grade.
5. In OvCa, miR-7, miR-103a, miR-107 and miR-4289 are likely to play a role of oncogenes, while, as suggested in other tumours, miR-34a and miR-10a of tumour suppressors.
6. To better understand the role of miRs in OvCa, functional studies are on-going.

Supported by the grant NN 401 050 138 of Polish MNiSW.

O7.4 Systemic inflammatory response and CA 125 and HE4 levels in patients with benign and malignant ovary diseases

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Background: It is increasingly recognized that the presence of systemic inflammatory response is associated with poor survival in various malignancies. In assessing the prognosis of patients with cancer are often used indicators of inflammation and nutritional status.

Aims: The aim of presented study was to analyze of the relationship between inflammation and the level of selected tumor markers in terms of differentiation between malignant and benign ovary diseases.

Methods: The studies of CA 125 and HE4 as well as CRP, albumin (ALB), prealbumin (PRE) and alpha-1 acid glycoprotein (AAG) were performed before treatment in group of 69 patients with ovarian cancer in different stages of disease, and in 41 women with benign ovary diseases. For each patient under study was calculated Cancer Serum Index (CSI) and Prognostic Inflammation and Nutrition Index (PINI)

Results: In ovarian cancer group were found significantly higher CA 125, HE4, CRP, AAG and lower ALB and PRE level, in comparison to patients with BGD. In patients with ovarian cancer were observed significant correlation only between CA 125 and the CSI, whereas there was no relationship between the HE4 and both indicators of inflammation. Between ovarian cancer patients in stage [I + IIA + IIB] and in stage [IIC + III + IV] were found significant differences in CA 125, HE4 concentrations and CSI and PINI values. Where significant differences in analyzed parameters were also observed between patients with ovarian cancer in stage [IIC + III + IV] and BGD, that there was no such difference between patients in stage [I + IIA + IIB] and BGD, except HE4. In both groups of patients there were no significant differences in the levels of the analyzed parameters, depending on hormonal status. Patients with ovarian cancer responds well to first-line chemotherapy (CR) had significantly lower pretreatment concentrations of CA125 and HE4 as well as lower values of PINI and CSI compared to other.

Conclusion: In patients with ovarian cancer a comprehensive evaluation of markers of inflammation and CA 125 and HE4 may request relevant information for differential diagnosis of malignant and BGD as well as for predicting response of ovarian cancer patients to first-line chemotherapy.

O7.5 CA 19.9 and useful complementary tumor marker in ovarian cancer: Comparison with HE4, CA 125 and ROMA in Gynaecological diseases

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Background: In some groups of patients with gynecological disorders the inclusion or exclusion of tumor markers and their combination may improve the sensitivity and specificity.

Aims: The aim of this study is to evaluate the inclusion of CA 19.9 as complementary tumor marker in ovarian cancer.

Methods: CA 125, CA 19.9 and HE4 serum levels were determined in 211 healthy women, 617 patients with benign gynaecological diseases (199 endometriosis, 95 myomas, 230 ovarian cysts, 54 endometriomas, and 14 with other diseases), 52 patients with non-active gynaecological cancer and 311 with active gynaecological cancer (238 ovarian cancers). CA 125, CA 19.9 and HE4 cut-offs were 35 U/mL, 37 U/ml and 120 pmol/L, respectively. ROMA

algorithm cut-off was 13.1 and 27.7 for premenopausal or postmenopausal women, respectively.

Results: HE4, CA 125, and CA 19.9 results were abnormal in 0 %, 8.1 % and 5.2 % of healthy women and in 1.9 %, 23.3 % and 14.3 % of patients with benign diseases, respectively. Among patients with cancer, HE4 (in contrast to CA 125) had significantly higher concentrations in ovarian cancer than in other malignancies ($p < 0.001$). Tumour marker sensitivity in ovarian cancer was 80.7 % for HE4, 77.9 % for CA 125, 90.1 % for ROMA and 20 % for CA 19.9. All tumor markers, HE4, CA 125 and CA 19.9 were related to tumour stage and histological type. The highest HE4, CA 125 serum concentrations were found serous malignancies and CA 19.9 in mucinous tumors (mean 22445 U/ml). A significantly higher area under the ROC curve was obtained with ROMA and HE4 than with CA 125 in the differential diagnosis of benign gynaecological diseases versus malignant ovarian cancer (0.952, 0.936 and 0.853, respectively).

Conclusion: ROMA algorithm improves the tumor marker sensitivity only in postmenopausal women. CA 19.9 improves the diagnostic sensitivity obtained with CA 125 or HE4 in mucinous tumors.

O8 LUNG CANCERS

O8.1 Non-invasive breath analysis of pulmonary nodules

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Background: The search for non-invasive diagnostic methods of lung cancer has led to new avenues of research, including the exploration of the exhaled breath. Previous studies have shown that lung cancer can in principle be detected through exhaled breath analysis.

Aims: This study aims to evaluate the potential of exhaled breath analysis for the distinction of benign and malignant pulmonary nodules (PNs).

Methods: Breath samples were taken from 72 patients with PNs in a prospective trial. Profiles of volatile organic compounds (VOCs) were determined by (i) gas chromatography/mass spectrometry (GC-MS) combined with solid phase microextraction (SPME) and by (ii) a chemical nanoarray.

Results: 53 PNs were malignant and 19 were benign with similar smoking histories and co-morbidities. Nodule size (mean $\pm$ SD) was 2.7 ± 1.7 vs. 1.6 ± 1.3 cm ($p=0.004$) respectively. Within the malignant group, 47 were NSCLC and 6 were SCLC. Thirty had early stage disease and 23 had advanced disease. GC-MS analysis identified a significantly higher concentration of 1-octene in the breath of lung cancer, and the nanoarray distinguished significantly between benign vs. malignant PNs ($p<0.0001$; accuracy 88 ± 3 %), between adeno- and squamous- cell carcinomas ($p<0.0001$; 88 ± 3 %) and between early stage and advanced disease ($p<0.0001$; 88 ± 2 %).

Conclusion: In this pilot study, breath analysis discriminated benign from malignant PNs in a high-risk cohort based on lung cancer related VOC profiles. Further, it discriminated adeno- and squamous- cell carcinoma and between early vs. advanced disease. Further studies are required to validate this non-invasive approach, using a larger cohort of patients with PNs detected by CT.

O8.2 Tumor markers utility in the diagnosis of lung cancer. Are they useful in screening?

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Background: The diagnosis of lung cancer is made late and is not always easy. Recently it was reported the utility of imaging techniques in screening, but the problem is the high cost. Previously our group reported the utility of a combination of 6 tumor markers (CEA, CA 15.3, SCC, CYFRA 21-1, ProGRP and NSE) in the diagnosis and differential diagnosis of patients with lung cancer.

Aims: 1) To confirm results previously published in a large cohort of patients. 2) To evaluate the diagnosis in early stages (Stage I and II).

Methods: Tumor marker serum levels were prospectively studied in: 1324 patients with suspicion of lung cancer but finally diagnosed with non-malignant diseases, and 2058 patients with cancer (1719 NSCLC: 863 stage I-III, 856 stage IV), 339 SCLCC (133 limited disease [LD], 206 extensive disease [ED], and 29 neuroendocrine tumors).

Results: Tumor marker serum levels were suspicious of malignancy (abnormal tumor marker levels) in 254 (9.4 %) of patients with non-malignancy, in 1515 (87.2 %) of NSCLC, 330 (97.9 %) of SCLC and in 27 (93.1 %) of patients with neuroendocrine tumors. Serial tumor markers were studied in 182 patients with false positive results, and similar or decreasing results were found in 98 % of them. Tumor markers were clearly related to tumor stage and histology. Tumor marker sensitivity ranged from 63.1 % in stage Ia to 94.3 % in stage IV. Combined tumor marker sensitivity was 69 % (189/274) in stage I, and 79.6 % (125/157) in stage II.

Conclusion: Tumor markers are useful tools as an aid in the diagnosis and may be considered in the evaluation of risk assessments in screening programs.

O8.3 Neuroendocrine neoplasms of the lung: distinct entities or a continuous spectrum reflecting oncodevelopment

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Background: Neuroendocrine neoplasms of the lung (NEL) are that part of the entire body neuroendocrine tumors (NET) originated in the embryonal foregut. NEL are located in the bronchopulmonary system, are characterized by definite immunohistochemical markers, circulating peptides and may have associated clinical syndromes. Currently, they consist of four entities: typical carcinoid (TC), atypical carcinoid (AC), large-cell neuroendocrine carcinoma (LCNEC) and small-cell lung cancer (SCLC).

Aims: To look for evidence supporting the hypothesis that NEL members share oncodevelopmental (OD) features and to what extent are they genomically similar.

Methods: Web, particularly PubMed, search. Selected clinical cases for illustration.

Results: Based on 2004 WHO classification, all 4 NEL members express tissue NE markers: synaptophysin, chromogranin and CD 56) and require mitotic rate and necrosis to define low- vs. high-grade neoplasm (LG and HG). However, post-2004 practice employs proliferation Ki67 as the marker scoring for this distinction. Thyroid transcription factor-1 (TTF1) is an important marker of pulmonary neoplasms. It is frequently expressed in LCNEC and SCLC, but rarely in TC and AC. In the latter, *menin* (encoded by *MEN-1*) may be expressed.. Somatostatin receptors (SSTR) are an additional NEL feature. They are better detected by PET/CT Ga68 DOTANOC octreotate scintigraphy and have therapeutic implications in TC, AC and, selectively, in LCNEC.patients. Notably, TTF1 encoding gene functions during embryonic life, involving brain development. Similarly, NOTCH1 shows up in that phase of life. SCLC is characterized by numerous peptide hormones (e.g. ACTH, ADH and calcitonin), occasionally associated with clinical syndromes. Comparative genomic hybridization (CGH) has recently been employed to show a partial overlap of NELs< irrespective of grade. SCLC, however, showed distinction expressed by TP53 and MYC copy numbers.

Conclusion: The hitherto described expression pattern of NELs coins them as a family. While Kultchitsky cell is regarded as the cell of origin, the stem-cell hypothesis in NEL needs further substantiation. There are clues for an underlying OD processes but this warrants continuing research.

O8.4 Early identification of lung cancer using autoantibody assays

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Background: Lung Cancer remains a major cause of cancer-related death worldwide. Early detection and diagnosis of lung cancer is paramount to improve prognosis, by enabling curative treatment. Early diagnosis has been proven to save lives however there is as yet no universally adopted screening strategy for individuals at risk of developing the disease. The NLST recently recommended screening heavy smokers between the ages of 55 and 74 with CT. However CT has an overall accuracy of only 50 %. Restricting screening to individuals within these guidelines will also mean a significant number of people, who would go onto develop a lung cancer, would still not be screened.

The literature widely documents the presence of an immune response, in the form of IgG antibodies raised to tumor-associated antigens (TAAs) in cancer patient's serum. Auto-antibodies (AABs) provide an *in vivo* amplification of carcinogenesis, months to years before the tumor bulk becomes otherwise clinically detectable, and their measurement may be used as an aid to early diagnosis in a number of solid tumors.

Aims: To characterise the autoimmune response in sera from individuals at high risk of developing lung cancer and to evaluate the performance of the EarlyCDT™-lung test in the clinical setting.

Methods: The presence of AABs to seven recombinant TAAs was assessed using the EarlyCDT-lung test in the sera of individuals, deemed by their clinician, to be at an increased risk of developing a lung cancer.

Results: The AAB assay in a clinical setting, maintained the previously reported specificity and sensitivity observed in case control studies. In the first 863 samples assessed, the specificity and sensitivity of the EarlyCDT-lung test was 91 % and 41 % (PPV: 1 in 8, overall accuracy: 92 %). Although absolute numbers of lung cancers were small, both non-small cell and small cell lung cancers, as well as early stage disease, were detected.

Conclusion: These results confirm that the performance of the EarlyCDT-lung test in the clinical setting matches the results obtained in retrospective case-control studies.

O8.5 Nitric oxide exposure results in cross tolerance

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Background: There is an interesting phenomenon observed in some cancer patients that become resistant to one form of therapy (like radiation); they often become resistant to therapies that they have not been exposed to (like chemotherapy). It is also known that the patients that have high Nitric Oxide producing tumors are more resistant to various therapies such as chemotherapy and radiation.

Aims: The goal of this study was to establish a laboratory model to study the cross tolerance phenomenon. Understanding the molecular mechanism(s) that drive this process would open opportunities for novel treatment strategies.

Methods: Using a previously developed High Nitric Oxide adapted cell line pair (A549 parent and A549-HNO), we tested differential survival response of these cells to radiation, chemotherapy, UV, heat, heavy metals, alcohol, and other toxic exposures using the cell viability assay, MTT.

Results: In brief, we found that A549-HNO was more resistant to all of the conditions tested than the parent cell line. While these cells were adapted only to Nitric Oxide, they demonstrated significantly increased resistance to other various toxic elements. This appears to be evident, even though they have not been adapted, or exposed to these same agents prior to testing.

Conclusion: This study provides the identification of one natural component (ie. Nitric Oxide), produced by tumors which results in cross tolerance. The A549 and A549-HNO model system can now be used to decipher molecular mechanisms responsible for cross tolerance phenomenon.

O8.6 Combined treatments against the evading tumors

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Background: Molecular analysis may now reveal major targets in tumors. However even when we do have a drug which can inactivate an oncogenic tumor driver almost universally a resistant tumor arises. We will discuss here a

few issues related to lung cancer treatments are not yet emerged as standard therapies but may be so in the near future .

Aims: Discussion of novel therapeutic combinations in resistant Tumors

Methods: We will demonstrate three case reports related to the use of advanced therapeutics- the last case involved molecular analysis of the original tumor and biopsies obtained from: tumors in nude mice.

Results: The first issue to be discussed is that of immunologic treatments with anti PD-1 antibodies. We will also describe a patient suffering from pancreatic cancer treated with an antibody against this target in combination with chemo in second line setting.

The next issue is that of c met inhibition. We will discuss the very interesting results obtained with this tyrosine kinase inhibitors of c met and monoclonal antibodies against this target and our experience with the use of cabozantinib - a dual inhibitor of VEGF and met with an emphasis on the patient stable for 14 months after failure of 5 previous anticancer therapeutics.

We will end our presentation with a short description of what may be termed our most detailed studied patient. He is probably the first patient in which a unique RET translocation was described in lung cancer. His case is unusual in that we have performed detailed genomic studies before and after treatments which reveal tumor escape mechanisms. Our studies with his implanted tumor revealed unique therapeutic options to overcome this resistance. In depth discussion of this case which include the use of several novel drug options reveals our current state of art in the treatment of lung cancer. Following failure of second line chemo combination we chose taxotere evorilumus combination based on the use of mouse models in which other genomically based targets such as RET inhibitors have failed- initial results with this treatment are promising.

Conclusion: We are in the dawn of a new area of cancer therapeutics. The choice of new combination therapeutics will depend upon complex use of genomic and mouse model data.

O9

GASTROINTESTINAL CANCERS

O9.1 Tumor markers in colorectal cancer relapse

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Background: While efficient surgical treatment is a key to prolonged survival of colorectal cancer, post-surgical follow-up is important for early detection of relapsing disease or disease progression. Current follow-up systems, typically based on imaging (CT, PET, MR), is frequently supported by observation of tumor markers CEA, CA19-9. Due to their limited sensitivity and selectivity, better tools for monitoring of the disease are desirable. Among possible biomarkers for early detection of primary or recurrent cancer cell-free DNA (cfDNA), microRNAs (miRNAs) or mucins that exhibit CRC specific expression may be applied.

Aims: To evaluate different tumor biomarkers in a long-term follow-up after radical surgery for colorectal cancer.

Methods: Three studies concerning different tumor markers performed on author's department and results of surgery for colorectal cancer relapse are reported. A total of 1790 patients were operated on for colorectal cancer, 70 % patients underwent R0 resection, 63 % of patients were under further surveillance, relapse was diagnosed in 20 % of patients. Oncofetal tumormarker CEA, mucin tumormarkers CA19-9, CA242, proliferative tumor markers Thymidine kinase, soluble cytokeratins fragments TPS, TPA, adhesive molecules ICAM - 1, VCAM -1, IGF-1, and adipocytokines Adiponectin, Leptin and cell-free tumor DNA were studied.

Results: The surveillance based only on CEA and /or CA19-9 was cost-effective, but failed to disclose 1/3 of patients suffering from relapse. Combination of CEA and either CA19-9 or CA242 can be recommended for preoperative investigation and follow-up. Our results indicate a potential for the detection of cell-free tumor DNA as a non-invasive test of metastatic liver disease. With intensive follow-up system and aggressive surgical approach the resectability of the relapse in all locations was 35 %.

Conclusion: Early diagnosis of local recurrence due to follow-up helps to improve resectability and may reduce the necessary extent of surgery. The cfDNA appears a viable tool for monitoring of clinical progression of colorectal cancer in patients with cfDNA positivity prior to the surgery. All patients having recurrences should be assessed by a multidisciplinary team in a cancer centre.

O9.2 Bevacizumab in therapy of patients with advanced metastasizing colorectal carcinoma

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Background: The objective of this paper was to demonstrate contribution of maintenance therapy by means of monoclonal antibody Bevacizumab in patients with chemo-resistant metastases of colorectal carcinoma.

Aims: The authors present results of a study which analyzes a group of 34 patients with advanced metastasizing colorectal carcinoma (mCRC) who underwent palliative bio-chemotherapy and were given Bevacizumab in the second and further lines of the therapy.

Methods: A data file of 34 advanced mCRC patients (24 men, 10 women; median age of 60 years) was evaluated. The primary aim of this study was to assess overall response rate (ORR) and time to progression (TTP), the secondary objective was to assess bio-chemical response rate through tumor markers (CEA, CA 19-9, TPS) assessment, to evaluate life quality of the patients and to assess toxicity of the therapy. ORR was assessed with all the patients in compliance with RECIST criteria after 6-9 cycles of the bio-chemotherapy.

Results: The therapy benefit was achieved with 59 % of the patients /CR-2 pts (6 %); PR-4 pts (12 %); SD-14pts (41 %)/. The overall response rate was achieved with 18 % of the patients. Median TTP was 21 weeks. In case of bio-chemical response rate assessment, the CEA and TPS tumor markers showed statistically significant decline of values comparing the values before the therapy and at the time of ORR achievement (CEA $p < 0.0372$; TPS $p < 0.0021$). Regarding the CA 19-9 tumor marker, no significant decline was recorded ($p < 0.1731$). The main undesirable effect appeared to be accelerated hypertension (grade II-III with 18 % of the patients) and bleeding (with 6 % of the patients), no serious infusion response occurred within our group of patients.

Conclusion: Bevacizumab can be used as a highly effective alternative therapy of mCRC with very good tolerance and minimum of undesirable effects. Assessment of the tumor markers: CEA and TPS is a suitable way to check effectiveness of the targeted biological therapy.

This study was supported by the grant IGA MZ CR 9731 and IGA MZ CR 12025.

O9.3 Tumor associated macrophages as a potential predictive biomarker of response to 5-fluoro-uracil in patients with microsatellite-stable colorectal cancer

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Background: Colorectal cancer (CRC) TNM staging system predicts outcome and is used to allocate patients (pts) with stage III tumors to chemotherapy (CHT). Microsatellite instability (MSI), a positive prognostic biomarker in CRC, would predict a poor response to 5-fluoro-uracil (5-FU) CHT. **Aims:** We aimed to explore the prognostic role of tumor associated macrophages (TAM) in CRC, in relationship with cancer stage, MS-status and response to CHT

Methods: The density of TAM in CRC specimens was measured by computer-assisted analysis at the CRC invasion front as the average percentage of CD68 immune reactive area (IRA), in 425 consecutive stage II/III CRC of known MS-status (MSI, n=66; 15.5 %) and in 136 metastatic lymph-nodes (LN) matched from the same stage III CRC pts

Results: CD68+ IRA was irrelevant to the prognosis of stage II pts (n=216, p=ns), whilst it was associated with better prognosis in pts with stage III CRC (n=209, p=0.02). Similarly, CD68+ IRA was significantly associated with no relapse only in pts with MSS CRC (n=359, P=0.009) and in those who received 5-FU CHT (n=238, P=0.009), but not in MSI and CHT untreated pts. Accordingly, restricting the analysis to pts with stage III MSS CRC, we found that 5-FU CHT was associated with better DFS only in pts with a high density of TAM (>4.12 % median) [n=67, HR=0.34 (95 %CI 0.16-0.75); P=0.007], but not in those with a low density of TAM (<4.12 % median) [n=63, HR=1.63 (0.16-15.87); P=0.67]. In metastatic LN, CD68+ IRA significantly correlated with that of the primary cancer [p<0.001 r=0.42, (0.12-0.27)], and a high TAM density had a positive prognostic impact in pts with stage III MSS CRC who were treated with 5-FU [n=94, HR=0.31, (0.15-0.64); p=0.001], not in those untreated [n=26, HR=1.02, (0.29-3.55); p=0.96].

Conclusion: The association of a high TAM density with good outcome of 5-FU-treated pts with stage III MSS CRC indicates that innate immune cells can be a predictive marker of response to 5-FU. Thus the hypothesis of a synergistic effect between CHT and macrophages against MSS CRC cells deserves mechanistic elucidation.

O9.4 Immunologic methods for defining Malignant Transformation occurring in benign appearing Colonocytes

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Background: We are presently employing immunologic methods using specific IgG1's to define malignant transformation occurring in benign appearing colonocytes. This has been accomplished by targeting immunogenic tumor proteins expressed in these cells. Several immunogenic proteins associated with colon cancer were described by Hollinshead and later characterized by Neogenix Oncology. Monoclonal antibodies (NPC-1, 31.1 and 16C3) produced against these corresponding colonic tumor proteins are highly effective in both in diagnosing and initiating an immune response to target these tumor antigens in colon cancer.

Aims: After showing that these immunogenic proteins could be detected with a high degree of sensitivity and specificity in those cells having undergone malignant transformation yet phenotypically appearing benign, a question raised was how early in the oncogenic process did tumor antigens first appear

Methods: The stool ELISA and IHC were employed

Results: By employing IHC to examine the possibility of a field effect around existing colon malignancies, benign appearing colonocytes expressing tumor antigen were seen extending 5 cm. or more beyond the site of the primary lesion. Such changes were found to occur about 6 months before the transforming colonocytes took on phenotypic features of malignancy. It also was important to note whether such premalignant cells shed tumor proteins in a similar way to when the fully malignant tumor sheds antigen into the serum. We therefore looked at stools samples for the accumulation of shed antigen. A stool ELISA was designed and did detect shed antigen from both tumor as well as premalignant lesions. Levels greater than 1000 nannogm of antigen indicated the presence of a malignant lesion, and could indicate whether the bowel was or was not free of a malignant process

Conclusion: Since the biopsy method is accurate in defining pre-existing malignancies, it can help the pathologist at the time of colon surgery define margins that are truly free of the potential for cancer to recur. In the primary care physicians office however, the stool ELISA can easily replace the stool hemoccult test to determine the existence of early malignancy

O9.5 Hyperthermia (HT) in exocrine pancreatic cancer (PaCa) - review and a new concept for evaluation

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Background: In spite of some progress in earlier diagnosis about 90 % of patients suffer from primarily non-resectable or recurrent disease with the need for palliative treatment strategies. In spite of the newer regimens gemcitabine + Erlotinib and FOLFIRINOX even nowadays the strategies follow our concept of an efficacy orientated sequential polychemo- or multimodal therapy.

Aims: In the present paper we now try to reevaluate again the potential role of HT in PaCa disease

Methods: On the one hand we reviewed again the literature and analysed the actual prospective randomized trial. On the other hand we tried to test the value of a combination of systemic palliative chemotherapy (using gemcitabine, oxaliplatin and/or 5-fluorouracil regimens) with capacitative HT (Oncotherm 2000) in an intraindividual setting: in actually 8 patients with diffus liver (n=6) and lung metastases(n=2) we applied HT only to the left liver or left lung (15 cm and 30 cm electrodes resp.) in order to analyse the effects of additional HT on these lobes by comparing the behaviour of the HT-treated left side metastases with the HT-untreated right side lesions, using modern generations of CT and/or MRT.

Results: Also our own actual review did not show any convincing prospective randomized studis. The actual HEAT study will also not allow definitive conclusions because of an -from our experience- non-correct planning and randomization. Our concept of intraindividual testing of the additional effects of external HT before a potential therapeutical application to the upper abdomen or both sides of the lung is well accepted by the patients and -done in a way as recommended by other HT users -tolerated without side effects, except from some slight skin reactions. The imaging data are just evaluated and discussed at the congress.

Conclusion: The results so far suggest that this kind of intra-individual testing of HT in combination with chemotherapy might allow a way to get reproducible and convincing data on the potential clinical value of HT in pancreatic cancer.

Background: For stage II colon cancer patients, risk of recurrence is 20-25 % and the benefit from adjuvant chemotherapy is 3-5 %. Treatment decisions are made by clinical and pathological parameters. We have today, a genomic tool that helps to improve treatment decisions in this group. For metastatic CRC patients, drug therapy is partially effective. New gene therapies are being developed in order to enhance response rates and improve the quality of life by more personalized treatment possibilities.

Aims: Oncotype colon: to develop a validated multi-gene expression assay which will help and improve treatment decisions for patients with stage II colon cancer. The test will define high risk group patients whom will benefit from adjuvant chemotherapy.

Target now and foundation: predictive information based on the genetic characteristics of individual tumors in order to use medications both chemotherapy and targeted therapies more effectively.

Methods: Oncotype colon: IHC for MMR protein and PCR for MSI. The MMR proficient tumor will continue for an additional 12 gene assays.

Target now molecular profiling service and the deep sequencing of tumor DNA, Such Foundation genome profiling detects point mutations, short deletions or insertions, copy number abbration and rearrangements leading to clinical a report.

Results: Oncotype colon: selection of 12 genes in 4 large studies and results of the independent prospectively designed QUASAR and CALGB 9581 validation studies led to a "recurrence score" which provided independently and quantitatively an individual recurrence risk for stage II T3 MMR proficient tumors which can be used for treatment decision. The Target Now is an evidence based molecular profiling service. Tumor specific biomarkers are measured and matched with available therapeutic agents using a database of the world's clinical literature. The result leads to a report which summarizes which drugs may or may not be associated with benefit, such as; ERCC-1, TOPO-1, MGMT or the Foundation which identifies driver mutations with clinical implications such as EGFR, BRAF, HER-2, MET, C-KIT, M-TOR.

Conclusion: Clinical decisions for CRC patients will become more individualized based on tumor gene profiling.

O9.6 Genomic profiling of the tumor in CRC and it clinical use

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O9.7 CD24 as Biomarker and therapeutic target in GI malignancies

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Background: CD24 is a GPI-anchored mucin-like protein proven as a valid target in GI malignancies. Anti-CD24 mAb treatment induces a significant growth inhibition of colorectal and pancreatic cancer cells, in a time- and dose-dependent manner as well as a reduction of tumor growth in mAb-treated mice.

Aims: Engineer anti-CD24 chimeric and humanized full-length, toxin-conjugated Abs fused to PE38, a truncated form of *Pseudomonas* exotoxin, and smaller derivatives (scFv and Fab).

Methods: Edman-degradation, cDNA synthesis, sequence and computational analysis revealed the entire sequence of the murine Ab. Fc replacement with the human IgG1 resulted in a mouse-human chimera. A human donor (Ig-blast) was chosen as a scaffold for grafting critical sequences of the murine Ab. The immunotoxins were first constructed by non-covalent linking via the ZZ protein. Then a bacterial system allowed an Ab-toxin covalent linkage. High-efficiency engineering technologies were established to construct, express, and purify novel anti-CD24 derivatives. Their binding affinity, selective targeting, and cytotoxic activities were confirmed in CD24-expressing GI tumor cells.

Results: The chimeric and humanized Abs have higher cytotoxic activity. Moreover, the immunotoxins are better than the naked Ab with lower IC₅₀ values by several magnitude orders. The scFv-toxin killed CD24-expressing cells with an IC₅₀ of 1 µg/ml, while the values for the IgG-ZZ-toxin were much lower. The values of the humanized anti-CD24 (di/tetra)-PE38 were similar to those of the scFv, suggesting that the absence of post-translational modifications in the E.coli-produced IgG, affected reactivity. MTD studies confirmed no toxicity at high doses. Antibody targeting and accumulation within the tumor and its clearance was demonstrated by direct imaging using an anti-CD24 mCherry-fluobody.

Conclusion: CD24 inhibition may be a promising treatment for GI malignancies in combination with chemotherapy. Humanized Abs to CD24, naked and toxin-conjugated are more effective than the murine parental Ab. We are currently in the process of toxicology studies aiming for a Phase I study in 2013.

O9.8 Immunochemical Fecal Occult Blood Test (FIT) or Guaiac-based fecal occult blood tests (GFOBT): What is the preferred test?

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Background: Guaiac-based fecal occult blood tests (FOBTs) are not specific for human hemoglobin. Automated-development, FIT is quality-controlled relatively new test. It is specific for human hemoglobin, and does not require diet restriction.

Aims: To review the line of data comparing the performance of FIT with GFOBT

Methods: 1000 consecutive ambulatory patients were offered FIT (three samples, 70 ng/ml threshold). Sensitivity and specificity were 94.1 % and 87.5 % respectively, for cancer and 67 % and 91.4 % respectively, for any clinically significant Neoplasia.

In other study, average-risk persons were offered FIT or G-FOBT. Participants with positive tests underwent colonoscopy. Participants were followed through the Cancer Registry 2 years after the study. Hemoccult SENSa and OC-MICRO (three samples, 70 ng/ml threshold) were used. FIT was offered to 4,657 persons (Group A) and G-FOBT to 7,880 persons (Group B).

Results: The sensitivity, specificity, negative and positive predictive value for cancer in Group A and B were 100 %, 85.9 %, 100 %, 3.9 % and 61.5 %, 96.4 %, 99.8 %, 9.1 %, respectively. There was increased detection of advanced adenomatous polyp (AAP) by FIT.

EGD was performed in 160 patients after a negative colonoscopy. FIT positivity was not correlated with the finding of lesions, which are likely to bleed on EGD. Thus, EGD is probably not indicated in patients with positive I-FOBT and unrevealing colonoscopy

Conclusion: The sensitivity of FIT and GFOBT for the detection of cancer is comparable. However, FIT has a better sensitivity for the detection of AAP. Upper endoscopy is not indicated after positive FIT and negative colonoscopy.

O10

UROLOGICAL CANCERS (PROSTATE, BLADDER)

O10.1 Lucap prostate cancer xenografts: clinically relevant models of prostate cancer and their use in preclinical testing

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Background: Prostate cancer (PCa) is a heterogeneous disease and generating clinically and biologically relevant data invariably requires the use of multiple models. Results obtained using a single model can provide misleading information because of heterogeneity of PCa in patients. A paucity of PCa models that replicate the diversity of this disease has limited our understanding of PCa biology and evaluation of novel therapeutics.

Aims: Our objective was to establish multiple PCa xenografts to facilitate studies into PCa biology and design more effective and translational preclinical studies.

Methods: Primary and metastatic PCa tissues were obtained from surgery and at autopsy. Tumors were implanted into SCID mice. Characterization of established LuCaP models includes: histology, PSA expression, immunohistochemistry, oligo arrays, expression of the androgen receptor (AR), bone response, and responses to therapy.

Results: We have established 28 LuCaP xenografts from which 24 are adenocarcinoma and 4 are neuroendocrine. We also developed 8 matched pairs of androgen-sensitive and castration-resistant xenografts. These models exhibit the characteristics and reflect the diversity of PCa in man. Unsupervised gene expression clustering analyses revealed an association between the xenografts and the original specimens, pairing of androgen-sensitive lines with their castration-resistant offspring and maintenance of the phenotype during continuous passage in vivo for 2–8 years. Parallel to PCa in humans, LuCaP xenografts exhibit considerable diversity in the AR and its variants, in PSA expression, and significant heterogeneity in all biomarkers evaluated. Responses to therapy are also heterogeneous. Advanced PCa is associated with osteoblastic bone metastases, and 7 of our LuCaP models elicit an osteoblastic reaction in the bone. Additional analyses revealed that 5 models are PTEN negative, and 8 lines have the TMPRSS2:ERG fusion.

Conclusion: LuCaP. PCa xenografts are highly diverse and clinically relevant models to study PCa biology and evaluate new treatment modalities.

O10.2 miRNAs as tumor stratification and noninvasive urinary diagnostic biomarkers for bladder cancer

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Background: Altered microRNA (miRNA) expression may occur early in bladder cancer and may play a role in carcinogenesis and tumor behavior.

Aims: We evaluated whether alterations in miRNA expression could improve disease stratification and outcome prognosis in bladder tumors and noninvasive diagnosis in urinary samples.

Methods: miR-143, miR-222, and miR-452 expression levels were analyzed by quantitative RT-PCR (RT-qPCR) in paired urinary and matching tumors and in two independent prospective series of tumors and urinary specimens.

Results: Differential expression of miR-143, miR-222, and miR-452 in urine were verified by in situ hybridization in matching tumors. Tumor miRNA expression by RT-qPCR correlated with tumor grade, size, and presence of carcinoma in situ for miR-222, recurrence (miR-222 and miR-143), progression (miR-222 and miR-143), disease-specific survival (miR-222), and overall survival (miR-222). Protein expression patterns of potential miRNA targets, including vascular endothelial growth factor, BCL2, v-erb-b2 erythroblastic leukemia viral oncogene (ERBB) homolog 3, and ERBB4, were evaluated by IHC in tissue arrays containing tumors for which miRNAs were assessed by RT-qPCR. Target expression correlated with expression of their predicted regulatory miRNAs, recurrence (ERBB3), progression (ERBB4), disease-specific survival (ERBB3 and ERBB4), and overall survival (ERBB3 and ERBB4). Furthermore, RT-qPCR of miR-452 (area under the curve, 0.848) and miR-222 (area under the curve, 0.718) in urine provided high accuracies for bladder cancer diagnosis.

Conclusion: Thus, bladder tumors were characterized by changes in miRNA expression that could aid in tumor stratification and clinical outcome prognosis, and miRNAs were detected in urinary specimens for noninvasive diagnosis.

O10.3 Tumor markers serum levels for early diagnosis of prostate cancer

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Background: Prostate cancer is the sixth leading cause of cancer-related death in men worldwide. Current diagnostic procedure using PSA and fPSA ratio may lead to an over-diagnosis and to an overtreatment. It is therefore currently essential to focus on optimal early diagnosis based on new approaches.

Aims: Aim of the study is to evaluate the use of PSA and PHI index for early diagnosis of prostate cancer and for optimal indication for prostate biopsy.

Methods: Sera from 75 patients of Urology Clinic were tested from November 2010 till April 2012 in the Laboratory of Immunoanalysis, Faculty Hospital in Pilsen. These patients were suspicious for prostate cancer and all of them were indicated for TRUS prostate biopsy. All the sera were tested for total PSA, free PSA, proPSA. PHI index and % free PSA ratio were calculated. The tests were performed using chemiluminescence method on Dxl 800 instrument (Beckman Coulter, USA). All statistical analyses were performed using SAS 9.2 software.

Results: No significant statistical differences were found for medians of total PSA and % free PSA between benign prostate hyperplasia and prostate cancer (histology verified); (PSA 7.1 vs 7.7 ng/mL. % free PSA 16 vs. 13.7 %. But there were significant differences found for proPSA and PHI index between benign hyperplasia and malign prostate cancer. (proPSA median 14 vs. 27 pg/mL, $p < 0.0002$ and PHI median 35 vs. 77 $p < 0.0001$).

Conclusion: Our pilot study clearly confirmed that proPSA and PHI index is very useful for more precise differential diagnosis of prostate cancer.

Acknowledgements: This study was supported by the project of Ministry of Health of the Czech Republic for conceptual development of research organization 00669806 - Faculty Hospital in Pilsen.

O10.4 PSA based prostate cancer screening- Good and bad news, how to continue?

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Background: Recently the European Randomized study of Screening for Prostate Cancer (ERSPC) reported the outcome of their final analysis on the effect of systematic PSA based prostate cancer screening on disease specific mortality. PSA based screening can reduce prostate cancer (PC) mortality by 21 % and 29 % if adjusted for non-compliance in the screening arm; so far so good [1]. A reduction comparable with e.g. breast cancer screening and opening the way for population based screening programs.

However this mortality reduction coincides with considerable unnecessary testing and over diagnosis. The number needed to screen and the number needed to diagnose to avoid one man dying from PC is 936 and 33 respectively. Men were PSA tested repeatedly without clinically significant changes in their PSA level and the positive predictive value of a purely PSA driven biopsy protocol is 24 %.

Aims: Optimizing a PSA based screening algorithm

Methods: Within ERSPC Rotterdam risk prediction tools have been developed incorporating the most important predictor in a PSA based algorithm, being prostate volume (PV) [2]. Assessment of PSA density (PSA-D = PSA divided by PV) can give an idea of the percentage of men eligible for biopsy (applying a purely PSA based biopsy protocol with PSA cut-off of ≥ 3.0 ng/ml) and PC detection within different PV ranges.

Results: Analyses per PV category show the following: In men with PV < 30 cc, the median PSA was 4.1, the % of men with PSA ≥ 3.0 (and thus eligible for biopsy) was 35 % and mean PSA-D was 0.17. Similar statistics for PV classes 30-40, 40-50 and ≥ 50 cc were 5.2, 5.9 and 8.2 ng/ml for PSA, 49 %, 65 % and 89 % men eligible for biopsy and 0.15, 0.13 and 0.12 for PSA-D. PC detection rate in men with a PSA-D < 0.15 and ≥ 0.15 was 15.6 % and 53 % res. The percentage of GS ≥ 7 PC cases was 30 % and 53 % res.

Conclusion: In a purely PSA based screening algorithm most biopsies are performed in men with the lowest PSA-D while potentially aggressive PC are more common in men with PSA-D levels ≥ 0.15 . A correction for PV can avoid unnecessary biopsies and decrease over diagnosis. There is no one size fits all in PC screening and prediction models including information on PV can help in decreasing the unwanted side effects of PSA based screening.

1. Schröder et al. NEJM 2012, 2. Roobol et al Eur Urol 2010, 2012

O10.5 The DRE based ERSPC prostate cancer risk calculator and the additional value of Phi

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Background: PSA based screening can reduce prostate cancer (PC) mortality by at least 21 % [1] but coincides with considerable unnecessary testing and over diagnosis. A multivariate approach has shown to be able to

reduce these potential drawbacks to a considerable extent [2].

Aims: To assess the additional value of the Prostate Health Index Phi (a combination of tPSA, free-PSA and -2proPSA) within a multivariate prediction tool: the ERSPC prostate cancer-risk calculator (RC) developed on ERSPC section Rotterdam screening results (www.prostatecancer-riskcalculator.com). Two RC's were developed (# 3 for initial biopsy and # 4 for men having a repeat biopsy) predicting biopsy outcome using information on PSA, DRE outcome and prostate volume without the necessity of a ultrasound (DRE assessed volume: 25,40 or 60 cc).

Methods: The development dataset consisted of 889 men with tPSA values between 1.6–8.0 ng/ml (WHO calibration). A total of 617 men were biopsied for the first time while 272 men had a previous negative biopsy. To assess the additional value of Phi in the ERSPC prediction tools we applied transformation of prior odds to posterior odds via the likelihood ratio.

Validation of the newly developed prediction tool was done on an additional cohort of 950 biopsied men (628 initial biopsies and 322 repeat biopsies).

Results: Refitting the DRE based RC # 3 and DRE based RC #4 with and without Phi on the development dataset resulted in a significant increase in AUC of 0.68 to 0.74 for RC#3 and 0.67 to 0.76 for RC#4. The newly developed RC with Phi was validated showing again a sign. increase in predictive capability with AUC's of 0.68 for RC#3 and 0.72 for RC#3 + Phi. In men having a repeat biopsy the AUC's were 0.64 and 0.69 res.

Conclusion: A PSA based prediction model including information on prostate volume has shown good performance in predicting prostate biopsy outcome. The addition of Phi to such a model increases predictive capability even further and hence can be of aid in avoiding unnecessary biopsies. Data on the prediction of potentially aggressive prostate cancer will be presented. 1.Schröder et al. NEJM 2012, 2. Roobol et al Eur Urol 2010,2012

O10.6 Prospective head-to-head evaluation of [-2]proPSA, PCA3 and TMPRSS2:ERG in patients referred for prostate biopsy

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Background: Prostate cancer (PCa) specific biomarkers are needed to improve the accuracy of diagnosis.

Aims: To evaluate the ability of urinary PCA3, TMPRSS2:ERG (T2:ERG:ERG) and serum [-2]proPSA

(p2PSA) based prostate health index (phi) to predict outcome of prostate biopsy.

Methods: Serum samples and first-catch urine samples (after digital rectal examination) were collected from consented outpatients with PSA 0.5–20 ng/ml scheduled for prostate biopsy in two prostate cancer centers. Transrectal ultrasound-guided prostate biopsies were performed (at least 10 cores). PCA3 Score (ProgenSA PCA3 Assay, Gen-Probe) and T2:ERG Score (Gen-Probe research assay) were determined. Serum prostate specific antigen, free PSA, p2PSA (Beckman Coulter Immunoassays) were measured and percent free PSA (%fPSA) and phi ($p2PSA/fPSA \times \sqrt{PSA}$) were calculated.

Results: Of 246 enrolled men 110 (45 %) were diagnosed with PCa. 70 of the 136 men with no evidence of malignancy (NEM) were confirmed with 1 to 7 repeat biopsies, while 66 underwent first time biopsy. PCA3, phi and T2:ERG differed significantly between PCa and NEM and these markers showed the largest area under the curve (AUC) of all parameters (0.74, 0.68 and 0.63) respectively. PCA3 had the largest AUC of all parameters although not statistically different from phi. Phi showed partially lower specificities than PCA3 at 90 % sensitivity. The combination of both markers enhanced the diagnostic power with a modest gain in AUC of 0.01 to 0.04. While PCA3 showed the largest advantage in the repeat biopsy cohort, phi had the largest AUC within the 2–10 ng/ml PSA range for DRE negative patients.

Conclusion: PCA3 and phi were superior to the other evaluated parameters but the combination of both only moderately enhanced their diagnostic accuracy to predict PCa at first or repeat prostate biopsy.

O10.7 Factors affecting prognosis of prostate cancer

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Background: The value of early detection and treatment of prostate cancer is hotly debated. European studies show that a clear, 20 – 30 %, reduction in mortality can be achieved by screening based on determination of serum PSA and biopsy of men with elevated PSA values. However, in screening studies performed in the USA, no reduction in mortality was observed, but when screening was limited to patients with minimal co-morbidity, a substantial reduction in mortality was obtained. In a Scandinavian study, SPCG-4, comparing radical prostatectomy with conservative therapy, a significant reduction in mortality was obtained in men under but not over 65 years of age. In a similar study in the USA, no

difference in mortality was observed between patients treated by surgery or observation. Taken together, these results indicate that we need prognostic factors that identify patients that benefit from early detection and curative therapy. Ideally, it should be possible to determine these prognostic factors in serum, plasma or urine before diagnosis and to calculate the probability of various outcomes before diagnosis.

Aims: To evaluate the utility of serum makers for prediction of prognosis of patients with prostate cancer.

Methods: Total PSA, free PSA, tumor-associated trypsin inhibitor, and the beta subunit of hCG were determined in serum samples from men participating in the Finnish Randomized Screening for Prostate Cancer. Mortality was monitored for up to 15 years.

Results: Elevated concentrations of TATI and a reduced proportion of free PSA were associated with adverse outcome.

Conclusion: Serum markers can be used to predict prognosis of patients with prostate cancer. This can be expected to reduce both mortality, over diagnosis, and harmful therapy.

O10.8 Recent prostate cancer intervention strategies directed at the target androgen receptor: Molecular aspects

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Background: The androgen dependency of prostate cancer is known for more than sixty years now and androgen deprivation became the mainstay of first-line therapy for advanced prostate cancer.

Aims: However, not before this decade the accurate molecular definition of the incurable state i. e. the castration resistant prostate cancer (CRPC) was coined. It became obvious then, that the term “hormone refractory” prostate cancer no longer was the exact description for this state of the disease.

Methods: Meanwhile two diverse anti-androgen strategies reached clinical application offering new options even after chemotherapy. Two new drugs, Abiraterone Acetate and MDV3100, still provide a lever to arrest aberrant androgen signalling for those patients who already underwent multiple therapies and eventual failure. The mode of action of these new compounds is either complete pharmacological castration also shutting down the residual adrenal and

intracrine androgen sources, or a full androgen receptor blockade which obviates its function as transcription factor in more than one way.

Results: With the advent of these new treatment options to the clinic data from trials as well as recent experimental results lead to an ongoing discussion on their potential and limitations. These include therapy sequence, side effects and putative resistance mechanisms.

Conclusion: This presentation will address the molecular background to these new treatment options and will consider the origin of conceivable resistance mechanisms.

O10.9 Radiotherapy for localized prostate cancer

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Background: The treatment of choice for patients with localized Prostate Cancer remains controversially. Treatment decisions should be based on tumor extension (stage), tumor differentiation (Gleason score) and serum PSA.

Aims: To review the role of radiotherapy in the treatment of localized Prostate Cancer.

Methods: Review of literature concerning radiotherapy for Prostate Cancer

Results: Radiotherapy, Brachtherapy and external irradiation are considered as standard treatments for patients with localized Prostate Cancer. Although there are no comparative trials, the results with external irradiation are not inferior to surgery. The treatment recommended to patients with good prognostic factors is brachtherapy, for those with intermediate prognostic factors is external radiotherapy with short-term hormonal therapy, and for those with poor prognostic factors- external radiotherapy and long-term hormonal therapy. Prospective randomized trials have demonstrated a significantly better local control and biochemical free-survival when using a high dose of radiotherapy. The new technology of irradiation such as Intensity modulated radiotherapy and Imaging guided radiotherapy, provide more precisely and conformal treatment and reduce the incidence and the severity of side effects that are consistent with radiation cystitis and proctitis. Radiotherapy is effective, in term of local control and disease-free survival, as surgery is in patients with localized Prostate Cancer.

Conclusion: Radiotherapy is effective and a well tolerated treatment for patients with localized Prostate Cancer.

O11

NEW ANTI - CANCER THERAPIES

O11.1 Ablation of malignant tumors by intra-tumoral Ra-224 loaded wires and its biological and immunological effects

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Background: Radiotherapy mainly with gamma radiation is an important treatment for benign and malignant tumors. Alpha particles are of interest in radiotherapy because of the high linear energy transfer (LET) of these emissions, which lead to a high frequency of irreparable double-stranded DNA breaks (DSB). Yet, the short range of alpha particles inside tissue (50-80 microns) limit their use in cancer treatment.

Aims: To develop a practical alpha radiation-based treatment for solid tumors, which is termed Diffusing Alpha-Emitters Radiation Therapy (DaRT).

Methods: DaRT treatment is based on insertion of ²²⁴Ra-loaded wire(s), which release by recoil into the tumor short-lived alpha emitting atoms. These atoms spread in the tumor and spray it with lethal alpha particles, which cause damage to the malignant tissue.

Results: in vivo experiments demonstrated that DaRT treatment was effective in the ablation of various mouse and human derived metastatic solid tumors and was significantly enhanced when used in combination with chemotherapy.

Various tumor cell types differ in their sensitivity to be killed by alpha radiation. Higher in vivo tumor ablation by DaRT is in correlation with a larger distribution of radioactive atoms inside the tumor, low clearance of the radioactive atoms from the tumor, and higher cell sensitivity to alpha radiation mediated cell killing. Increased cell sensitivity was in correlation with a higher number of DSBs formed and a reduced ability of the cells to repair DNA damage.

Ablation of the primary tumor may result in spillage of antigenic components and the tumor becomes its own antigenic vaccine. Tumor destruction by DaRT was able to trigger systemic anti-tumor immune responses, which can be further enhanced by immunomodulation strategies.

Conclusion: DaRT was proven to be an efficient ablation treatment for solid tumors that can be used in combination with surgery, chemotherapy and immunostimulation for the treatment of solid metastatic tumors.

O11.2 de novo design of bioactive peptides: a novel approach for cancer therapeutics

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Background: Small molecular weight peptides have been applied to develop cancer therapeutics, mostly for their ability to easily penetrate cellular membranes and to interfere with enzymatic functions or protein-protein interactions within cells. In this study we have employed a physico-mathematical model, called the Resonant Recognition Model (RRM) for the design of small therapeutic antitumor peptides.

Aims: The study investigates the suitability of the RRM to design short bioactive peptide analogues for mammalian proteins known to possess antitumor activity, such as interleukin12 and TNF α , as potential candidates for cancer therapy.

Methods: The RRM approach was used to design 18-22 mer bioactive peptide analogues for murine interleukin12 and TNF α . In addition, a non-bioactive peptide lacking the specific RRM frequencies was designed as a negative control. The cytotoxic effects of RRM-IL, RRM-TNF and RRM-C were investigated on B16 mouse melanoma cells, on mouse normal skin fibroblasts, and on CHO cells. The effects were detected by cellular cytotoxic changes of cancer and normal cells using confocal laser scanning microscopy and quantitatively by LDH assay. Quantitative analysis of data was performed using ANOVA

Results: The CLSM images clearly indicated that the RRM-IL and RRM-TNF produced time and dose-dependent apoptotic and necrotic effects on the B16 cancer cells while no such effects were detected on the normal cells. The quantitative validation of RRM-IL on the B16 cells indicated that cell cytotoxicity levels increased significantly after 3 h with 0.2 μ g/ml of RRM-IL (25 %) compared with the non-treated cells (7 %) and with cells treated with RRM-C (2 %). No significant effect was observed on CHO cells (2.3 % without treatment vs 2.8 % after treatment). Moreover, RRM-TNF showed similar effects on B16 cells after 3 h at 0.2 μ g/ml (16 % with treatment vs 4 % without treatment). No cytotoxic effects were detected on normal cells after treatment with any of the peptides.

Conclusion: These findings indicate the suitability of the RRM approach to design bioactive peptides with the desired biological functions such as the selective anti-tumor effect of RRM-IL and RRM-TNF on melanoma cells.

O11.3 Feasibility och technical principles of t-cell-based adoptive immunotherapy in the first 12 patients with advanced urothelial urinary bladder cancer. preliminary data on a new immunologic treatment based on the sentinel node concept

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Background: Expected 2-yr survival for patients with urothelial urinary bladder cancer (UBC) with lymphnode involvement (pN2) is approximately 20 %, regardless of standard neoadjuvant/adjuvant oncologic treatment. Tumor-reactive lymphocytes are present in Sentinel nodes (SNs) draining UBC and display immunologic function on restimulation in vitro. Metinell nodes (MNs) drain secondarily from metastatic tumors and also possess tumor-reactive lymphocytes, which might be a source for adoptive T-cell immunotherapy.

Aims: To determine if MN detection and subsequent expansion of autologous T-helper cells with subsequent reinfusion was feasible and safe in patients with metastatic UBC.

Methods: In an open trial, the first 12 included patients are described. Patients were prospectively selected and had metastatic UBC. All patients were preoperatively staged with transurethral resection of the bladder and routine computer tomography. All 12 patients were preoperatively staged as T2-T4b N1-2 and/or M0-M1 or MX.

MNs were excised in conjunction with intended cystectomy. T lymphocytes were extracted with enhancement and expansion of tumor specific T-helper cells, followed by reinfusion of expanded T cells. Intended detection of MNs was performed intraoperatively with intended cystectomy. Harvested T cells were evaluated and cell cultures were established. Assessment of reinfusion of expanded, autologous, tumor-specific T-helper cells to six of the patients was performed, focusing on adverse effects.

Results: In six patients, it was feasible to administer the treatment. Reinfusion of these T cells was performed without any major adverse effects. In six other patients, we encountered technical failures but no adverse effects.

Conclusion: A novel adoptive immunotherapy based on T cells from tumor-draining lymph nodes in advanced UBC is displayed. Technical aspects and feasibility of the method is described.

O11.4 An IGF-Trap for the treatment of cancer

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Background: The IGF-I receptor (IGF-IR) plays an important role in the progression of malignant disease and has become a target for anti-cancer therapy. Clinical trials to evaluate IGF-IR inhibitors provided evidence of therapeutic efficacy, but have also exposed several problems with IGF-targeted therapy due mainly to the high degree of homology between the IGF and insulin systems.

Aims: Our aim was to engineer an IGF-IR inhibitor with specific, anti-cancer activity that could bypass limitations of current IGF-IR-targeting strategies.

Methods: We engineered a novel recombinant fusion protein comprising a soluble IGF-IR and the Fc portion of IgG-1 (the IGF-Trap) and characterized its properties using functional assays in vitro and animal models of metastatic cancer.

Results: Analysis of the IGF-Trap by Surface Plasmon Resonance (Biacore) technology revealed that it had a very high affinity for hIGF-I (4-10 nM) and hIGF-II (8-16 nM) but a very low affinity (1000-2000 fold less) for human insulin. In vitro, the IGF-Trap blocked the proliferation, anchorage-independent growth and invasion of highly metastatic lung carcinoma cells in response to IGF-I, confirming its ability to trap the ligand with high efficiency. In therapeutic models of metastatic colon and lung cancers, a reduction of up to 70 % in hepatic metastases was observed when the Trap was administered at clinically relevant doses.

Conclusion: The results identify the IGF-Trap as a specific, highly efficacious inhibitor of metastatic cancer growth and provide a rationale for further exploration of its clinical utility.

O11.5 Overcoming drug resistance to chlorambucil by its conjugation to a cancer cell targeting peptide

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Background: Targeted Drug Delivery (TDD) via carriers specific to receptors on target cancer cells can overcome the

propensity of chemotherapy to induce outgrowth of drug resistant cancer clones. The development of carrier-drug conjugates with engineered deviations in time release of different drugs from multi-drug conjugates could enhance their clinical efficacy.

Aims: As a step towards preparing such conjugates, our aim here was to prepare multi-drug conjugates and to test their cytotoxic potential in vitro versus free drug.

Methods: A 13mer peptide derived from the Myelin Basic Protein (pMBP) was used as the carrier peptide to target Chlorambucil (Chlor) to murine MBP B-cell leukemic cells bearing antibodies to pMBP. Solid-phase synthesis was used to prepare dendrone conjugates of MBP peptide (pMBP) coupled to either 1, 2 or 4 molecules of Chlor. MBP, non-relevant BCL-1 hybridoma cells and normal murine spleen cells were exposed to different concentrations of pMBP-Chlor conjugates or free drug and cell growth (metabolism) was measured.

Results: MBP cells and BCL-1 were initially resistant to free Chlor. Their exposure to pMBP-Chlor-1 completely reversed the resistance for MBP cells but not for BCL-1 cells. Dose response experiments in which MBP cells were exposed to pMBP-Chlor-1, pMBP-Chlor-2 or pMBP-Chlor-4 at equivalent doses of Chlor showed a statistically significant enhancement in cytotoxic activity of multi-drug copy conjugates versus single-drug copy conjugates.

Conclusion: First, the results demonstrate that TDD can overcome previous resistance of cancer cells to a drug, thus expanding the opportunity for more effective use of drugs with a proven clinical history in cancer chemotherapy. Second, the cytotoxic efficiency of Chlor was significantly enhanced by the use of conjugates comprising multiple copies of the drug. These data suggest that the increased efficacy of multi-dose conjugates is not only related to the amount of drug delivered to the cell but also to the mode of delivery. Drug transport through receptor-mediated endocytosis is more efficacious than delivery of free drug through non-specific cell uptake.

O11.6 Biomarker-guided clinical development of multi-peptide cancer vaccines – from discovery to phase III

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Background: Distinct biomarker classes may be used to guide the development of cancer immunotherapies: immune response biomarkers reflecting biological response, immune status biomarkers reflecting immune suppression and biomarkers predictive for response to therapy.

Aims: Two immunotherapy candidates were tested in three phase I-II studies for signs of clinical efficacy, immunogenicity, biological efficacy of two further immunomodulators and to discover predictive biomarker candidates.

Methods: IMA901 is composed of 10 peptides associated with renal cell cancer (RCC). Two phase I-II studies with a total of 96 HLA-A*02+ advanced RCC patients were conducted. IMA901 is currently being developed in a global phase III study.

IMA910 is composed of 13 peptides associated with colorectal cancer (CRC). A phase I/II study in 92 HLA-A*02+ advanced CRC patients was conducted.

Results: For both candidates, in all three studies, multiple vaccine-induced immune responses were significantly associated with clinical benefit confirming the proposed mechanism-of-action.

Additional biomarkers reflected immune suppression. Increased numbers of circulating myeloid-derived suppressor cells were significantly associated with a reduction of biological response and shorter survival.

Randomization of RCC patients to single-dose cyclophosphamide was associated with a significant decrease of circulating regulatory T cells. A cohort of CRC patients receiving imiquimod showed an significant increase in the number of immune responses.

Serum APOA1 and CCL17 were candidates for the prediction of response to therapy in RCC and are currently being validated in a subsequent clinical trial.

Finally, we present data indicating that peptide immunogenicity can be predicted from assessing the repertoire of CD8+ T cells in healthy donors.

Conclusion: These distinct biomarker classes reflect relevant mechanisms of immune induction and tolerance and assist in defining treatment regimen and patient subpopulations with a higher prospect of response.

O11.7 One step ahead toward personalized medicine in oncology: The WINTHER project - a change in the paradigm of cancer therapy

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Background: Current cancer therapeutics is based on standard protocols prescribed in consecutive lines. Today there are several models for selecting molecular targeted therapies based on tumoral molecular abnormalities (mutations / translocations / or amplifications of genes). Oncogene-driven therapy can be exemplified with the following couples: EGFR mutation/gefitinib, HER2 amplification/trastuzumab, B-RAF mutation/PLX4032, etc. However, most of the patients will not benefit of such advances and therefore therapeutic decision will still be done according to decade-old protocols which are unsatisfactory at the individual level.

WINTHER clinical trial is using an innovative method to predict efficacy of therapeutics that addresses these unmet needs and explores a rational choice of therapeutics beyond actual limitations.

Aims: To improve the individual outcome of chemo resistant metastatic solid tumors, in which classic concept of targeted therapy does not apply, by comparing the progression-free survival (PFS) using a treatment regimen selected by a biological approach of a patient's tumor with the PFS for the most recent regimen on which the patient had experienced progression

Methods: The method is based on three elements:

1. The investigation of the tumor and of the matched normal tissue from strictly the same patient and the same histological origin, using a combination of technologies, including sequencing (to detect mutations, translocations), Comparative Genomic Hybridization gene expression profiling, and miRNA profiling.
2. Building of a comprehensive database containing information about drugs and associated target genes, for all drugs known and drugs under clinical trials.
3. An algorithm providing a weighted genomic score to predict efficacy of drugs using the database and individual molecular complex profile.

Results: The plan is to complete a 4 center 200 patients study in 2 years. Investigations of tumoral and matched normal tissue biopsies will be centralized in two laboratories: Foundation Medicine, and IGR. The bio-informatics analysis will be performed by IGR and Ben Gurion University, Israel.

Conclusion: The novelty of the WINTHER trial lies in the use of tumor and matched normal tissue biopsies together with the use of a novel method for predicting efficacy of therapies.

O12

NEW EMERGING THERAPY TARGETS

O12.1 The splicing factor hnRNP A2/B1 is a biomarker and a driving oncogene in glioblastoma and metastatic breast cancer R. Golan-Gerstl¹, A. Shilo¹, I. Loinger¹, R. Reich², B. Davidson³, J. Jacobs⁴, K. Cesarkas⁴, G. Rechavi⁴, F. Rigo⁵, F. Bennett⁵, and R. Karni¹

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Background: The process of alternative splicing is widely misregulated in cancer, but the contribution of splicing regulators to cancer development is largely unknown.

Aims: To evaluate the splicing factor hnRNP A2/B1 as a new diagnostic and prognostic marker in glioblastoma, and metastatic breast cancer and as a new target for cancer therapy.

Methods: We examined the expression of hnRNP A2/B1 in glioblastoma and metastatic breast cancers; manipulated the expression of hnRNP A2/B1 in immortal and in cancer cells and examined their oncogenic properties. We performed RNA-seq analysis to identify hnRNP A2/B1 splicing targets.

Results: We found that the splicing factor hnRNP A2/B1 is upregulated in glioblastoma and in breast cancer metastasis. Moreover, patients with extra copies of the gene HNRNPA2B1 in their tumors show poor prognosis, while patients in which the HNRNPA2B1 gene is deleted, show better prognosis than the average. Overexpression of hnRNP A2/B1 can induce transformation and tumorigenesis while knockdown of hnRNP A2/B1 inhibits anchorage-independent growth and tumors in mice in glioblastoma and metastasis of breast cancer cells. Deep RNA sequencing analysis identified many targets, among them several tumor suppressors and oncogenes which probably contribute to the oncogenic properties of hnRNP A2/B1. Finally, we show that hnRNP A2/B1 targeting by antisense oligonucleotides has a potential therapeutic value in treating these cancers.

Conclusion: Our results suggest that hnRNP A2/B1 is a new biomarker for glioblastoma and metastatic breast cancer diagnosis and prognosis, and a new important target for glioblastoma and metastatic breast cancer therapy.

O12.2 Genome-wide DNA copy number alterations in human cancer are not random and reveal distinct molecular tumor groups

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Background: Cancer is characterized by a plethora of altered chromosomal regions. Currently, tumor classification, therapeutic decisions and determination of prognosis relies mainly on the investigation of single chromosomal loci for amplified or homozygously deleted genes. High-throughput, next-generation sequencing efforts are meant to unravel further novel genes, facilitating a more targeted decision for personalized medicine. As these approaches neglect the impact of the entire plethora of genomic aberrations on cancerogenesis, we believe that a multitude of markers will be identified, but they will not contribute to the overall understanding of functional strategies followed by human cancers for guided drug therapy.

Aims: Gene- or proteinexpression patterns reflecting the sum of genetic aberrations in individual tumors have not been explored to date. We hypothesized that similar sets of common, tumor-specific properties may exist, independent of any histologic, clinical or single molecular parameters.

Methods: We first analyzed matched ctrl/tumor DNA of 45 renal cell carcinoma patients, unbiased from current clinicopathological, genetic and biological valuations, by SNP microarrays. We integrated results into a novel workflow by combining data from in silico data mining, gene expression and tissue microarrays. Finally, we used our unique Artificial Intelligence (A.I.) platform, Powered by Qlaym GmbH, to screen for identified systematics of prognostic or predictive value in 11 additional human tumor indications, encompassing more than 3000 individual samples. Eventually, we retranslated A.I. suggested systematics into applied pathology.

Results: Five constellations of genome-wide systematics were identified; information is encoded by the global relationship amongst expression products (defined as “Cancer States”). Pathology has confirmed the novel nature of identified systematics - Cancer States are of prognostic and therapeutic relevance.

Conclusion: We propose a novel concept for molecular cancer classification and suggest that carcinogenesis is acquired through “goal-oriented” strategies. “Cancer States” present a relevant novel type of biomarkers in personalized medicine; they are also expected to lower costs for drug R&D as well as in clinical testing and adaptive clinical trials.

O12.3 High-throughput sequencing of H3K9me3- and H4K20me3-associated circulating nucleosomal DNA

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Background: Modified histone tails in nucleosomes circulating in the blood bear the potential as cancer biomarkers. Recently, using chromatin immunoprecipitation (ChIP)-related quantitative PCR we described reduced plasma levels of the two pericentric heterochromatin-specific histone methylation marks H3K9me3 and H4K20me3 in patients with colorectal cancer (CRC).

Aims: Here, by utilizing ChIP-related high-throughput sequencing we further characterized these modifications in circulation. The total number of nucleosomes, of sequence reads and of diverse DNA repetitive elements were statistically compared between the study groups.

Methods: Plasma DNA from nucleosomes immunoprecipitated by H3K9me3- and H4K20me3-specific antibodies from patients with CRC (N=15) and healthy individuals (N=15) was subjected to the Roche 454 FLX sequencing, and the generated array of ChIP-enriched sequences were compared to the human reference genome.

Results: Total nucleosome amount was not different in both groups. Concerning both histone modifications, lower numbers of sequence reads were detected in CRC patients as compared with healthy controls (medians in H3K9me3: 32 vs 61; $p<0.01$; in H4K20me3: 54 vs 88; $p<0.01$). Most abundant sequences were repetitive LINE and SINE elements while simple repeats, LTR, DNA, SAT and low complexity elements were less frequent. Best discrimination between both groups was achieved by total number of H3K9me3 reads (AUC 0.90) and H3K9me3 LINE elements L1 (AUC 0.93) and L2 (AUC 0.91).

Conclusion: The present results confirm earlier findings of lower H3K9me3 levels in CRC and show LINE elements to be the most frequent and best discriminative markers on modified histones.

O12.4 Recombinant human alpha-fetoprotein (rHuAFP) - a novel cancer targeting agent

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Background: AFP is a carrier protein produced by embryo. It enters the cells by endocytosis after binding to the specific AFP receptor. Embryonic cells express the receptor but its expression disappears in adult cells. Most cancer cells revert to expressing the receptor. Thus loading cytotoxic drugs onto AFP offers an attractive opportunity to selectively target cancer cells.

Aims: The study was undertaken to provide proof-of-principle that rHuAFP can be used as an effective targeting platform for selectively delivering cytotoxic drugs to cancer cells.

Methods: We have developed a complex of rHuAFP with paclitaxel (AFP-P) which can be reconstituted in saline without addition of solvents typically used for highly hydrophobic, cytotoxic compounds, such as Paclitaxel. The efficacy of this complex was then tested in vitro and in xenograft model.

Results: IC50 in COLO-205 cells was 9.7 ± 2.2 nM, and 1.7 ± 0.1 nM for P and AFP-P, respectively. In U937 cells IC50 was 3.4 ± 0.8 nM, and 0.5 ± 0.6 nM for P and AFP-P respectively suggesting that AFP-P may improve the delivery of the drug to cancer cells.

In mouse COLO-205 xenograft study control animals survived for maximum of 20 days following implantation of cancer cells. Mice treated with low dose of P (180 ug/week for 3 weeks) lived on average 2 days longer, while mice that received a high dose of P (300 ug/week) had median survival of 36 days with longest survival of 39 days. Mice that received a low dose of AFP-P (180 ug/week of P for 3 weeks) lived significantly longer than mice in control or either of the 2 P groups. Median survival in AFP-P group was 42 days, with longest survival to day 50.

Conclusion: AFP-P provided superior efficacy using lower total amount of P. In addition P-treated mice had significant vascular damage at the site of drug injection while no such damage was seen in AFP-P mice indicating safety advantage of AFP-P. Ability to constitute AFP-P in saline, and higher efficacy of AFP-P suggest that this approach could provide significant benefits in the clinical setting.

O12.5 The SNP -592 C/A of human il-10 gene promoter region as a risk predictor biomarker of premalignant lesions and cervical cancer in mexican women

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Background: An immunosuppressive state has been identified in women with persistent HPV infection, characterized by high levels of interleukin (IL)-10 in cervix.

Aims: In the present study the association of single nucleotide polymorphisms (SNPs) in the promoter region of IL-10 gene with the risk of developing squamous intraepithelial lesions of the cervix (SIL) and cervical cancer (CC) were analyzed and evaluated. The level of mRNA expression of IL-10 (mRNA-IL10) systemically and in the cervix and the IL-10 protein level in serum were determined.

Methods: Using a cross-sectional design, samples of peripheral blood of patients with SIL (n=204), with CC (n=80) and without SIL (n=166) were used to evaluate SNPs at loci -592A/C (rs1800872), -819C/T (rs1800871), -1082A/G (rs1800896), and -1352A/G (rs1800893) by allelic discrimination and measuring the mRNA-IL10. Cervical swabs and cervical biopsies were used for HPV typing and determination of mRNA-IL10. The genotype and allele frequencies of SNPs were analyzed using logistic regression.

Results: No significant differences were found between the frequencies of genotypes at loci -819, -1082 and -1352. Individuals who carried at least one copy of the risk allele A of SNP -592, showed an OR of 2.02 (95 % CI 1.26-3.25 p<0.003) for SIL and OR of 1.70 (95 % CI 1.06-2.71 p<0.02) for CC. We found a significant difference in mRNA-IL10 both systemically and in cervix in women with SIL and CC compared to women without SIL, and in the serum IL-10 protein (p<0.0001), was higher in patients carrying the risk allele A.

Conclusion: The -592 C/A SNP in the IL-10 promoter is associated with increased risk of SIL and CC and can serve as a biomarker predictive of risk for development of SIL in the cervix and CC in Mexican women. The mRNA-IL10 and IL-10 protein were progressively higher in SIL and CC cases, so that the presence of IL-10 can be a relevant factor for viral persistence and progression of disease.

O12.6 The predictive role of ERCC1 status in oxaliplatin based neoadjuvant treatment for metastatic colorectal cancer (mCRC) to the liver

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TASMC

Background: Colorectal cancer (CRC) is the third leading cancer diagnosis. The subgroup of patients who present with, or progress to, liver-only metastases can achieve a 5-year survival of 30 to 40 % when surgery is combined with perioperative oxaliplatin chemotherapy. Changes in the nucleotide excision repair pathway have been strongly associated with resistance to oxaliplatin. The mechanism of resistance is thought to be via increased excision repair cross-complementing 1 (ERCC1) expression.

Aims: This study aims to explore the possible predictive role of ERCC1 status in this setting.

Methods: Consecutive mCRC patients who underwent liver metastasectomy and received preoperative oxaliplatin based chemotherapy in our center were studied. ERCC1 mRNA expression levels were evaluated in primary bowel and post resection liver parenchyma when available. A single pathologist evaluated margins, necrosis in tumor and changes to the healthy liver parenchyma. Demographic and treatment response variables were also collected.

Results: 51 patients were evaluated. ERCC1 expression levels from the primary tumor were obtained for 25 patients and from the liver metastasis tissue for 41 patients, with 24 patients having matched primary and metastatic samples available. A significant negative correlation was detected between primary tumor expression of ERCC1 and both the extent of clear surgical margins (R=-0.61308, p=0.0011) and the percent of liver metastasis necrosis (R=-0.47393, p=0.0167). No relationship was observed between ERCC1 expression and response (defined by RECIST) or disease free or overall survival in this cohort.

Conclusion: High ERCC1 expression levels in the primary tumor correlate with extent of clear surgical margins and tumor necrosis in liver mCRC resected after neoadjuvant oxaliplatin based chemotherapy. Further study is needed, in a larger patient cohort, to assess the role of ERCC1 expression as a predictive marker benefiting subgroup for oxaliplatin in this setting.

O12.7 The iap livin plays a dual role in tumorigenicity and may serve as potential target for cancer therapy

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Background: The inhibitor of apoptosis protein (IAP) family can inhibit apoptosis induced by a variety of stimuli. We and

others previously described the IAP Livin. We found that Livin encodes two splicing variants, Livin α and Livin β , with diverse antiapoptotic effects. We further found that Livin is unique among IAP members as upon a strong apoptotic stimulus it is specifically cleaved by caspases to produce a truncated protein with paradoxical proapoptotic activity (tLivin).

Aims: We investigated the role of Livin in tumorigenicity and response to anti cancer therapy.

Methods: An animal model was established and the effects of Livin α and Livin β on initiation and development of tumors were compared.

Results: In the animal model, Livin α promotes tumor initiation in comparison with control. Interestingly, tumors originating from cells expressing Livin β did not develop. In these tumors, Livin β was cleaved to produce the proapoptotic tLivin β that repressed tumor development. In terms of mechanism, we showed that Livin β tumors developed only in mice lacking natural killer (NK) cell activity.

Patients with advanced melanoma usually do not benefit from conventional chemotherapy. We found that unlike chemotherapeutic drugs NDVHUIJ, a one-cycle replicating virus, overcomes the resistance to apoptosis of melanoma primary cultures that over express the Livin protein. In contrast, melanoma tumor cells that do not express Livin are relatively resistant to NDVHUIJ. Furthermore, we show that NDVHUIJ-induced oncolysis is attributed to the truncated proapoptotic tLivin that induces tumor cell death. Taken together, our data offer the possibility of a new viral oncolytic treatment for chemoresistant melanoma.

Conclusion: From the animal model, we conclude that Livin plays a major role in tumorigenicity and that NK cells induce cleavage of Livin to proapoptotic tLivin that in turn inhibits tumor growth. Our results demonstrate the potential of tLivin in inducing tumor cell death. Therefore, Livin and tLivin may serve as potential targets in cancer therapy.

O13

GUIDELINES for TUMOR MARKERS

O13.1 Preparing Guidelines on the Clinical Utility of Tumor Markers: Experiences from the Gastrointestinal Focus Group of the European Group on Tumor Markers (EGTM)

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Background: Practice guidelines are systematically developed statements to assist practitioners and patients in making decisions about appropriate health care for specific clinical circumstances. The availability and implementation of well-prepared guidelines should result in improved patient outcome, provide criteria self-evaluation by medical professionals, assist with reimbursement and offer criteria for use in accreditation.

Aims:

- a) To describe how the EGTM GI Focus Group guidelines are prepared.
- b) To present a draft version of updated guidelines on markers in colorectal cancer (CRC), gastric cancer and GI stromal tumors.

Methods: Literature review

Results: Over the last 10-15 years, the Gastrointestinal Focus Group of the EGTM has published several different guidelines on markers in gastrointestinal cancers, especially colorectal cancer (CRC). Currently recommended markers for CRC include FOBT in screening, CEA in postoperative surveillance and monitoring therapy in advanced disease, MSI in prescreening for Lynch syndrome and KRAS mutation testing for identifying patients with advanced disease likely to benefit from anti-EGFR antibodies. For patients with advanced gastric cancer under consideration for systemic treatment, measurement of HER2 is recommended while determination of KIT protein may be used as a diagnostic aid in patients with suspected gastrointestinal stromal tumors (<http://www.egtm.eu/recommendations.html>).

Conclusion: Additional Reading

1. Duffy MJ, van Dalen A, Haglund C, et al. *Eur J Cancer* 2003;39:718-27.
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O13.2 Evidence based guidelines in laboratory medicine

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Background: Guideline development has changed rapidly the last years. More and more, the principles of evidence based medicine govern the field, and guidelines are currently defined as systematically developed evidence-based statements.

Sound systematic reviews of randomized controlled trials are viewed as the highest level of evidence, and expert opinion is valued as a low level of evidence. Methods that together define evidence based medicine have been well developed, and are being tested and improved; together this might be regarded now as a science on its own.

Aims: Compared to clinical specialties, laboratory medicine as a diagnostic specialty has its own characteristics. These will express themselves also in the type of guidelines in this field. The different phases of guideline development will be studied against this background.

Methods: Illustrated with practical examples from studies and experience, the process of guideline development will be explained, taking into account both theoretical and practical aspects.

Results: The process starts with prioritizing guideline subjects, based on criteria from the profession or other parties. When the decision is made to develop a guideline, the next step is the formation of a guideline working group that includes all stakeholders. The topics are identified that require recommendations. This allows for formulating the questions, that form the bases for structured literature searches. Performing the literature research includes applying proper quality criteria and selection of adequate studies. The data are analysed and where possible summarized. An important step is the weighting of the results and writing the recommendations, a process that might involve not only evidence but also expert opinion. A good guideline is also expected to give attention to aspects of implementation.

Conclusion: The theory of guideline development includes several well defined steps. It might however not always be possible to translate theory into practice. Due to the special characteristics of the laboratory profession, other methods sometimes need to be applied. In many cases the lack of evidence, or lack of evidence of sufficient quality, will make it difficult to make strong recommendations.

O13.3 The Italian Experience on Clinical Governance of biomarker use. From a pilot approach of the Venice Centre to a national committee operating under the Health Ministry commitment

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Background: Available evidence shows that the use of tumor markers (TMs) in the clinical practice remains largely inappropriate. Clinical practice guidelines (CPGs) are certainly useful in clinical decision making, but their impact is limited by several barriers.

Aims: The evaluation of the true impact of CPGs is a key point in appraisal of the effectiveness of EBM in clinical practice.

Methods: From 2006 to 2008 the Regional Center for Biomarkers (CRIBT) in association with the Italian Society of Clinical Chemistry (SIBioC) carried out a survey on TMs request in Italy. The matching of TMs requests for target malignancies with epidemiological data showed that the number of the requests were higher than expected on the basis of cancer prevalence data.

From 2009 to 2010 CRIBT leaded and coordinated a multidisciplinary working group to produce a synthesis of TMs recommendations in 17 tumors. A comparison of CPGs was presented in a tabular form containing a synthesis of the recommendations associated with their AGREE scores. The tables were published in a book which was distributed throughout Italy.

In 2011 CRIBT presented to the policy makers of Veneto Region a feasibility plan of validating the two initiatives on a wider scale.

Results: The health care representative of Veneto Region proposed to other Italian Regions the activation of an interregional committee for the improvement of appropriate request of TMs, under the Italian Health Ministry commitment. Eleven out of 20 Italian Regions – representing 70.9 % of the entire Italian population – accepted to participate.

Conclusion: The interregional working group plans to reach the following goals over the next two years: 1) to optimize a model for an indirect monitoring of appropriateness surveying TM requests on a national scale; 2) to provide physicians with the best available evidence, preparing an updated version of the “Guide” and implementing it on a national scale; 3) to analyze the relation between the TMs actually requested and the expected

figures in order to identify priority areas that require immediate attention for improvement.

O13.4 Personalized treatment options for Patients with colorectal cancer:

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Background: For physicians facing patients with organ limited metastases from colorectal cancer, tumor shrinkage and sterilization of micro-metastatic disease is the main goal, giving the opportunity for secondary surgical resection.

Aims: For the majority of patients not achieving a sufficient tumor response, disease control remains the predominant objective. Since FOLFOX or FOLFIRI have similar efficacies, the challenge is to define which could be the most effective targeted agent

Methods: The prior molecular identification of patients that could benefit from anti-EGFR or anti-VEGF monoclonal antibodies is of critical importance.

Results: The KRAS mutational status is the first identified predictive marker of response to anti-EGFR and is required prior treatment with anti-EGFR therapy. For KRAS wild-type patients, the choice is in between anti-VEGF and anti-EGFR. In the future, in addition to clinical considerations, molecular markers may provide critical information for selecting patients who might benefit preferentially from one of these drugs. Indeed, the tumor’s pathology and patient’s metabolism is driven by genetic make-up, influencing the individual response and toxicity. Identification of mutations/alterations such as BRAF, PI3k, PTEN, AKT and others, we have the challenge in designing molecular driven clinical trials in small patient subgroups. The challenge of treatments’ optimization through specific biomarkers gain special value for a potentially curable disease as mCRC with metastases confined to the liver, as we do have the opportunity of mutually exclusive therapeutic choices in this setting.

Researchers have to perform clinical validation of the most promising factors. In the future, to make translational research productive, affordable and more concrete it will be necessary to have a world wide effort from the scientific community. One of the most important challenges is the inherited vs acquired resistance.

Conclusion: Resistance to therapy may already exist at the time of diagnosis and will emerge over time such as KRAS mutations and amplification suggesting the existence of subclones which may have stemness like features.

O14

NEW BIO MARKERS 2

O14.1 Vitamin D and cancer

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Background: Most epidemiological studies have assessed the effects of dietary intake only, although endogenous production after sun exposure is the main source of vitamin D.

Aims: Study aim is to present an overview of currently published papers related to the D vitamin and etiopathogenesis of cancer diseases and to compare it with the data of a study realized in Pilsen region in 2010-2012.

Methods: The control group consisted of 250 healthy individuals. Pathological groups of patients included 300 patients with different cancer types (50 patients with prostate cancer, 120 patients with breast cancer, 49 patients with colorectal cancer, 50 patients with melanoma and 31 with lung cancer). All of the patients were enrolled in the early clinical stage of cancer I - III. Advanced stages were not included into the study. Vitamin D serum levels were measured using CMIA method (Abbott, USA).

Results: All the results for serum vitamin D from pathological groups were significantly lower compared to the levels of the control group. All the cancer types had a high incidence rate of very low serum levels of vitamin D. Lung cancer had the highest incidence rate of very low vitamin D serum levels.

Conclusion: We found a high incidence of hypovitaminosis D in cancer patients compared to a healthy control group among a Czech population. This incidence rate is higher in comparison to data found in literature from the other parts of the world.

Acknowledgements: Supported by the project of Ministry of Health of the Czech Republic for conceptual development of research organization 00669806 - Faculty Hospital in Pilsen and Charles University project 264801

O14.2 The scent fingerprint of hepatocarcinoma: in-vitro metastasis prediction with volatile Organic Compounds (VOCs)

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Background: Hepatocellular carcinoma (HCC) is a common aggressive cancer with poor prognosis, due to a high rate of postoperative recurrence. Subclinical metastases are the major causes of tumor recurrence and patient mortality. There is currently no reliable prognostic method of invasion.

Aims: To investigate the feasibility for in-vitro prediction of metastasis with fingerprints of volatile organic compounds (VOCs).

Methods: Headspace gases were collected from 36 cell cultures (HCC with high and low metastatic potential and normal cells) and were analyzed using nanomaterial-based sensors. Predictive models were built employing discriminant factor analysis pattern recognition, and the classification success was determined using leave-one-out cross-validation. The chemical composition of the headspace samples was studied using gas chromatography coupled with mass spectrometry (GC-MS).

Results: Excellent discrimination was achieved, using the nanomaterial-based sensors, between (i) all HCC and normal controls; (ii) low metastatic HCC and normal controls; (iii) high metastatic HCC and normal controls; and (iv) high and low HCC. Several HCC-related VOCs were identified through GC-MS analysis that could be associated with biochemical cellular processes.

Conclusion: The presented results constitute a proof-of-concept for the in-vitro prediction of the metastatic potential of HCC from VOC fingerprints, using nanotechnology. Further studies on a larger number of more diverse cell cultures are needed to evaluate the robustness of the VOC patterns. These findings could hold future potential for the development of a fast and potentially inexpensive laboratory test of subclinical HCC metastasis.

O14.3 Long time oral vitamin D supplementation in patients suffering from exocrine pancreatic insufficiency

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Background: Recently we were able to demonstrate that, in general, oral short time supplementation (suppl.) of vitamin D (vit. D) allows normalization of decreased 25(OH)D (vit. D) serum levels in most of our patients (pat.) suffering from pancreatic diseases.

Aims: We now report on studies in 24 pat. compared to 7 controls (C = individuals without upper gi-diseases) suppl. with oral vit. D for 18 up to now more than 30 months.

Methods: In 7 controls (C) and 24 pat. (P) suffering from chronic pancreatitis (CP,n=6), pancreatic cancer (PC,n=7), curative resective pancreatic surgery (PR,n=11) and gastrectomy for gastric cancer (GE,n=2) total serum 25 (OH)D was measured (Diasorin (R)) before and during an oral vit. D suppl. period of 18 up to now more than 30 months. After an initial dose finding to increase the serum levels to values between 30-60 ng/ml a continuous oral vit. D dose was given in each pat. and the serum levels were controlled by 3-8 determinations/year (med. 5) per pat. For oral suppl. were given 1 000 IE up to 20 000 IE vit. D daily.

Results: Total serum 25(OH)D levels before vs during vit. D suppl. (mean value of 3 - 8 determinations/year): 12,3±7,8 vs. 41,9±9,8 (med 10,7 vs 41,3) for group C, 6,5±3,9 vs 45±11.8 (med 4,0 vs 38,6) for group PC, 6,9±4,3 vs. 41,3±8 (med 4,0 vs 39,0) for group PR, 49,3±11,1 vs 9,3±6,3 (med 47,9 vs 6,9) for group CP and 8,5±5 vs. 44,5±10,3 (med 4,6 vs 44,1) for the whole group of patients (P). Oral suppl. dose: 7000 up to 40000 IE (med 20 000) per week for the single pat. of group C and 7 000 up to 140 000 IE (med 40 000) per week for group P. The difference between the highest and the lowest 25 (OH)D serum levels in the course of a 12 months period amounted to 14,7+/-9.2 ng/ml (med 13).

Conclusion: Oral vit. D suppl. allows a long time normalization of serum 25(OH)D also in pat. suffering from pancreatic diseases and/or pancreatic surgery in the case of adequate substitution with pancreatic enzymes and adequate nutrition advice as done in all of our pancreatic pat.. These findings might be of interest for these pat. not only with respect to bone metabolism in order to avoid metabolic bone diseases, but also with respect to recent discussions suggesting potential relevance of vit. D metabolism also for immunologic diseases, cancer prevention and/or treatment of diabetes mellitus.

most promising is the use of genetically engineered mouse models.

Aims: To develop and characterize a mouse model for aggressive human breast cancer

Methods: A pre-clinical breast cancer stem cell model is based on the properties of cancer stem cells (CSCs) derived from transgenic MMTV-PyMT mice. To identify putative CSCs, a systematic study of cell marker expression related to tumor initiation (TI) was carried out. Properties of the CSCs that might be used for nano-therapy were identified.

Results: Transplantation of as few as 10 cells cultured from primary breast cancers of MMTV-PyMT mice can recapitulate the morphological and physiological components of a primary tumor upon transplantation to syngeneic immune competent recipients. Four cell populations were recovered from primary breast cancers. Two of these cell populations, one epithelial and one mesenchymal (MSCs), were able to proliferate in vitro and then differentiate in selective media, but only the epithelial population exhibited TI. In addition, only epithelial cancer cells expressing CD49f could initiate tumors, indicating that CD49f is a more stringent CSC marker than is expression of CD24. Co-injection of TDMSCs with tumor initiating cells (TICs) from cultured CSCs significantly increases TI of CSCs after subcutaneous injection and allows TI with a single TIC. Expression of Bmi-1 was found to be necessary for the self-renewal of the breast CSCs. Treatment of breast cancer cells lines with shRNA to Bmi-1 significantly and specifically reduced growth of BCA stem cells in vitro and blocked TI.

Conclusion: We now propose to use our characterized mouse model of breast cancer stem cells for pre-clinical testing of nano-particle directed therapy of breast cancer. Both transplanted and primary breast cancers in the MMTV-PyMT male and female mice will be treated with specifically synthesized nano-particles coated with CD49f (marker for BCa stem cells) containing placitaxol and/or shRNA to Bmi-1.

O14.4 A pre-clinical breast cancer stem cell model based on MMTV-PyMT Mice for testing nanoparticle directed therapy

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Background: Every feasible approach possible should be used to develop ways to prevent and treat human cancer. One of the

O14.5 Intravenous iron : supplement or substitute to ESAs?

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Background: Therapeutic options for the treatment of anemia in cancer patients include iron therapy and erythropoiesis-stimulating agents (ESAs, only in patients receiving chemotherapy). Intravenous (I.V.) iron is more effective than oral or no iron; it improves haematopoietic response to ESA therapy. quality of life and, reduces transfusion needs .

Aims: Can hematopoiesis be turned on if iron is made available by the intravenous route to produce red blood cells?

Methods: Review of two small randomised, controlled trials in patients with gynaecological malignancies (N=44, N=75) receiving chemotherapy or radiochemotherapy (1,2) and a third one, an observational study including 420 anaemic cancer patients (3).

Results: In the first two trials, I.V. iron as anaemia monotherapy increased haemoglobin (Hb) and significantly reduced the number of required blood transfusions by 40,9 and 24 % respectively (1,2). The third trial was an observational study including 420 anaemic cancer patients and it provides the as of yet largest subset of data on patients treated with I.V. iron alone 3. In this study, 83 % received ferric carboxymaltose alone and only 17 % also received ESA. Ferric carboxymaltose alone was as effective as in combination with ESA3. Five weeks after the first ferric carboxymaltose treatment, stable median Hb levels >11 g/dL were achieved (3) which compares well to the only transient increase reported for blood transfusions (4).

Conclusion: In the three clinical trials, I.V. iron increased haemoglobin and, in two, it reduced the need of transfusions. These results are consistent with the notion of functional Iron deficiency where iron is not available due to Hepcidin mediated iron sequestration. It appears that soluble iron is necessary for effective erythropoiesis in cancer patients. Further studies are required to confirm these results and to help to identify I.V. iron responders. Moreover, ESA dose reductions as achieved with I.V. iron can reduce ESA-related treatment costs.

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O14.6 Vitamin D and cancer -associations and therapeutic potentials

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Background: It has been observed that regular sun exposure is associated with substantial decrease in certain cancer death rates as well as a reduction in overall cancer mortality rates, which may be related to body's vitamin D metabolic pathway. In cloudier Northern latitudes of the USA, prostate, breast and colon cancer have been documented to occur 2-3 times more often than the sunnier regions. The anticancer action of vitamin D at cellular level has not been fully understood.

Aims: To evaluate the correlation between Vit D levels and Cancer.

Methods: A literature search.

Results: The non-calcium mobilizing functions of vit D are associated with inhibition of cell differentiation, inhibition of cancer-cell proliferation, and exhibits anti-inflammatory, proapoptotic, and antiangiogenic properties. Colorectal cancer was hypothesized to be related to vit D status and it is most strongly associated with it. A recently meta-analysis of Colorectal cancer studies, found that individuals with ≥ 82 nmol/l (33 ng/ml) serum 25(OH) level had 50 % lower incidence of Colorectal cancer compared to those with relatively low levels, less than 30 nmol/l (12 ng/ml) ($P < 0.01$). Another meta-analysis of Colon cancer found that for each 20 ng/mL increase in serum 25(OH)D levels, the odds of Colon cancer were reduced by 40 % (OR, 0.57; 95 % CI, 0.43-0.76). Most studies comparing Prostate cancer incidence and vitamin D levels do not show clear associations with risk for Prostate cancer, although some of the studies suggest weak inverse associations. Two studies conducted in Nordic countries, clearly supported an inverse association. Relatively few studies of 25(OH) D level and risk of Breast cancer have been reported, some of them with conflicting results. An inverse correlation has been observed between regionally estimated UVB and Breast cancer mortality. In the Nurses' Health Study, 701 Breast cancer cases and 724 controls were included. Women in the highest quintile of 25(OH) D had a RR of 0.73 (P trend = 0.06) compared with those in the lowest quintile. The consecutive trial (NHS II) found no correlation between the two.

Conclusion: Although these data are not highly validated they certainly imply that vitamin D metabolism might be related to cancer biology.

O14.7 MESOTHELIN- a new Biomarker for Mesothelioma

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Background: Mesothelioma is an aggressive malignant tumor that surrounds the serous layer of different body organs, as the pleura of the lung and the peritoneum, which carries a poor prognosis. The survival rate of these patients (pts) who are diagnosed at an advanced stage, is 15 % over 5 y. There are no other available Biomarkers for detection of this disease.

Aims: To distinguish Mesothelioma from Lung Cancer, to early detect Mesothelioma in a high risk population and to assess response to therapy in treated pts.-by Mesothelin levels.

Methods: Levels of Mesothelin were compared between a Control group (n=50), Refinery workers exposed to

respiratory risk factors (n=94), 32 Mesothelioma pts (71 samples), taken during times of active disease (verified by imaging, biopsy and clinical correlations) before and after treatment and Lung Cancer pts.(42 samples).

Results: Mesothelin levels in Mesothelioma pts ($4.92 \text{ nM} \pm \text{SD } 5.65$) were significantly higher ($P < 0.0001$) than in each of the other groups. There was no significant difference in Mesothelin levels between the Lung Cancer ($0.469 \text{ nM} \pm \text{SD } 0.383$), a group exposed to risk factors ($0.325 \text{ nM} \pm \text{SD } 0.412$) or the group controls ($0.399 \text{ nM} \pm \text{SD } 0.376$). In the Lung Cancer pts, Mesothelin levels in

stage IV ($0.569 \text{ nM} \pm \text{SD } 0.388$) were significantly higher ($P < 0.016$) than those of stage III ($0.305 \text{ nM} \pm \text{SD } 0.08$). A cutoff of 1.092 nM Mesothelin, showed specificity of 95.4 % and sensitivity of 65.3 %, PPV of 88.7 % and NPV of 83.2 %, while using a cutoff of 0.7 will provide a 74.6 % sensitivity for the diagnosis of Mesothelioma pts and a specificity of 73.8 %.

Conclusion: Mesothelin levels can be used to distinguish between Lung cancer and Mesothelioma pts, as well as indicate treatment response. High Mesothelin levels are a clear indication for pursuing a comprehensive and rapid workup of pts, because of the high probability of Mesothelioma.

O15

NEW BIOMARKERS and APPLICATIONS

O15.1 The Cancer Immunotherapy Immunoguiding Program (CIP): An international network for the harmonization of cellular immunological biomarkers

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Background: The CIMT Immunoguiding Program (CIP) was founded in 2005 as a working group of the Cancer Immunotherapy Association (CIMT). This network currently comprises 52 academic and industry participants in Europe and the USA.

Aims: CIP has pioneered the concept of assay harmonization: by supporting technical validation of in vitro assays, CIP strives to establish high-quality biomarker assessment for guiding the development of new and efficient immunotherapeutics on the basis of cellular immunological markers (immunomonitoring).

Methods: CIP has recently enlarged its proficiency panel program including service panels for quality control purposes and exploratory panels for addressing new aspects and methods. To facilitate inter-laboratory comparisons, CIP is establishing and providing cellular reference samples that can be reproducibly used among different labs. In large-scale proficiency panels, CIP has followed a two-step approach to identify parameters critical for assay sensitivity, specificity, precision and accuracy. These panels also deliver direct feed-back to individual laboratories on technical performance as compared to that of other participants and to internal benchmarks. CIP is sharing its results and data-driven guidelines by peer review publications and on its homepage (www.cimt.eu/workgroups/CIP).

Results: Results of the most recent proficiency panels will be presented, as well as an update on the activities of the group. While primary effort was focused on the measurement of antigen-specific CD8⁺ T cells, CIP is currently extending its area of interest to other immune cells of relevance for immunotherapy, in particular immunosuppressive cells such as regulatory T cells and myeloid-derived suppressive cells. Strategies for harmonizing the monitoring of these subsets will be discussed.

Conclusion: Assay harmonization by inter-laboratory comparisons using the two-step approach identifies critical assay

parameters that are key to harmonization guidelines. Furthermore, direct feedback is provided to each lab as compared to a peer group.

O15.2 Early detection of solid tumors by infrared biochemical analysis of mononuclear cells – a preliminary study

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Background: Detection of malignancies such as breast, colorectal and lung at early stages can improve survival rates from 20 % up to 80 %. In the last few years, great efforts were taken to develop novel, non invasive and effective methods for early diagnosis of cancer. However, having a low sensitivity or specificity, withheld their admission as methods for early detection of cancer in population at risk.

Aims: In the present study we present a different concept for cancer detection. Our principle was to observe the immune system response to tumor presence somewhere in the body rather than looking for the tumor cells themselves. Furthermore, we analyzed the entire biochemical signature of the effected immune cells using infrared spectroscopy, instead of focusing on a single specific protein as a biomarker.

Methods: The clinical study was conducted at Todos Medical Ltd., Ben-Gurion University of the Negev and Soroka University Medical Center. 2 ml of peripheral blood was collected from 45 cancer patients and 46 healthy controls. Peripheral blood mononuclear cells (PBMC) were extracted from whole blood and measured under Fourier transform infrared microscope (FTIR-MSP). The differences between spectra of cancer and controls subjects' PBMCs were analyzed by Analysis.

Results: The results show significant differences between healthy and cancer subjects at multiple bands along the infrared spectra of the PBMCs. Furthermore, we observed a distinct spectral pattern for each type of cancer which may indicate for a difference in composition and activity of PBMCs population. Several absorption bands may indicate for the stage of the disease.

Conclusion: This study presents the potential of FTIR spectroscopy analysis as a simple, cheap and minimal invasive method for cancer detection. The present research is preliminary and additional clinical trials are required for further validation and verification.

O15.3 Volatile fingerprints of lung cancer specific genetic mutations

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Background: Gene profiling became a key role for accurately classifying tumors in individual patients, for predicting the response to the available treatments and for personalizing cancer treatment. Common techniques require tissue specimen and thus are invasive, time consuming and may delay therapy.

Aims: Here, we report on a new, non-invasive, highly sensitive test for profiling of genetic mutations of lung cancer cells.

Methods: Headspace samples of several cell lines with specific well-defined genetic mutations (EGFR and KRAS mutations and ALK fusion) were studied and compared. The headspace VOCs from the cell lines was collected using passive sampling badges throughout the whole incubation time then exposed to an array of cross-reactive gold nanoparticles (GNPs) coated with organic capping. The signals collected from the GNP sensors were analyzed using a discriminant factor analysis (DFA). The nature and chemical composition of the VOCs associated with each of the investigated genetic mutations was explored and determined through complementary chemical analysis and discussed in terms of metabolic pathways.

Results: Based on the new profiling approach, we present a volatile fingerprint assay based on nine binary tests in a form of a barcode. Best-match classifications in the over-defined system lead to the correct identification of 76 % of the blind samples.

Conclusion: While the reported concept was obtained through in vitro studies, as a way to eliminate confounding factors, it is reasonable to assume that similar findings could be detected directly from blood or the exhaled breath samples. Those will allow immediate testing to predict a clinical benefit from targeted therapy. The reported approach could help guiding treatments by tracking genetic alterations, so that patients could benefit from safer, more timely and effective interventions that improve survival and quality of life while reducing hospitalizations caused by unnecessary invasive procedures.

O15.4 Optimising treatment outcomes in GI cancer with biomarker research

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Background: Predictive biomarkers have helped to define treatment strategies for various cancers. At Amgen,

we have an extensive biomarker programme in GI cancers, including metastatic colorectal (mCRC), gastric (mGC) and pancreatic cancer (mPC). Panitumumab (pmab), ganitumab (gmab) and rilotumumab (rmab) are fully human, monoclonal antibodies targeting key cancer pathways: EGFR; IGF1-R and HGF/SF, respectively. Biomarker research with these compounds has aimed at improving patient selection and thus efficacy.

Aims: We summarize published data from the Amgen biomarker program in GI cancers.

Methods: Data from three randomized studies will be presented

Results: The 408 study of pmab as third-line monotherapy in mCRC was the first to show that KRAS status was predictive for pmab – patients with WT status could benefit while those with MT status did not. This finding was confirmed in two phase III studies of pmab combined with chemotherapy (PRIME [FOLFOX4 ± pmab, first line] and 181 [FOLFIRI ± pmab, second line]). In both studies, patients with WT status showed PFS and ORR (and a trend for improved OS) with chemotherapy plus pmab vs chemotherapy alone.

Gmab: GAMMA was randomised phase III trial comparing gemcitabine ± gmab as first-line treatment in patients with mPC. Investigation is ongoing as to whether high levels of total IGF-1, IGF-2, and IGFBP-3 may predict an enriched treatment effect when gmab is added to gemcitabine. Rmab: A double-blind, placebo-controlled, phase II study randomised 121 patients with locally advanced or metastatic gastric or esophagogastric junction cancer 1:1:1 to 15 mg/kg rilotumumab + ECX, 7.5 mg/kg rilotumumab + ECX, or ECX alone. The addition of rilotumumab to ECX increased PFS compared with ECX. Analysis by immunohistochemistry of MET-expression showed that in patients who had tumours with high levels (>50 %) of MET PFS and OS were longer with rmab + ECX vs ECX alone.

Conclusion: KRAS was the first biomarker shown to have a predictive effect for a targeted agent in mCRC. KRAS testing is now standard practice and part of the label indication for panitumumab. Following in its wake, other biomarkers for ganitumab and rilotumumab show promising results. These discoveries lead the way for the future of personalised medicine in GI cancer.

O15.5 Receptor for advanced glycation end-products and its ligands in cancer.

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Background: Receptor for advanced glycation end-products (RAGE) and its ligands – advanced glycation end-products (AGEs), S100 proteins and high mobility box group protein 1 (HMGB1) are involved in the pathogenesis of cancer. sRAGE, S100 proteins and HMGB1 levels are altered in many diseases including nephropathy, diabetes mellitus or inflammatory diseases.

Aims: The aim of the study was to focus on the significance of new biomarkers in patients with cancer.

Methods: sRAGE, several S100 proteins and HMGB1 are determined in serum immunochemically by enzyme linked immunosorbent assay.

Results: sRAGE is decreased in patients with cancer – we demonstrated decrease in patients with breast cancer as well as in pancreas cancer. However, higher sRAGE levels were observed in patients in stage III breast cancer. In patients with pancreas cancer sRAGE further decreases in the presence of diabetes mellitus. sRAGE is influence by renal function which is of high significance mainly in patients with advanced kidney disease. Presence of infection/inflammation also decreases sRAGE levels but the increase in patients with sepsis was associated with adverse outcome.

Conclusion: sRAGE and RAGE ligands are new potential biomarkers. Although they are not cancer specific, they seem to be promising in patients with cancer. Further studies are required to demonstrate their utility in patients' monitoring.

Acknowledgement: The study was supported by PRVOUK-P25/LF1/2.

O15.6 The breathprint of gastric cancer.

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Background: Gastric cancer (GC) is often diagnosed late, leading to very poor prognosis. Upper digestive endoscopy/biopsy, the standard for diagnosing GC, is invasive, slow and costly. Recently the analysis of volatile organic compounds (VOCs) in exhaled breath has attracted much interest as a possible method to rapidly diagnose diseases. Breath biomarkers may provide a safe and elegant solution for GC diagnosis/screening.

Aims: To identify the VOC pattern of GC with a new nanomaterial-based breath test.

Methods: Alveolar breath samples were collected from 160 patients with gastric complaints (GC/ulcers/less severe conditions) that underwent endoscopy/biopsy, and were analyzed using an array of 11 nanomaterial-based sensors; 25 % of the samples were randomly blinded for

determining the sensitivity/specificity of the method. Predictive models were built employing discriminant factor analysis (DFA) pattern recognition; insensitivity of the models against three major confounding factors (alcohol/tobacco consumption; *Helicobacter pylori*) was ensured, using leave-one-out cross validation. The chemical composition of the breath samples was studied using gas chromatography/mass spectrometry.

Results: Three DFA models were developed that achieved excellent discrimination between the sub-populations: (i) GC vs. benign gastric conditions, among all the patients (based on a subset of 7 sensors; 83 % sensitivity; 88 % specificity); (ii) early stage GC (I-II) vs. late stage (III-IV), among GC patients (5 sensors; 88 % sensitivity; 93 % specificity); (iii) ulcer vs. less severe, among benign conditions (6 sensors; 75 % sensitivity; 73 % specificity). The models were insensitive against the tested confounding factors. Chemical analysis found statistically significant concentration differences of 12 VOCs (alcohols/aldehydes/esters/ketones/alkenes) among the sub-populations.

Conclusion: A proof-of-concept was delivered for the differential diagnosis & staging of GC among patients with gastric complaints. Additionally the method provided differential diagnosis of benign gastric conditions. In the future this technology could provide a low-price high-scale screening tool for identifying patients that should be referred to endoscopy.

O15.7 Biomarkers to monitor mtor signaling in hepatocellular carcinoma cell lines

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Background: Hepatocellular carcinoma (HCC) has become the third leading cause of cancer-related death worldwide, a trend paralleled by an increase in advanced hepatitis C virus (HCV)-related liver disease. Targeted therapies in HCC have a stellar importance in identifying novel molecular targets for treatment.

Aims: Our aim was to evaluate mTOR- activation in human HCC cell lines, as well as the anti-tumoral effect of blockade of the mTOR-pathway by everolimus.

Methods: : The mTOR-pathway was assessed using molecular immune and electron microscopy markers. The effects of mTOR-signaling using Everolimus were evaluated in normal human hepatocytes and liver cancer cell lines HepG2 cells and HuH7 in the presence or absence of replicon (HCV active).

Results: Aberrant mTOR-signaling (phosphorylated RPS6) was present in cancer cell lines and was associated with insulin-like growth factor pathway activation, epidermal growth factor up-regulation, and changed the tumor cell viability in vitro. Blockage of mTOR-signaling with Everolimus shows a higher apoptotic activity in replicon cells when compared with the control HuH7 and HepG2 cells by immunological and biochemical biomarkers as well as electron microscopy. MTor-inhibitor reduced vascular endothelial growth factor (VEGF) and tumor necrosis factor (TNF α) cellular expression and their levels in the media. Transmission electron microscopy revealed apoptotic cells and apoptotic bodies in replicon treated cells. Also replicon cells treated with everolimus presented a distinct

VEGF-immunohistochemistry pattern of more intensive foci. Moreover the replicon cells treated with everolimus show lower HCV RNA viral load when compared with the replicon cells untreated.

Conclusion: VEGF and TNF α represent essential biomarkers in monitoring HCC progression or regression in vitro. mTOR signaling has a critical task in the pathogenesis of HCC, supporting that everolimus cancer-blockade is effective especially in the HCV-related tumor cells. These findings establish a rationale for targeting the mTOR-pathway in HCC induced specifically by HCV. Stratifying the patients based upon their primary liver disease and monitor the everolimus efficacy in reducing tumor burden using VEGF and TNF levels will translate in personalised efficient results.

O16 LEUKEMIA & LYMPHOMA

O16.1 Role of microenvironment in myeloma: focus on bone disease

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Background: Multiple myeloma (MM) is characterized by accumulation of malignant plasma cells in the bone marrow and induction of bone disease, induced by activation of osteoclasts and deactivation of osteoblasts (Yaccoby 2010). We demonstrated that inhibition of osteoclasts attenuated growth of medullary MM. Our group also discovered that MM cells secrete the canonical WNT signalling inhibitor, DKK1, which suppresses osteoblastogenesis. Osteoblast activation by DKK1 neutralization, WNT3A, PTH or mesenchymal stem cell (MSC) cytotherapy promoted bone formation and inhibited MM (Yaccoby et al., 2007; Qiang et al., 2008; Pennisi et al., 2010; Li et al., 2012). We further demonstrated that while osteoclasts stimulate MM cell survival through overexpression of growth factors such FAP (Ge et al., 2006), osteoblasts have negative effect on MM by secreting decorin (Li et al., 2008).

Aims: Our goals are to dissect molecular mechanisms that mediate MM-microenvironment interaction and develop effective therapies to simultaneously control MM and associated bone disease.

Methods: MSC cytotherapy in the SCID-hu model for MM was used to assess the consequences of increased bone formation on MM. Gene microarray was used to dissect molecular mechanisms and identify microenvironmental targets.

Results: Human microarray was performed in nonmyelomatous and myelomatous MM-bearing SCID-hu mice cytotherapeutically treated with human MSCs. 60 genes were found to be upregulated and 50 genes downregulated in myelomatous bones after cytotherapy. Among the top upregulated genes we identified HMOX1, whose expression was lower in myelomatous compared to nonmyelomatous bones and was restored to normal level after cytotherapy. Clinical biopsies from MM patients (n=369) also had lower HMOX1 expression than biopsies from healthy donors (n=20) or patients in remission (n=92, $p<0.0001$), and reduced HMOX1 expression was associated with poor survival. The HMOX1 inducer, hemin, prevented MM-induced bone disease in experimental model.

Conclusion: We identified dysregulated microenvironmental biomarkers in MM. Increased bone formation creates an inhospitable environment for MM through increased production of anti-MM and anti-inflammatory mediators (e.g. HMOX1).

O16.2 Phenotype of High-Risk relapsed myeloma

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Background: Recent treatments for multiple myeloma (MM) result in high rates of complete remissions and extended survival from a median of 5.5 to 10 years, with patients living disease-free over 20 years. Yet as protocols include more effective agents, relapsing patients experience significantly shorter post relapse survival (PRS) than those who relapsed after earlier treatments

Aims: To identify genes that are associated with and can predict survival of relapsed myeloma patients

Methods: Using Affymetrix microarrays (GEP), we analyzed changes in GEP in patient MM cells after co-culture with osteoclasts (OC) or mesenchymal stem cells (MSC) to identify genes associated with MM cell survival. We then determined if these genes are also associated with post-relapse survival, using GEP obtained at first relapse from 127 patients treated with high-dose therapy protocol Total Therapy 2 (TT2-RL), 33 with TT3 (TT3-RL), and 98 patients treated with TT-like (TTL-RL) protocols

Results: Expression by MM cells of 887 and 365 Affymetrix probe sets changed after co-culture with OC and MSC, respectively, with only 72 overlapping probe sets. Expression of 33 of the 887 probe sets was associated with PRS with a false-discovery rate (FDR) ≤ 0.01 . Models were developed with these probe sets. The best model, as determined by hazard ratio (HR) of the training set (24.3), identified 13 high-risk relapses in TT3 (HR = 26, median PRS 5.8 vs. 26.4 months), and 17 high risk relapses among TTL-RL patients (HR = 20, median PRS of 7 vs. 46 months). 30 of the 365 probe sets changed following MSC co-culturing and were associated with PRS with a FDR ≤ 0.01 , and predicted PRS of the training set with median survival of 14 vs. 46 months (HR = 24), and of the test sets with HR = 9 and HR = 4 for TT3-RL and TTL-RL, respectively. Only 8 genes used in the models were among those commonly changed following OC and MSC co-cultures.

Conclusion: Studying the effects of co-culturing with osteoclasts and osteoblast progenitors on GEP of MM cells revealed a pattern of gene expression associated with and predictive of post relapse survival of patients treated with autologous stem cells supported high dose therapy. The 8 genes common to both models may be attractive targets for directed therapy. We are currently testing the effects of altering expression of some of these genes during therapy

O16.3 New developments in flow cytometry analysis of cellular markers for diagnosis and prognosis of hematological malignancies

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Background: Flow cytometry immunophenotyping has become a routine methodology in conjunction with morphology, cytochemistry, molecular and chromosomal analyses for characterization of malignant cells of the lymphohematopoietic system.

Aims: To incorporate flow cytometry for assisting in diagnosis, prognosis, evaluation of the efficacy of the treatment, the expression of cellular antigens available for immunotherapy, and detection of Minimal Residual Disease. Other cellular properties studied include evaluation of the Multi-Drug Resistance phenotype, the cell cycle and ploidy status, oxidative status and the expression of various oncogenes and anti-tumor proteins.

Methods: Single-cell suspensions derived from the peripheral blood, bone marrow, lymph nodes and, occasionally, from other sources (pleural and cerebrospinal fluids) are stained with a combination of fluorochrome-conjugated antibodies or other fluorescence agents such as propidium iodide for analysing the cell cycle and ploidy, rhodamine-123 for evaluations of Multi-Drug Resistance, and dichrolofluoresceine diacetate for determining the oxidative status). Flow cytometry allows simultaneous measurement of cell size, granularity as well as four (or more) different fluorochromes.

Results: The acute leukemia panel is designed primarily to quantify leukemic blasts and determine their cellular origin - myeloid or lymphoid, and subclassification. For lymphomas/lymphoproliferative disorders phenotyping includes their characterization as B, T or An acute leukemia panel was designed primarily to quantify leukemic blasts and determine their cellular origin - myeloid or lymphoid, and subclassification. For lymphomas/lymphoproliferative disorders, phenotyping includes their characterization as B, T or NK cells. For B cell malignancies, the presence of a monoclonal population expressing restricted kappa or lambda immunoglobulin light chain, and for T cell malignancies, the existence of an abnormal T cell phenotype.

Conclusion: Flow cytometry allows multi-parameter analysis of different subpopulations of peripheral blood, bone marrow and lymph nodes cells.

O16.4 Hijacking of an inflammatory pathway as a driving event and a biomarker of high risk acute lymphoblastic leukemia

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Background: The receptor to thymic stromal lymphopoietin (TSLP) exists on subpopulations of T-lymphocytes, on Dendritic cells and on Basophils. It is a heterodimer between CRLF2 and IL7R that signals through JAK-STAT pathway and participates in inflammatory and allergic reactions.

Aims: To describe the consequences of abnormal activation of this pathway in lymphoid leukemias

Methods: mutational analysis, functional studies in-vitro and in-vivo, epidemiological studies

Results: Over the last five years we and others have uncovered the aberrant expression and activation of this inflammatory pathway in B cell precursor ALL (BCP-ALL). Genomic aberrations lead to overexpression of CRLF2. Consequently there are activating mutations in components of the TSLP pathway, most commonly in JAK2 but also in JAK1 or in CRLF2 or IL7R. We also found activating mutation in IL7R in about 10 % of T-ALLs. The abnormal expression and the mutations cause constitutive ligand independent signaling that enhance cytokine independent survival in-vitro and in-vivo. Several international studies demonstrated that the existence of these aberrations is a biomarker for poor prognosis leukemias.

Conclusion: Importantly the same aberrations are potential targets for drugs originally developed for treatment of allergic and inflammatory disorders and clinical trials are in progress.

O16.5 The increase in serum thymidine kinase 1 subunit levels is more significant than the TK1-activities in samples from patients with solid tumors; implications for disease detection and monitoring efficiency.

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Background: Thymidine kinase (TK1) is a DNA precursor enzyme and its levels correlate with cell proliferation. Measurements of TK activity, with the TK-REA and TK LIAISON (DiaSorin Inc.) assays, are used clinically in different hematologic malignancies. Antibody based methods have recently been introduced to study the concentrations of the TK1 protein.

Aims: To determine the TK1 subunit and activity levels in sera from humans and dogs with different tumor diseases.

Methods: Serum TK1 activities were measured with a ³H-Thd phosphorylation assay and in some cases with the TK-REA method. The TK1 protein levels were measured with an immune-affinity and western blot assay with anti-TK1 antibodies and in some cases with the TK 210 ELISA (AroCell AB). Serum samples from dogs with leukemia, lymphoma, solid tumors, and human sera from patients with myelodysplastic syndrome, prostate, colorectal and ovarian malignancies were tested. Sera from healthy dogs and human blood donors were used as controls.

Results: Serum TK1 is composed of multimers of 25/26 kDa subunits. The levels TK1 subunits in sera from humans and dogs with hematologic malignancy, solid tumors and healthy subjects were measured. The result showed a significant increase in the TK1 protein levels in sera from both humans and dogs with solid tumors and hematologic malignancies relative to the controls. However, the level of TK1 activity was only increased (about 20-fold) in patients with hematologic malignancies and not in patients with solid tumors.

Conclusion: Sera from patients with solid tumors contains high levels of enzymatically inactive TK1. These results have important implications for in vitro diagnostic methods for determination of serum TK1 in clinical practice, of relevance for both veterinary and human medicine.

O16.6 Loop-Mediated Isothermal Amplification (LAMP) as a rapid, sensitive and reliable screening tool for molecular diagnosis of leukemias

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Background: the diagnostic work-up of acute leukemias consists in an initial morphologic evaluation of the peripheral blood and bone marrow, followed by

detection of several molecular markers for diagnosis confirmation and prognosis assessment. For this purpose, PCR-based molecular methods are widely employed with important limitations such as time-consuming and multi-steps procedures.

Aims: to develop novel isothermal molecular assays, not PCR-based, to overcoming the limitations of conventional PCR in detection of chromosome translocations.

Methods: we have developed a novel molecular technique, based on the LAMP (Loop mediated AMplification) reaction, for specific and sensitive detection of fusion transcripts originated by gene translocations. Our assay, called Retro-Transcription (RT)-LAMP detects in a multiplex format up to two fusion transcripts and the mRNA of the housekeeping gene GUS β , that act as internal control for validation of negative results. Both retro-transcription and amplification are achieved by incubation at constant temperature for less than one hour, thus resulting in a close-tube, rapid single step procedure.

Results: We established the RT-LAMP performances on two molecular markers: the BCR-ABL (p190 and p210) and the PML-RAR α (Bcr1, Bcr2 and Bcr3) fusion transcripts. The level of sensitivity was established on serial dilutions of mutant RNA into wild type RNA and resulted between 10⁻⁵ and 10⁻². The specificity was 100 %, established on RNA from wild type cell lines and on 30 samples from healthy donors. The RT-LAMP assays have been validated on clinical samples previously diagnosed by conventional RT-PCR (n=186), obtaining perfect agreement.

Conclusion: The isothermal LAMP technique for detection of translocations demonstrated an improved level of sensitivity, specificity and simplicity of assay set up in respect to PCR and can therefore significantly improve the diagnostic routine of clinical labs.

O16.7 TK-Thymidine kinase- A new Biomarker for Lymphoma and Leukemia

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Background: Thymidine Kinase (TK) is an enzyme in the salvage pathway for pyrimidines, involved in the nucleic acid synthesis and considered to be a proliferation biomarker. TK, in its old version, was shown previously to be a significant prognostic marker in a number of malignant diseases, both in solid and hematological diseases. Today it is not used routinely in clinical settings.

Aims: Our study aimed to assess the usefulness of a new kit that measures TK serum level in a variety of Hematological Malignancies.

Methods: We studied correlations between TK levels and: The routine marker β 2-microglobulin, patients survival and a longitudinal follow-up of TK levels in Multiple Myeloma. 194 patients with Hematological Malignancies and 40 Controls were included in this study. The upper limit of the normal values range was determined to be 7.7U/l, based on average + 2SD.

Results: No significant correlation was found between TK and β 2-microglobulin serum levels. However, in some pts,

there was a similar pattern of changes in levels, correlated to clinical parameters. In a long term follow-up of patients with Multiple Myeloma, increases in TK levels were accompanied by recurrence or increase in severity of disease. A statistically significant decreasing trend was found in the TK levels in Multiple Myeloma pts., during the first 3 m of treatment ($P < 0.005$).

Conclusion: This preliminary study of the new TK Bio-marker in Hematological malignancies shows prognostic significance of increasing levels and correlation to survival of those patients.

O17 MELANOMA

O17.1 Interesting possibilities to improve the safety and efficacy of ipilimumab (Yervoy)

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Background: As predicted, an anti-CTLA-4 mAb (ipilimumab) on binding to T lymphocytes breaks down immunetolerance. Thereby, it was hoped, tumor-specific-T cells would be freed for a sustained attack on cancer cells. Data on ipilimumab treatment in 1498 patients with advanced melanoma (in 14 phase I–III trials) has shown immune-related adverse events (irAEs) in 64.2 % of the patients consistent with tolerance breakdown. However, there is no evidence that the antitumor effects via a CTLA-4 blockade are attributable to T cells specifically targeting tumor cells. In fact, several trials indicate a possible correlation between grade 3 and 4 irAEs with clinical efficacy of ipilimumab; tumor regression may be associated with autoimmunity development.

Aims: Therefore, we suggest a new treatment paradigm. The non-tumor specific pan-lymphocytic activation should be exploited by a ‘pretargeting’ approach proven successful in radioimmunodetection and radioimmunotherapy.

Methods: First, an anti-tumor mAb conjugated with streptavidin (StAv) should be administered to be followed by the delivery of biotin-labeled anti-CTLA-4 mAb. This schedule has the virtue of endowing T cells with the ability to travel to tumor sites without prematurely succumbing to apoptosis, while streptavidin’s ultra-high affinity for biotin (KD, 10(–15) M) ensures capturing all T cells binding biotin labeled anti-CTLA-4.

Results: Using the law of mass action, we calculated that following administration of ipilimumab at >1 mg/L concentration (~5 mg/patient ~70 kgbw), the strong forces of the immune system liberated by the anti-CTLA-4 antibody blockade would then be focused with laser sharp accuracy on tumor cells without collateral damage to normal host cells.

Conclusion: Despite the dramatic effects in a subgroup of patients, the majority of patients with metastatic melanoma do not obtain long-lasting clinical benefit from ipilimumab. Thus, further studies are required to optimally exploit the immune-stimulating properties of ipilimumab for cancer treatment.

O17.2 Self-renewal and oncogenic pathways in a stem cell-derived cat melanoma

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Background: We have studied a highly malignant melanoma of a domestic cat which produces highly metastatic tumors in young cats and nude mice. A remarkable feature of this rare tumor is that it transforms into neuronal cells within 48 hours after the exposure to an endogenous cat retrovirus RD114 in vitro. Since both the neural cells and melanocytes arise from the neural crest of the embryo we questioned whether the cat melanoma was a stem cell tumor?

Aims: This study aims to identify biomarkers of a highly metastatic cat melanoma.

Methods: Proteins were extracted from the melanoma, transformed neuronal cells and normal cat embryo fibroblasts, and each extract was fractionated by 2-dimensional gel electrophoresis. All in-gel-protein spots were compared with corresponding spots (if present) from identically prepared gels from the counterpart neuronal cells and cat embryo fibroblasts. Differentially expressed protein spots were tested for peptide fingerprints by Matrix Assisted Laser-Desorption Ionization Time Of Flight (MALDI-TOF) mass spectrometry.

Results: Comparisons of genome-wide proteins from the melanoma with those expressed in the normal cat embryo fibroblasts and the transdifferentiated neuronal cells indicated that 90 % of all proteins expressed in this melanoma were of stem cell origin. By the use of multiple bioinformatics tools, we have identified 33 proteins in the melanoma that have been normally associated with the growth, self-renewal and development of mammalian embryos. Functional categorization of each protein revealed that the bulk of these proteins belonged to the extracellular matrix, membrane receptors, cytokines, transcription factors, kinases, hormones, oncoproteins and other signaling proteins.

Conclusion: Our studies demonstrate that the cat melanoma cells originated from embryonic stem cells. These cancer stem cells can be distinguished from other tumor cells by their ability to self-renew and their capacity to transdifferentiate into neuronal cells. Further, both the melanoma and transformed neuronal cells express germ-cell proteins not present in the normal cat embryo fibroblasts. These proteins will now help in classifying biomarkers of human melanomas that may distinguish clinically similar tumors that respond to therapy versus those that do not.

O17.3 Tumor markers as an aid in the histological diagnosis of lung cancer

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Background: The treatment and survival of lung cancer patients is clearly related to the histological type of tumor. Discrimination between Adenocarcinoma and other NSCLC is also important because responses to chemotherapy seem to be different.

Aims: The aim of this study was to evaluate the utility of 6 tumor markers in the differential diagnosis of lung cancer (NSCLC vs. SCLC and Adenocarcinoma vs. Squamous tumors).

Methods: Tumor marker serum levels were prospectively studied in: 1719 patients with NSCLC (863 stage I-III, 856 stage IV) and 339 patients with SCLC (133 limited disease [LD], 206 extensive disease [ED]). We also included 23 patients with neuroendocrine tumors, 6 carcinoid tumors and 8 patients with mixed tumors.

Results: Significantly higher CEA and CA 15.3 concentrations were found in Adenocarcinomas, SCC and CYFRA 21-1 in squamous tumors and ProGRP and NSE in SCLC than in NSCLC. Combinations of tumor markers were useful to distinguish NSCLC (SCC > 2 ng/ml, CEA > 5 ng/ml or CYFRA > 3.3 ng/ml and ProGRP < 100 pg/ml, and NSE < 35 ng/ml) and SCLC (ProGRP > 100 pg/ml or NSE > 40 ng/ml) in 86 % of the patients (1800/2093 patients), with a low number of discrepancies with the pathologist (2.6 %, 51 patients). Likewise, tumor marker combinations were also useful to distinguish Adenocarcinomas vs. Squamous tumors in 82 % of NSCLC with discrepancies with the pathologist in only 8.4 % of the patients.

Conclusion: Tumor markers are useful tools as an aid in the histological diagnosis, mainly in patients with advanced tumors.

O17.4 Clinical responses of metastatic melanoma using adoptive transfer of “young” tumor infiltrating lymphocytes

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Background: Adoptive cell therapy (ACT) utilizing Tumor Infiltrating Lymphocytes (TIL) has developed into a promising approach for the treatment of metastatic melanoma patients.

We have modified the standard TIL generation procedure and used short-term cultured, unselected TIL (“Young-TIL”) for treatment, as short culture periods were shown to correlate with clinical response and significantly simplify the laboratory process. In addition, patients received lymphodepleting chemotherapy and high-dose bolus Interleukin-2.

Aims: Study the clinical outcome of treatment of metastatic melanoma patients with adoptive immunotherapy.

Methods: Ninety patients were enrolled to the study and Young-TIL cultures were successfully generated for 81 patients (90 %). Twenty-three patients were dismissed from the study, mainly due to clinically deterioration, resulting in a total drop-out rate of only 26 %, compared to 70 % in previous TIL ACT. Fourteen patients will complete their treatment shortly and 53 patients have been evaluated. At the time of treatment all patients were refractory to prior IL-2 based therapy and mainly staged M1c (76 %). Eight patients had CNS involvement.

Results: Twenty-three treated patients (=43 %) achieved clinical responses, including six complete remissions (CR) and 17 partial responses (PR). Five patients with brain metastasis showed responses (1 CR and 4 PR). Thirteen additional patients demonstrated disease stabilization (SD). There was a significant correlation between clinical response and overall survival (OS) ($p < 0.001$). Non-responding patients had a median OS of 5.4 months, whereas the median OS for complete and partial responders has not been reached yet. 70 % of the responders are alive two years after treatment. Side effects were transient and manageable. The positive correlation between short TIL culture duration and clinical response was clearly demonstrated.

Conclusion: To summarize, we report here that treatment with Young-TIL can mediate tumor regression in 43 % of metastatic melanoma patients with manageable toxicity. Multi-institutional trials are crucial to support the widespread application of this promising approach.

O17.5 Identification of a novel circulating microRNA signature for diagnosis of melanoma based on a comparative analysis of total serum expression profiles

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Background: The quest for accurate, simple and low-cost technologies for cancer diagnosis is the focus of intensive

global efforts. Quantification of serum microRNAs (miRNAs) has only recently become feasible and reliable, with most efforts focusing on miRNAs overexpressed by cancer cells.

Aims: To identify a combination of serum miRNAs unique for melanoma

Methods: We conducted an unbiased comparison of serum miRNA expression profiles between metastatic melanoma patients and age- and sex- matched healthy donors. The significant miRNAs were further tested in age- and gender- matched colon and renal cancer patients. Computerized hierarchical clustering and ROC analyses were used to test the value of the miRNAs.

Results: For the first time, we show that loss of two normal serum miRNAs, miR-29c and miR-324-3p, is highly indicative of metastatic melanoma. Computerized hierarchical clustering analysis clearly distinguished melanoma patients from healthy donors, metastatic colon and metastatic renal cancer patients. ROC analysis confirmed >90 % of specificity, sensitivity and accuracy.

Conclusion: These results imply that this specific alteration might reflect a host response or reaction to melanoma, which is mostly independent of tumor-specific phenotype. This approach therefore offers a substantial advantage, as it is independent of tumor heterogeneity and is expected to have superior biomarker performances.

O17.6 The use of biomarkers in the era of novel Melanoma therapies.

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Background: Serum markers can be important tools for treatment monitoring in melanoma patients. Recently, novel melanoma treatments have shown promising results in metastatic melanoma patients. The specific BRAF inhibitor Vemurafenib has shown more than 50 % response rate, but unfortunately most patients will become resistant to the treatment in a matter of months. Finding a serum marker to predict resistance can be an important clinical tool. Immunotherapy has also shown benefit in metastatic melanoma patients with a small percentage of patients having long term endurable responses. Having a serum marker to predict early disease progression may help plan immunotherapeutic regimens

Aims: The purpose of this study was to assess the use of serum levels of S100 in stage IV melanoma patients treated with Vemurafenib in predicting resistance to therapy and stage IV melanoma patients treated with the monoclonal anti-PD1 to help predict minimal disease progression.

Methods: S-100B were measured in serum samples of patients with metastatic melanoma treated with Vemurafenib (n=6) or anti PD1 (n=10). serum levels of S100 were correlated with clinical parameters of the disease.

Results: S100 in serum showed correlation with disease progression in some of the cases but was not consistent in all.

Conclusion: there is need to further study the role of serum levels of S100 to evaluate its role in predicting disease resistance to Vemurafenib and disease progression in patients treated with anti PD1

O17.7 Biomarkers for uveal melanoma –metastasis prediction

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Background: Uveal Melanoma (UM) sends metastasis mainly to the liver. Therefore, it is essential to detect early metastasis formation and improve survival. We have previously demonstrated elevated serum biomarker levels as Osteopontin (OPN), S-100, Melanoma-inhibitory activity (MIA), Tissue Polypeptide-Specific Antigen (TPS), VEGF, in metastatic UM vs disease-free (DF) patients (pts).

Aims: To study which Biomarkers will best predict and detect early Liver Metastasis, enabling treatment with impact on survival.

Methods: Levels of serum biomarkers were analyzed for 95 Uveal Melanoma pts, 75 metastatic UM, compared to 98 controls.

A matched-pairs analysis was used to compare pre and post treatment marker levels. Differences in biomarker levels were analyzed for each treatment group and correlated to metastasis vs DF status of pts.

Results: Of the 95 pts, 62 underwent brachytherapy and 13 were enucleated.

A significant decrease in S-100 levels in the enucleated pts (0.09 to 0.06 µg/l, p=0.0492) and in the brachytherapy pts (0.11 to 0.08 µg/l, p=0.0251) after 10 m were found. Significant increasing levels (5-25 fold) (p=0.002) in the BioMarkers OPN, MIA and S -100 were predictive of liver metastases, as shown later by CT.

TPS and VEGF changes were less significant, due to a large pts heterogeneity.

Conclusion: Local treatment (enucleation) of UM resulted in significant decreases in S-100 after 1 m and in OPN levels, after 4 m. Significant increases in all 5 markers detected earlier than CT and liver enzymes the development of UM liver metastases.

We introduced S100 and OPN as routine biomarkers for UM pts.

O18

RADIOLOGY and RADIOBIOLOGY

O18.1 Genome wide expression analysis of radiation induced DNA damage responses in isogenic HCT 116 cell lines

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Background: The fate of a tumor cell following irradiation is pleomorphic and related to dose (quality and rate), origin of cell and mutations within the cell cycle regulating genes.

Aims: To investigate the response of isogenic human colon carcinoma cells to irradiation and the mechanisms behind the response.

Methods: The gene expression profile was performed on isogenic tumor cell lines HCT 116 p53 (+/+) and HCT 116 p53 (-/-) following exposure to 5 Gy of ionizing radiation ((60) Co) using Illumina arrays. Signaling pathways were explored using MetaCore. Cellular viability, activation of cell cycle checkpoints (FACS) and induction of senescence (β -galactosidase) were used to correlate the biological responses to gene expression changes.

Results: Gene expression analysis revealed genes with significantly altered expression in cell cycle regulation, senescence and apoptosis for (+/+) cells. Cell cycle progression linked genes, like Cdc25A and Cyclin A were found to be down regulated at 12 h. This could also be seen by FACS analysis in which a G2/M arrest could be seen at 6 hrs followed by accumulation in both subG1 and G2/M. Rb, which is an important regulator of cell cycle progression and senescence was upregulated. 3-4 days after radiation there was an upregulation of apoptosis genes like BAX, TRAIL, Gadd45, p21 and Fas. β -galactosidase assay confirmed that (+/+) cells undergo senescence 5 days following irradiation in contrast to (-/-) cells with no senescence. (-/-) cells had an altered gene expression coupled to cell cycle regulation and apoptosis which appeared at 72 h. A transient G2/M arrest could be seen at 24 h after radiation followed by an increase of cells in both G1 and subG1 phase. A comparison of cell viability between cell lines revealed that (+/+) cells were more sensitive to lower doses of radiation than (-/-).

Conclusion: This study indicates that gamma-radiation induces senescence and apoptosis in HCT 116 p53+/+ cells. HCT 116 p53 -/- do not undergo senescence but arrest in G2/M and later undergo delayed apoptosis, similar to cells that undergo mitotic catastrophe.

O18.2 Targeting vascular endothelium for radiosensitization and tumor cure

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Background: While there is significant interest in combining anti-angiogenic therapy with conventional anti-cancer treatment, clinical trials have as of yet yielded limited therapeutic gain, mainly because mechanisms of anti-angiogenic therapy remain to a large extent unknown. Currently, anti-angiogenic tumor therapy is conceptualized to either “normalize” dysfunctional tumor vasculature, or to prevent recruitment of circulating endothelial precursors into the tumor. Genetic data indicate an acute wave of ceramide-mediated endothelial cell (EC) apoptosis, initiated by acid sphingomyelinase (ASMase), regulate tumor stem cell response to single dose radiotherapy (SDRT), obligatory for tumor cure.

Aims: Elucidating mechanism of action of anti-angiogenic agents when combined with SDRT.

Methods: EC were irradiated in the presence or absence of anti-VEGFR2 or anti-VEGF antibodies added at different times before or after SDRT. ASMase activation, ceramide levels and apoptosis were measured. MCA/129 fibrosarcoma tumors were implanted in *asmase*+/+ mice or *asmase*-/- littermates and irradiated in the presence or absence of anti-VEGFR2 or anti-VEGF antibodies injected at different times before or after SDRT. Tumor response was correlated to EC apoptosis.

Results: Here we show that VEGF prevented SDRT-induced ASMase activation in cultured EC, occurring within minutes after SDRT exposure, consequently repressing apoptosis. Anti-VEGFR2 acts conversely, enhancing ceramide generation and apoptosis. In vivo, MCA/129 fibrosarcoma tumors were implanted in *asmase*+/+ mice or *asmase*-/- littermates and irradiated in the presence or absence of anti-VEGFR2 or anti-VEGF antibodies. These anti-angiogenic agents, only if delivered immediately prior to SDRT, de-repressed SDRT-induced ASMase activation, synergistically increasing the EC apoptotic component of tumor response and tumor cure. Anti-angiogenic radiosensitization was abrogated in tumors implanted in *asmase*-/- mice that provide apoptosis-resistant vasculature, or in wild-type littermates pre-treated with anti-ceramide antibody, indicating that ceramide is necessary for this effect.

Conclusion: Understanding the temporal sequencing of anti-angiogenic drugs and radiation enables optimized radiosensitization and design of innovative radiosurgery clinical trials.

O18.3 Placental cells based treatment for radiation syndrome as an indirect cell therapy versus stem cell based tissue engineering

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Background: Though a wide range of cell types are now available for cell therapies, their effective application in tissue regeneration and tissue engineering is limited. Our findings in previous studies suggested indirect a “bystanders” effect of cell therapy, based on the secretion profile of a wide range of factors that may contribute to the repair of the target tissues, while the role of the “stemness” of such cells for their therapeutic effect was not straightforward.

Aims: We investigated a new approach to mitigate lethal high dose of irradiation in conditions of acute radiation syndrome we used 3D expanded adherent human derived placental stromal cells with adequate profile of chemokines and cytokine secretion (PLX-RAD).

Methods: Due to the expected systemic effect, 2 million cells could be delivered IM per injection to the muscles of irradiated C3H/HeN mice at day 1 and 5 after 7.7 Gy irradiation. The injected PLX-RAD cells were traced only within the injected muscle for 6–10 days with no observed adverse effects.

Results: The IM PLX-RAD treatment increased the survival rate of the mice exposed to high radiation dose of 7.7 Gy from ~27 % to ~98 % ($p < 0.001$). Delaying the initiation of the treatment to 48 hrs after irradiation yielded similar effect. Further delay to 72 hrs after irradiation, was found to be less effective, with some better survival than the injection of just a vehicle control. These findings were supported by the fast recovery of the bone marrow (BM) and reconstitution of all blood cell lineages in the PLX-RAD treated mice during the follow-up. The proportion of CD45+/SCA1+ hematopoietic progenitor cells within the nucleated BM population in the irradiated mice was increased in the animals treated by PLX-RAD.

Conclusion: IM treatment with PLX-RAD cells may serve as a safe complication free “off shelf” therapy to treat bone marrow failure following exposure to high heterogeneous doses of radiation with possible application for nuclear disasters, as well as for clinical practice in cases of BM aplasia, as already successfully tested at Hadassah Hospital in a compassionate treatment for 2 terminal patients.

O18.4 Early cytokines levels dynamics may predict severity of acute mucositis during chemo-radiation Head and Neck Cancer—a prospective pilot study

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Background: Oral mucositis during chemo-radiation for Head and Neck (H&N) Cancer is a severe side effect that may affect the treatment continuity. In severe cases the patients may need active interventions like insertion of percutaneous gastrostomy (PEG), however there is no test that can predict the severity of mucositis early in the treatment, and thus far, only clinical parameters are used.

Aims: This study aims to evaluate whether early cytokine markers dynamics can predict the severity of mucositis during chemo-radiation for H&N Cancer.

Methods: This is a Helsinki committee approved study. Fifteen patients with H&N Cancer who received chemo-radiotherapy using standard protocols were included. Patients were evaluated for mucositis according to WHO common toxicity criteria, and by quality of life questioners at baseline and weekly during the treatment. Blood samples for inflammatory cytokines (IL-1, IL-6, IL-8, TNF-alpha, IL-2R) and anti-inflammatory cytokines (IL-10) were drawn prior and weekly during the treatment.

Results: A correlation between IL-6 and IL-2R serum levels and severity of mucositis and dysphagia was found; specifically, high IL-6 and IL-2R levels at week 2 were correlated with a need for PEG tube installation. A seemingly reverse correlation was found between low IL-8 serum levels and the severity of mucositis. No correlation was found between IL-1, TNF-alpha and IL-10 levels and the severity of mucositis.

Conclusion: These pilot study results, demonstrate a correlation between IL-6, IL-2R and a reverse correlation of IL-8 serum levels and severity of mucositis during chemo-radiation for H&N Cancer. These results should be tested in a large scale study.

O18.5 Toxicity of induction chemotherapy with docetaxel, cisplatin, & 5-fluorouracil for advanced head and neck cancer

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Background: induction chemotherapy followed by radiotherapy alone or concurrent chemo-radiotherapy is an option for the treatment of locally advanced H&N carcinoma. the

best combination for induction chemotherapy is the (docetaxel-cisplatin-5-fluorouracil) TPF combination

Aims: The aim of this study was to evaluate the feasibility of induction chemotherapy with (TPF) followed by external radiotherapy (EBRT) with concomitant chemotherapy (CRT) or cetuximab (ERT) in the treatment of patients with advanced squamous cell carcinoma of the head and neck (SCCHN).

Methods: The data of all patients with advanced SCCHN, stage III and IV, treated in 2007-2010 were reviewed. Tolerability was assessed and scored by the proportion of patients completing the planned study protocol. Toxicity was scored using the NCI common Toxicity Criteria (version 4) for classification of adverse events.

Results: Fifty-three patients were included in the study. TPF was initiated in reduced dose in 13 (25 %) patients. Twenty-two (41.5 %) patients received primary prophylaxis with GCSF and 42 (77 %) patients completed treatment according to schedule. During the induction phase, one (2 %) patient died and 24 (45 %) patients had one or more grade 3-4 complications. The number of patients who developed neutropenia was lower in the group who received primary GCSF prophylaxis. Secondary dose reductions were required in 21 % of patients.

Conclusion: It was found that induction TPF was associated with grade 3-4 toxicity and prophylaxis with GCSF should be part of the treatment regimen.

O19

ROCHE SYMPOSIUM

O19.1 A european multicenter study for the evaluation of the analytical and clinical performance of the Elecsys® HE-4 assay on Cobas e411 from roche diagnostics

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Background: HE-4 is known as a valuable marker for the management of ovarian cancer.

Aims: To evaluate analytical and clinical performance of Elecsys HE4 on Cobas e411.

Methods: The analytical and clinical performance of Elecsys HE4 on Cobas e411 (Roche Diagnostics) was evaluated at three different European sites and compared with a reference system, defined as HE-4 on the Fujirebio HE4 EIA (Fujirebio Diagnostics Inc).

Results: Total imprecision (%CV) of Elecsys HE4 ranged between 1.99 % and 8.3 %; inter-laboratory reproducibility between 1.1 %-5.1 %. Linearity upon dilution showed a mean recovery of 98.5 % (SD +9.1 %). Endogeneous interferences had no influence on HE4 levels. There was no high-dose hook effect up to 12.786 pmol/l.

Clinical performance was investigated in sera from 1,395 individuals showed similar results with both methodologies (Roche vs Fujirebio) in all groups: healthy (mean 50.2 vs 45.9 pmol/l; 95 % percentile 96.5 vs 93.6 pmol/l), pregnant women (mean 35 vs 35.3 pmol/l; 95 percentil 55.7 vs 54.9 pmol/l), benign gynaecological diseases (49.7 vs 46.5 pmol/l; 95 percentil 112.5 vs 104.3 pmol/l) or in ovarian cancer (media 302.2 vs 303.7 pmol/l; 95 percentil 3244 vs 3610 pmol/l) Significantly higher concentrations of HE4 were found in patients with benign disease than in healthy women, but ROC curves showed a significantly higher specificity, mainly in premenopausal women, that those obtained with CA 125. HE-4 serum levels were clearly related to age, race (highest in Caucasians), pregnancy (lower concentrations). Clinical value of HE4 results (benign gynaecological vs ovarian cancer) HE-4 results were similar with both methodologies, Roche vs Fujirebio (AUC 0.926 vs 0.920).

Conclusion: Elecsys HE-4 provides accurate methodological characteristics and demonstrates an analytical and clinical correlation with Fujirebio HE-4. HE-4 sensitivity and specificity was significantly better than those obtained with CA 125.

O19.2 A predictive model combining HE4 and ultrasound features for ovarian cancer

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Background: Ovarian cancer is the leading cause of death of gynecologic cancer among women. The majority of women are diagnosed with advanced stage ovarian cancer. CA125 is the most commonly used biomarker for the diagnosis of ovarian cancer, but is limited in both sensitivity and specificity. Therefore, research has been done to evaluate the use of novel biomarkers alone or in combination with CA125. In the Netherlands, the RMI (Risk of Malignancy Index) is mostly used to distinguish patients with benign ovarian lesions from patients with malignancy. The potential role of Human Epididymal secretory protein 4 (HE4) as an ovarian cancer biomarker has been explored intensively. Combining the results of serum HE4 and CA125 in an algorithm named ROMA (Risk of Malignancy Algorithm), may provide better accuracy in the risk stratification for ovarian cancer than either measurement alone. However, the (additional) predictive value of ultrasound is not included in this algorithm.

Aims: In the present study HE4 was evaluated for the pre-operative diagnosis of ovarian cancer and a predictive model including both biomarkers and ultrasound features was developed.

Methods: HE4 and CA125 were measured in 361 patients (34 benign, 147 ovarian cancer, 180 healthy controls). Clinical data and ultrasound and CT results were collected. Sensitivity, specificity and area under the curve (AUC) for CA125, HE4, RMI and the ROMA score for the diagnosis of ovarian cancer were calculated using receiver operating characteristic (ROC) analysis. A predictive model was constructed including ultrasound and CT-scan.

Results: The sensitivity and specificity of CA125 and HE4 in our study population is 89.8 % and 89.1 % and 64.7 % and 97.1 % respectively. The pre- and postmenopausal ROC-AUCs of ROMA and RMI were very similar, with ROMA being slightly better than RMI (AUC 0.94 vs 0.91 and 0.95 and 0.91 respectively). We found that HE4 adds significantly to the predictive value of the RMI ($P < 0.0001$).

Conclusion: In the diagnostic work up for ovarian cancer, ultrasound is indispensable. Therefore, ultrasound features should be included in a predictive algorithm. HE4 adds significant predictive value to RMI. In a further study a predictive model will be prospectively validated.

O19.3 Serum biomarkers in the prediction and early estimation of anticancer therapy

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Background: Serum biomarkers are valuable for supporting diagnosis, estimation of prognosis and therapy monitoring in many cancer entities. As increasing therapeutic alternatives are available nowadays – even in the advanced tumor stage – the pretherapeutic selection of the patients for the appropriate therapies becomes more and more important. In addition, early estimation of therapy efficacy after the first application of systemic therapy is necessary to modulate the therapy plan more efficiently, avoid unnecessary toxicity, morbidity and costs.

Aim: Therapy targets, such as Her2-neu, are relevant biomarkers for selecting subgroups of patients for biological therapies. Here, we investigated tumor-, organ-, cell death- and inflammation-related serum biomarkers in groups of non-selected cancer patients for their potential to predict or anticipate response to systemic cytotoxic therapy.

Methods: Diverse setting of around 1000 cancer patients, including lung, breast, colorectal, liver, and pancreatic tumors in limited or advanced tumor stages, who

underwent systemic or local chemo- or radiation therapy, were tested. In serially collected serum samples of these patient groups, the relevant established oncological biomarkers, the cell death markers CYFRA 21-1 and nucleosomes, the immunogenic markers HMGB1 and RAGE, as well as further organ- and inflammation related markers were investigated.

Results: Depending on the clinical setting, we found pretherapeutic, tumor and non-tumor-related biomarkers that indicated highly significantly the later therapy outcome. In addition, the biomarker concentrations obtained after only one application of the therapy showed an even higher potential to estimate therapy efficacy in many studies. Especially, the cell death biomarkers CYFRA 21-1 and nucleosomes were very informative markers in this context, for example in lung and gastrointestinal cancers. In addition, the new immunogenic markers HMGB1 and RAGE were meaningful in breast and gastrointestinal cancers. Multivariate analyses, including biomarkers and clinical factors, revealed that combinations of several independent biomarkers enhanced the power of early therapy response estimation.

Conclusion: Serum biomarkers are highly relevant for the prediction and early estimation of anticancer therapy. The biomarker panels for each tumor type, therapy and the time points of monitoring have to be standardized and validated to make them a valuable tool in the clinical patient management.